

Log abundance in characteristic zero

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September 24, 2026

Abstract

We prove the rational-boundary log abundance conjecture in every dimension over algebraically closed fields of characteristic zero: every nef $\mathbb{Q}$ -Cartier log canonical divisor on a projective log canonical pair with effective rational boundary is semiample. Over $\mathbb{C}$, the proof also establishes canonical nonvanishing for smooth projective varieties in every dimension.

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1 Introduction

Let (X, B) be a projective log canonical pair with rational boundary. Its log canonical divisor $K_X + B$ records the canonical class together with the boundary. A rational Cartier divisor is *nef* if its degree on every curve is nonnegative. It is *semiample* if a positive Cartier multiple is generated by global sections. Such a multiple defines a morphism, so semiamplicity turns numerical positivity into a fibration described by sections. The log abundance conjecture asks whether a nef log canonical divisor is semiample. We give a positive answer in characteristic zero.

Theorem 1.1 (Log abundance). *Let (X, B) be a normal projective log canonical pair over an algebraically closed field of characteristic zero. Assume that B is an effective rational divisor and $K_X + B$ is $\mathbb{Q}$ -Cartier. If $K_X + B$ is nef, then some positive Cartier multiple of it is generated by global sections.*

The proof proceeds by dimension induction over $\mathbb{C}$. Its inductive assertion is stronger than the displayed theorem: every projective lc pair with effective real boundary and pseudo-effective real Cartier adjoint has a good log minimal model, on which the adjoint is semiample. Smooth canonical nonvanishing asks whether a pseudo-effective canonical class on a smooth variety has a nonzero pluricanonical section. We establish this statement in each dimension before that good-model assertion is used in the same dimension. At the end, descent of the evaluation map of an actual Cartier line bundle gives the theorem over every algebraically closed field of characteristic zero.

We write $\kappa(L)$ for the Iitaka dimension of a rational divisor L , with value $-\infty$ if all positive integral systems are empty. The consequences fall into three groups. Good models, rational-linear

triviality when $\kappa = 0$, and gluing and extension across boundaries are treated in Section 11. Finite generation of the algebras of rounded pluricanonical sections is proved in Section 12, both absolutely in characteristic zero and for proper morphisms over algebraic bases over $\mathbb{C}$. Section 13 relates canonical nonvanishing to rational curves, cotangent tensors, fundamental groups, and quasi-projective covers. These sections give the full statements, additional hypotheses, and derivations from the work of Fujino–Gongyo, Boucksom–Demailly–Păun–Peternell, Lazić–Peternell, Brunebarbe–Campana, Campana, Taji, and Claudon–Hörling–Kollár.

There is also a relative analytic application of Fujino’s theorem: Section 14 treats projective surjective morphisms of normal complex analytic varieties with rational lc adjoint nef over the whole base. It obtains a relatively generated Cartier multiple near each compact base subset; the multiple may depend on that subset. This is distinct from the compact Kähler symmetric-pluriform criterion in Section 13; the cotangent-invariant and fundamental-group consequences there remain projective.

Historical development

The threefold proofs show why nonvanishing and boundary geometry are central to abundance. Miyaoka proved the numerical-dimension-one case by studying an effective pluricanonical divisor and its infinitesimal neighborhoods after suitable coverings [50]. Kawamata proved the remaining numerical-dimension-two case for minimal threefolds [38, Theorem 3.1]. His Section 4 gives an alternative proof of Miyaoka’s case through finite covers, Hodge theory and compatible infinitesimal deformations. Keel, Matsuki and McKernan established the logarithmic theorem [39, 48, 40]. Fujino extended abundance to semi-log-canonical threefolds [21], where sections on the normalization must agree along the conductor before they define sections on the original space. This compatibility is also needed in higher-dimensional induction: the coefficient-one boundary can be reducible even when the ambient variety is normal.

In higher dimensions, Birkar, Cascini, Hacon and McKernan established minimal models for klt pairs in the big-boundary or log-general-type range [6]. Their finite-generation theorem for projective rational klt pairs does not require bigness [6, Corollary 1.1.2]. The rational lc finite-generation consequence below goes beyond that klt setting. Two distinctions remain essential on the boundary of the positive cone. First, nonvanishing produces sections but does not by itself assert that a nef adjoint is semiample. Hashizume reduces real-boundary log canonical nonvanishing and ordinary minimal-model existence through dimension n to smooth canonical nonvanishing in dimension n [36, Theorem 1.4]. Our induction must then make these models good. For positive Kodaira dimension, the required step is supplied by lower-dimensional good models [28, Lemma 3.5]. Second, Kodaira dimension zero and numerical dimension zero are different hypotheses. Gongyo proves that a numerically trivial rational log canonical adjoint is $\mathbb{Q}$ -linearly trivial [30, Theorem 1.2]. In the zero-Iitaka argument below, numerical triviality forces an already available effective representative to vanish.

Recent work of Liu and Xu treats numerical dimension at most one. Their Theorem 5.1 gives good minimal models for projective log canonical pairs of dimension at most five with nonnegative invariant Kodaira dimension and numerical dimension at most one; their Theorem 5.4 gives a reduction in that numerical range to smooth nonvanishing [47]. The induction here has no numerical-dimension restriction. The ordinary-pair specialization of the companion generalized-minimal-model paper [53, Corollary 1.2] concerns existence of a nef model. It is not an input to this good-model induction. After abundance has been proved, we use that ordinary-pair result in the absolute finite-generation argument of Section 12.

The two tasks in the induction

Assuming good models in dimensions below n , we first prove smooth canonical nonvanishing in dimension n , and then make the resulting log minimal models good. Both tasks use a criterion for a *signed* representative on a reduced boundary. The lower-dimensional hypothesis supplies its whole-boundary semiample assumption. We describe this common ingredient first. For a nef divisor L , write $\nu(L) = \max\{j : L^j H^{n-j} > 0\}$, where H is ample and $n = \dim X$. The criterion is

$$\begin{aligned} L = K_X + D \text{ nef}, \quad L - cD \text{ pseudo-effective } (c > 0), \\ L \sim_{\mathbb{Q}} \sum_i a_i D_i, \quad L|_D \text{ semiample} \implies \kappa(L) = \nu(L). \end{aligned}$$

Here (X, D) is projective $\mathbb{Q}$ -factorial dlt, and $D = \sum_i D_i$ is the decomposition of the reduced boundary into its prime components. The rational coefficients a_i may be positive, negative, or zero. Semiample is required on the entire reduced scheme D . Normalization and conductor gluing therefore belong to the induction, as in the slc interface of Fujino–Gongyo [26, Theorem 1.5]. For the good-model task, positive Kodaira dimension is handled by lower-dimensional good models; the signed criterion addresses the remaining zero-Iitaka-dimension case in Proposition 2.5.

The signed criterion is proved geometrically. A generated multiple of $L|_D$ defines a morphism from D to projective space. Over a general point of the image of a positive-coefficient component, a root construction separates that component from the negative and coefficient-zero boundary. A filtered Hodge-module calculation lifts local parameters on this image, together with a parameter transverse to the lifted boundary, through all infinitesimal neighborhoods. Fixing the lifted base parameters and varying the transverse one deforms a boundary fiber into a projective subvariety disjoint from D . Varying the fiber gives a dominating family on which L is trivial. Section 5 explains the final geometric step: these subvarieties bound the dimension of the nef reduction, and vertical descent identifies L with the pullback of a big divisor on its base.

This formal deformation method has a direct precedent in Miyaoka’s construction; Miyaoka credits Reid with the suggestion to analyze a pluricanonical divisor [50, p. 220]. Boundary extension provides a second related approach. Demailly, Hacon and Păun use a singular-metric refinement of Ohsawa–Takegoshi extension to extend pluricanonical sections in the purely log terminal setting [18, Theorem 1.7 and Corollary 1.8]. Their nef corollary requires an effective representative whose support contains the coefficient-one divisor and lies in the boundary. Chan and Choi subsequently removed the containment of that representative in the boundary [13, Theorem 1.6.1]. Here the representative may have both signs, and compatibility is required across the whole reducible boundary. We therefore construct compatible formal lifts on the whole reduced boundary. Semiample on that boundary is supplied by the normalization theorem of Fujino and Gongyo [26, Theorem 1.5].

The construction combines familiar tools with an additional compatibility argument. Normalized cyclic covers and their eigenspaces follow the formalism of Esnault and Viehweg [20, Section 3]. Deligne’s logarithmic Hodge theory supplies the underlying framework [16]. On the singular boundary, we use Saito’s projective strictness and graded de Rham vanishing, together with his comparison with the Du Bois complex [56, Theorem 2.14 and Proposition 2.33] [57, Theorem 0.2 and Corollary 0.3]. The proof must still identify the obstruction to lifting and show that these ordinary-variety results apply to the covering construction.

Smooth nonvanishing. To prove smooth nonvanishing, suppose instead that a counterexample exists. The zero-boundary case of logarithmic Iitaka subadditivity, stated below as Theorem 1.2, first excludes positive irregularity through the Albanese map. This is its only use in the proof. The geometric preparations rule out two ways it could be covered by lower-dimensional subvarieties:

families that carry a positive current pulled back from a smaller space, and covering families of curves with bounded genus and bounded degree after normalizing the nef divisors used in the construction to a fixed small positive volume. The first exclusion uses positive currents and algebraic webs. The second uses an ascending chain condition for the singularity multiplicities of one fixed current, measured by Lelong numbers at valuations of bounded discrepancy. The signed criterion has a second role here: it forces the full reduced support of any signed canonical representative to have big logarithmic adjoint, as proved in Proposition 6.10. This will control the multisections in the two-slot construction below.

These exclusions produce two incompatible estimates for vanishing at a moving point. At a very general point, the first estimate bounds the vanishing order of every nonzero section of a Cartier multiple kP of a suitable nef divisor P by k times a small normalized constant. To obtain sections with stronger jet separation, we use the projective bundle associated to the sum of the two pullbacks of a line bundle to the product of a smooth model with itself. Its two summands allow the argument to move in either factor.

Failure of the required jet separation would produce a moving base component. Slice estimates and the curve exclusions leave only components that project generically finitely onto both base factors, hence give correspondences between them. Their branch divisors cannot sweep either factor; a fixed branch complement and bounded degree therefore leave only finitely many covers. After fixing the covers, the correspondences give a family of birational maps whose graphs dominate their product. Hanamura’s birational-group theorem then makes a fixed cover birational to an abelian variety [35, Theorems 2.1–2.2]. The ramification and positive-current argument in Section 6 would give a pluricanonical section on the counterexample, again a contradiction.

Once the projective bundle, its divisors, and the required finite collection of jet sections are fixed, reduction modulo a large prime gives an evaluation map on the product of two copies of the Frobenius-twisted model. The map has full generic rank. Its restriction to the ordinary diagonal is controlled by sections on the Frobenius thickening of the diagonal, where the two relative Frobenius maps agree. A filtration of those sections gives a much smaller rank bound there. The determinant is a section of an external product of two line bundles. The first estimate, expressed by a fixed movable curve on a blowup, bounds vanishing in each factor and hence the order of the determinant at the selected diagonal point. The rank deficit gives a larger lower bound on that order, a contradiction.

Theorem 9.1 states this incompatibility for a fixed smooth variety, fixed jet sections, and a fixed movable-curve bound. Its proof opens Section 9 and uses neither abundance nor subadditivity. The geometric application follows in the same section, using the fixed inputs supplied by the preceding curve exclusions and jet constructions. The numerical subadjunction and moving-kernel results needed for those constructions are proved in Section 7.

The numerical part combines two established methods. The argument that follows a moving base component uses differentiation in the parameter space, as in Ein–Küchle–Lazarsfeld [19, Proposition 2.3]; we give the tracking and adjunction estimates needed here. After reduction to positive characteristic, we compare ordinary jets with Frobenius jets using the local containment recorded by Mustaș–Schwede [52, Equation (2.8)]. The resulting evaluation map is studied through Sun’s canonical Frobenius filtration [59, Theorem 3.7], whose curve case is treated by Joshi–Ramanan–Xia–Yu [37, Section 5.3]. We use its description by powers of the diagonal ideal due to Kitadai–Sumihiro [41, Definition 3.1 and Remark 3.2]. Langer’s instability estimate [45, Corollary 2.5] then controls the slopes of the graded bundles.

The subadditivity input

The all-dimensional argument uses the following completed companion result. Its application occurs only in Lemma 6.1.

Theorem 1.2 (Logarithmic Iitaka subadditivity, [54, Corollary 6.2]). *Let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective complex varieties. Let D_X, D_Y be reduced effective simple normal crossing divisors, allowing zero boundaries, such that*

$$\text{Supp}(f^*D_Y) \subseteq \text{Supp}(D_X).$$

For a very general smooth fiber F , put $D_F = D_X|_F$. Then

$$\kappa(X, K_X + D_X) \geq \kappa(F, K_F + D_F) + \kappa(Y, K_Y + D_Y).$$

Here D_F is reduced with simple normal crossings and $K_F + D_F \sim (K_X + D_X)|_F$. The convention $(-\infty) + b = -\infty$ applies also when $b = -\infty$, and the zero divisor on a point has Iitaka dimension zero.

The support inclusion allows additional components in D_X , and the morphism need not be smooth away from the boundary. The theorem has no nefness, abundance, bigness, or good-model hypothesis. Its zero-boundary case is used in the Albanese reduction of Section 6; this is the sole subadditivity input in our proof. The separate characteristic-zero specialization [54, Corollary 6.3] uses a geometric generic fiber. We keep that field-and-fiber statement distinct from the complex, very-general-fiber statement above. No fractional-boundary, orbifold, nonprojective, Hodge-conjectural, or arithmetic extension is assumed here.

The particular zero-boundary application in Lemma 6.1 also follows from the external maximal-Albanese-dimension theorem of Hacon, Popa and Schnell [34, Theorem 1.1]: the smooth base there maps generically finitely to a subvariety of an abelian variety. Cao and Păun prove the underlying abelian-base case [12, Theorem 1.1]. These results cover that application, not the full logarithmic statement above.

Reading order. Section 2 sets out the dimension induction. Sections 3–5 prove the signed-boundary criterion: first the root and adjunction construction, then infinitesimal lifting, and finally compact fibers and descent. Section 6 derives the geometric exclusions for a hypothetical smooth counterexample. After the local tools of Section 7, Section 8 constructs its scalar and two-slot jets. Section 9 proves the independent Frobenius comparison and then constructs its fixed witnesses to establish smooth nonvanishing. Sections 10 and 11 finish the induction, descend actual global generation, and record good models, linear triviality, semi-log-canonical abundance, and supported dlt extension. Section 12 then proves finite generation of rational log canonical rings in the absolute and complex proper relative settings. Section 13 combines canonical nonvanishing and abundance with classical results on rational curves, cotangent tensors, fundamental groups, and quasi-projective covers. Section 14 records relative analytic abundance near compact subsets of the base for projective morphisms. A reader using only the finite-data comparison can start with Theorem 9.1 and its proof in Section 9; neither assumes the counterexample geometry.

1.1 Conventions

A variety is integral. Canonical divisors are chosen compatibly on birational models. For a divisor E on a smooth model $p : W \rightarrow X$, our log discrepancy is

$$a(E; X, B) = 1 + \text{ord}_E(K_W - p^*(K_X + B)).$$

Log canonicity means nonnegative log discrepancies, and klt means strictly positive log discrepancies. We use the usual dlt and slc conventions. For non-nef pseudo-effective divisors, statements about numerical dimension use Nakayama's κ_σ , as specified in the relevant reduction. These notions are not interchanged without a hypothesis that justifies doing so.

For nonempty systems, the Iitaka dimension is the maximum dimension of their images. Numerical and rational-linear equivalence are denoted by $\equiv$ and $\sim_{\mathbb{Q}}$, respectively. Unless a different base field is specified, the constructions are over $\mathbb{C}$. A very general point avoids a countable union of proper closed subvarieties. The transfer to arbitrary algebraically closed characteristic-zero fields is made only after the complex argument has been completed.

2 The inductive framework

The induction has two outputs in each dimension: smooth canonical nonvanishing and good minimal models for real-boundary lc pairs. This section proves the passage from the first output to the second, assuming good models in smaller dimensions. All varieties here are projective over $\mathbb{C}$.

For an effective real boundary Δ , a good log minimal model of (X, Δ) is a log minimal model on which the adjoint divisor is semiample. For real divisors, semiampleness means real linear equivalence to the pullback of an ample real divisor by a contraction. We allow the usual definition of a log minimal model in which extracted prime divisors are included in its boundary with coefficient one. For a pseudo-effective divisor A , we use $\kappa_\sigma(X, A)$ for Nakayama's numerical dimension. When A is nef this agrees with the intersection-theoretic numerical dimension $\nu(X, A)$.

Fix an integer $n \geq 1$.

Assumption 2.1 (Lower-dimensional good models). Every projective log canonical pair of dimension less than n , with real boundary and pseudo-effective adjoint divisor, has a good log minimal model.

The base case is a point. At the next step, smooth nonvanishing in dimension n will be proved from Assumption 2.1 in Theorem 9.6. It will then become an input to Proposition 2.5. The boundary-restriction argument below and the special-termination argument in Section 6 are available before this same-dimensional nonvanishing step; they use only the stated lower-dimensional hypothesis.

For clarity, let $\text{GLM}_{\leq d}$ denote good-model existence for all projective complex lc pairs with effective real boundary and pseudo-effective adjoint in dimensions at most d . Let NV_n denote smooth canonical nonvanishing in dimension n , and let SA_n denote semiampleness of nef rational lc adjoints in that dimension. The proof proceeds in the following order; the real-boundary conclusion in the third row supplies the next dimension's hypothesis.

Stage	Input	Output and location
Base	Dimension zero	$\text{GLM}_{\leq 0}$: a point
Smooth step	$\text{GLM}_{\leq n-1}$ and Theorem 1.2	NV_n , Theorem 9.6
Good-model step	$\text{GLM}_{\leq n-1}$ and NV_n	$\text{GLM}_{\leq n}$, Proposition 2.5
Nef conclusion	$\text{GLM}_{\leq n}$	SA_n , Lemma 2.2

The signed-representative theorem used in both steps is proved in Sections 3–5; its whole-boundary semiample hypothesis comes from the lower-dimensional row, via Proposition 2.3.

2.1 Comparison and boundary restrictions

We begin with two forms of descent. The first compares the adjoint on a pair and its minimal model as actual divisors on a common resolution. The second joins the sections obtained by adjunction on

the components of a reduced dlt boundary.

Lemma 2.2 (Comparison with a nef model). *Let (X, Δ) be log canonical and let (X', Δ') be a log minimal model. On a common resolution*

$$X \xleftarrow{u} W \xrightarrow{v} X'$$

there is an effective v -exceptional divisor F such that

$$u^*(K_X + \Delta) = v^*(K_{X'} + \Delta') + F. \quad (1)$$

If $K_X + \Delta$ is nef, then $F = 0$. Consequently a nef rational adjoint is semiample whenever it has a good log minimal model. If (X, Δ) is klt, its log minimal model is klt and extracts no divisors.

Proof. Put $F = u^*(K_X + \Delta) - v^*(K_{X'} + \Delta')$. Discrepancy improvement on divisors of X gives $u_*F \geq 0$, while $-F$ is u -nef because $K_{X'} + \Delta'$ is nef. The negativity lemma therefore gives $F \geq 0$. At a prime of W that is not v -exceptional, the coefficient of F is zero unless that prime is extracted on X' . In the latter case it equals the negative of its log discrepancy over (X, Δ) . It is therefore nonpositive, and hence zero. Thus F is v -exceptional. If the source adjoint is nef, then F is also v -nef, so the negativity lemma gives $F = 0$.

The equality of pullbacks transfers semiampleness of a rational adjoint. Indeed a proper birational morphism onto a normal variety has direct image of its structure sheaf equal to that structure sheaf. Projection formula therefore identifies the global sections of each Cartier multiple with the sections of its pullback. If every such pulled-back section vanished over a point downstairs, it could not generate upstairs over that point. Generation upstairs thus implies generation downstairs. Finally, for a klt source the coefficient at an extracted prime would be strictly negative, which is impossible. The remaining discrepancy inequalities show that the model is klt. $\square$

Proposition 2.3 (Semiampleness on the reduced boundary). *Assume Assumption 2.1. Let (V, D) be a projective $\mathbb{Q}$ -factorial dlt n -fold with D reduced and $M = K_V + D$ nef. Then $M|_D$ is semiample as a rational line bundle on the reduced scheme D .*

Proof. The ambient variety V is klt. The reduced floor of a $\mathbb{Q}$ -factorial dlt pair is S_2 , and it has ordinary double crossings in codimension one; its irreducible components are normal. For the S_2 assertion one can work locally with the cyclic class cover of the $\mathbb{Q}$ -Cartier divisor D . This cover is quasi-étale and klt, hence Cohen–Macaulay. The eigensheaf $\mathcal{O}_V(-D)$, being a direct summand of its finite direct image, is Cohen–Macaulay. The divisor sequence then shows that D is Cohen–Macaulay.

Divisorial adjunction on the normalization $\coprod D_i \rightarrow D$ gives lc pairs $(D_i, \text{Diff}_{D_i}(D - D_i))$. The different includes the conductor with coefficient one. Removing that conductor contribution defines an effective boundary on D ; together with D it is an slc pair whose adjoint line bundle is $M|_D$. The divisible residue identifications hold in codimension one, with even powers eliminating residue signs at double crossings, and extend by S_2 .

Each normalized adjoint is nef and semiample by Assumption 2.1 and Lemma 2.2. Semiampleness descends from the normalization by the slc gluing theorem [26, Theorem 1.5, equivalently Theorem 4.3]. This gives the required statement on the whole reduced scheme, not merely on its individual components. $\square$

2.2 A uniform trivial-perturbation argument

We will also perturb a nef adjoint into a klt program. A single Cartier index must survive every step: otherwise the smallness required of the perturbation could change along the program. The following estimate and line-bundle descent give that uniformity.

Lemma 2.4. *Let (Z, Δ) be a projective $\mathbb{Q}$ -factorial dlt rational pair of dimension n , and suppose $L = K_Z + \Delta$ is nef. Set $P = \Delta$ if the pair is klt and $P = \lfloor \Delta \rfloor$ otherwise. Fix $m > 0$ such that mL is Cartier. For every rational*

$$0 < \epsilon < \frac{1}{4nm + 2},$$

every step of an LMMP for $L - \epsilon P$ is L -trivial. On all its models, the transform of L is nef and its multiple by the same integer m is Cartier.

Proof. Both $\Delta - \epsilon P$ and $\Delta - \frac{1}{2}P$ are effective klt boundaries. If R is a ray negative for $L - \epsilon P$, nefness of L shows that R is also negative for $L - \frac{1}{2}P$. Choose a rational curve C spanning this ray and satisfying the length bound

$$-(L - \frac{1}{2}P) \cdot C \leq 2n.$$

Writing $a = L \cdot C \geq 0$ and $p = P \cdot C$, we have

$$p \leq 2a + 4n, \quad a < \epsilon p, \quad (1 - 2\epsilon)a < 4n\epsilon.$$

Our choice of ϵ gives $a < 1/m$. Since ma is a nonnegative integer, $a = 0$.

For the driving klt contraction $c : Z \rightarrow B$, the line bundle $\mathcal{O}_Z(mL)$ is numerically trivial over B . The line-bundle clause of the contraction theorem gives its descent to a line bundle $\mathcal{A}_B$ on B , with no change of m [22, Theorem 3.74(3)]. One may also see the unchanged index directly from relative basepoint freeness: two consecutive sufficiently large powers descend, so their quotient descends. The line bundle $\mathcal{A}_B$ is nef because its pullback $\mathcal{O}_Z(mL)$ is nef. For a flip $c^+ : Z^+ \rightarrow B$, its pullback $(c^+)^*\mathcal{A}_B$ is the line bundle of the transformed mL . This proves separately that the transformed L is nef and that mL remains Cartier with the same m .

The driving boundary remains klt throughout its LMMP. The transformed $\Delta - \frac{1}{2}P$ remains effective and is smaller than the driving boundary, so it too is klt. The same estimate and the same integer m apply inductively at every subsequent step. $\square$

2.3 The good-model implication

We now assume smooth nonvanishing in the current dimension and prove the good-model implication. The proof first reduces real nef boundaries to rational ones. The case of positive Kodaira dimension is then handled by lower-dimensional good models, leaving the zero-dimensional Iitaka case and its klt perturbations.

Proposition 2.5 (Inductive reduction to smooth nonvanishing). *Assume Assumption 2.1. Assume in addition that every smooth projective n -fold with pseudo-effective canonical divisor has nonnegative Kodaira dimension. Then every projective lc n -fold with real boundary and pseudo-effective adjoint has a good log minimal model. In particular, every nef rational lc adjoint in dimension n is semiample.*

Proof. Hashizume's reduction gives nonvanishing and log minimal models for projective lc pairs with real boundary in dimensions at most n [36, Theorem 1.4]. Thus we may pass to a $\mathbb{Q}$ -factorial dlt log minimal model. Fix the support of its boundary, impose the rational affine constraints that its coefficient-one components remain equal to one, and take a sufficiently small rational polytope around the given boundary within those constraints. On a fixed log resolution witnessing dlt, the strict discrepancy inequalities remain strict in this neighborhood; thus its boundaries remain dlt. The rational-polytope theorem for nef adjoints [4, Remark 3.1 and Proposition 3.2(3)], applied to all extremal rays and intersected with this neighborhood, expresses the given real boundary in a

finite rational simplex of dlt boundaries with nef adjoints. It is enough to prove semiample for these rational adjoints. Their positive real combinations are semiample, using the product of the associated contractions. For a rational adjoint, real semiample can also be rationalized: the finite linear equations expressing its divisor and principal-divisor coefficients have rational data, so a real solution with positive coefficients has a rational solution with the same positivity.

We therefore work with a rational dlt pair (Z, Δ) and $L = K_Z + \Delta$ nef. Real nonvanishing gives rational nonvanishing in this case: retain the finitely many divisors and principal divisors in an effective real representative and solve the resulting rational linear system with nonnegative rational coefficients. If $\kappa(Z, L) \geq 1$, lower-dimensional good models give a good minimal model by [28, Lemma 3.5]. Lemma 2.2 transfers semiample back to Z . It remains to treat

$$\kappa(Z, L) = 0.$$

The non-pseudo-effective perturbation. Put $P = \Delta$ in the klt case and $P = \lfloor \Delta \rfloor$ in the other case. Suppose $L - \epsilon P$ is not pseudo-effective for every sufficiently small $\epsilon > 0$. Choose rational ϵ as in Lemma 2.4, and run the klt LMMP with scaling to a Mori fiber space

$$(Z, \Delta - \epsilon P) \dashrightarrow (Z', \Delta' - \epsilon P') \xrightarrow{f} T.$$

Existence and termination in the non-pseudo-effective case are the established klt results of [6]. The program is crepant for L . The line-bundle descent in Lemma 2.4, applied also to the final contraction, gives a nef rational divisor A on T with

$$K_{Z'} + \Delta' \sim_{\mathbb{Q}} f^* A.$$

The base has dimension less than n .

If (Z, Δ) is klt, the transformed original pair (Z', Δ') is klt as well. Ambro's descent theorem [1, Theorem 0.2] gives an effective rational boundary Δ_T such that (T, Δ_T) is klt and

$$A \sim_{\mathbb{Q}} K_T + \Delta_T.$$

All its hypotheses hold: the total pair is globally klt and effective, f is a projective contraction, and its adjoint is rationally linearly pulled back. In particular, this use requires no conjecture about semiample of a moduli divisor. Assumption 2.1 and Lemma 2.2 make A semiample.

Otherwise P' is effective and relatively ample, since $K_{Z'} + \Delta' - \epsilon P'$ is relatively antiample. Some component of the floor therefore dominates T . Take a crepant dlt blowup $b : (\tilde{Z}, \tilde{\Delta}) \rightarrow (Z', \Delta')$, and choose the strict transform S of such a component. The restriction $f_S = (f \circ b)|_S : S \rightarrow T$ is surjective. Crepancy and adjunction give an lc pair on S whose nef adjoint is rationally linearly equivalent to $f_S^* A$. It is semiample by the lower-dimensional hypothesis. Semiample of a rational line bundle descends along a proper surjection: use the equality of sections for its connected-fiber Stein factor, and norms for the remaining finite morphism. For the latter assertion, choose a section of a basepoint-free multiple nonzero simultaneously at all points of the chosen finite fiber. Such a section exists because each vanishing condition is a proper linear subspace, and finitely many proper linear subspaces do not cover a complex vector space. Its norm is nonzero at the point downstairs. Hence A is semiample also in this case. Crepancy transfers the conclusion back to L .

The klt pseudo-effective perturbation. We next settle the klt case when $L - \epsilon \Delta$ is pseudo-effective for some small rational $\epsilon > 0$. Nonvanishing gives

$$L \sim_{\mathbb{Q}} G_0 := H_0 + \epsilon \Delta, \quad H_0 \geq 0.$$

Take a resolution $p: W \rightarrow Z$ with reduced SNC divisor D consisting of all exceptionals and the strict support of $H_0 + \Delta$. We have

$$K_W + D = p^*L + R, \quad R \geq 0.$$

Here every component of D has positive coefficient in

$$G_W := p^*G_0 + R \sim_{\mathbb{Q}} K_W + D.$$

Indeed, at strict boundary components the difference between coefficient one and a klt boundary coefficient is positive, and at exceptional components the coefficient contributed by the adjoint comparison is the positive log discrepancy. The pushforward p_*G_W is bounded coefficientwise by a fixed multiple of G_0 . Section spaces inject under birational pushforward, so

$$0 \leq \kappa(W, K_W + D) \leq \kappa(Z, G_0) = 0.$$

Take a $\mathbb{Q}$ -factorial dlt log minimal model (V, D_V) of (W, D) , and fix a common resolution

$$W \xleftarrow{q} U \xrightarrow{v} V.$$

Its boundary is reduced. On U , the comparison divisor in Lemma 2.2 is exceptional over V . Pushing q^*G_W forward by v therefore gives

$$K_V + D_V \sim_{\mathbb{Q}} G_V = \sum_i b_i D_i, \quad b_i > 0, \quad \text{Supp } G_V = D_V.$$

For completeness, positivity at a component extracted on V also holds: its log discrepancy over (W, D) is zero, its center lies in D , and the pullback of the full-support effective G_W has positive order there. There is no comparison-divisor coefficient at a prime retained on V . Effectivity and exceptionality in the comparison preserve the adjoint section ring, so the nef adjoint on V still has Iitaka dimension zero.

The small-perturbation hypothesis of Theorem 3.1 is now explicit. If $D_V \neq 0$, choose rational c with $0 < c < \min_i b_i$. Then

$$K_V + (1 - c)D_V \sim_{\mathbb{Q}} \sum_i (b_i - c)D_i \geq 0.$$

If $D_V = 0$, pseudo-effectivity is immediate. Proposition 2.3 supplies the required semiampleness on D_V . Theorem 3.1 thus makes $K_V + D_V$ abundant. Its Iitaka dimension is zero, so its numerical dimension is zero as well. For an ample divisor H_V on V , we obtain

$$G_V \cdot H_V^{n-1} = (K_V + D_V) \cdot H_V^{n-1} = 0.$$

Effectivity forces $G_V = 0$, and its full support then gives $D_V = 0$.

Recall that $G_W = p^*G_0 + R$ with $R \geq 0$. On the same common resolution,

$$0 \leq q^*p^*G_0 \leq q^*G_W, \quad v_*q^*G_W = G_V = 0.$$

Thus $q^*p^*G_0$ is effective and v -exceptional. It is nef because it is rationally linearly equivalent to $(p \circ q)^*L$. The negativity lemma for v forces it to vanish. Therefore $G_0 = 0$ and $L \sim_{\mathbb{Q}} 0$. Together with the preceding cases, this proves the klt good-model assertion in dimension n : for a non-nef pseudo-effective klt adjoint, first take its klt log minimal model and apply what was just proved.

The remaining non-klt case. Finally suppose the original pair is not klt and a small floor perturbation $L - \delta P$ is pseudo-effective. Choose rational $\epsilon > 0$ with $2\epsilon \leq \delta$ and small enough to keep the perturbed boundaries effective and klt. Convexity of the pseudo-effective cone, applied between L and $L - \delta P$, then makes both

$$L - \epsilon P, \quad L - 2\epsilon P, \quad P = \lfloor \Delta \rfloor,$$

pseudo-effective klt adjoints. Their Iitaka dimensions are zero: nonvanishing gives the lower bound, and adding the effective perturbation bounds each by $\kappa(Z, L) = 0$. The preceding klt case gives good models for both of these perturbed adjoints. Consequently their Nakayama numerical dimensions are zero. Since

$$2(L - \epsilon P) = L + (L - 2\epsilon P)$$

and the last summand has an effective rational representative, monotonicity and homogeneity of κ_σ give

$$\kappa_\sigma(Z, L) \leq \kappa_\sigma(Z, 2(L - \epsilon P)) = 0.$$

The nef divisor L is therefore numerically trivial. Nonvanishing is already available in this finishing step: write $L \sim_{\mathbb{Q}} E \geq 0$. For an ample divisor H , the equality $E \cdot H^{n-1} = 0$ forces $E = 0$. Thus $L \sim_{\mathbb{Q}} 0$, without any additional nonvanishing or abundance premise.

This completes the remaining zero-Iitaka-dimension case, so every rational nef dlt adjoint considered above is semiample. Returning to the finite rational simplex proves the real-boundary good-model assertion. Lemma 2.2 then descends semiampleness to every original nef rational lc pair. $\square$

3 Geometry of a signed boundary representative

The induction requires an abundance criterion for a nef log canonical divisor whose restriction to the reduced boundary is semiample. The representative supported on that boundary may have either sign. We first state the result, then construct the geometry needed to lift functions from the positive part of the boundary.

Theorem 3.1 (Signed representative). *Let (X, D) be a projective $\mathbb{Q}$ -factorial dlt pair over $\mathbb{C}$, with $D = \sum_i D_i$ reduced. Suppose that*

$$L = K_X + D \quad \text{is nef}, \quad L - cD \quad \text{is pseudo-effective for some } c > 0,$$

and that

$$L \sim_{\mathbb{Q}} G = \sum_i a_i D_i, \quad a_i \in \mathbb{Q}.$$

If $L|_D$ is semiample as a rational line bundle on the reduced scheme D , then L is abundant:

$$\kappa(X, L) = \nu(X, L).$$

Put $n = \dim X$ and $v = \nu(X, L)$. In the nontrivial range $0 < v < n$, the geometric goal is a dominating family of $(n - v)$ -dimensional projective subvarieties disjoint from D . We will show that a suitable positive-coefficient component has semiample image of dimension $v - 1$. Over a general point of that image, the construction in this section separates the positive boundary from the negative and coefficient-zero parts and makes it Cartier, with specified adjunction data. Proposition 4.1 then lifts the image parameters together with a transverse parameter through all infinitesimal neighborhoods. Fixing the former and varying the latter will deform a boundary fiber off D . Section 5 algebraizes these deformations and descends L along its nef reduction to prove abundance. This argument uses no Iitaka subadditivity assumption.

3.1 The positive boundary and its small intersections

If $D = 0$, the signed relation gives $L \sim_{\mathbb{Q}} 0$, so the theorem is immediate. For the construction below we may therefore assume $D \neq 0$; its semiample restriction then defines the stated morphism to a projective space.

Write

$$G = G_+ - G_-, \quad D_0 = \sum_{a_i=0} D_i,$$

where G_+ and G_- are effective and have disjoint prime supports. Take a positive integer m such that $mG_{\pm}$, mD_0 , and mL are integral Cartier divisors and

$$N_0 = \mathcal{O}_X(mG) \simeq \mathcal{O}_X(mL).$$

After increasing m , we may assume that $N_0|_D$ is generated by global sections. Fix the morphism it defines,

$$\phi : D \longrightarrow P = \mathbb{P}^s, \quad N_0|_D \simeq \phi^* \mathcal{O}_P(1),$$

and write s_0 for the rational section of N_0 whose divisor is mG . Fix an ample Cartier divisor H . The next lemma locates the part of the boundary over which the lifting construction will take place.

Lemma 3.2. *If $v = n$, then L is big. If $v = 0$, then $D = 0$ and $L \sim_{\mathbb{Q}} 0$. In the remaining case $0 < v < n$, put $r = v - 1$. Every component of D has ϕ -image of dimension at most r , and some component of G_+ has image of dimension r . Moreover,*

$$\dim \phi(\text{Supp } G_+ \cap (\text{Supp } G_- \cup D_0)) < r;$$

when $r = 0$, the intersection in this formula is empty.

Proof. For $v = n$, the numerical criterion for a nef divisor gives bigness. Assume $v < n$. Intersect the pseudo-effective class $L - cD$ with the indicated product of nef classes:

$$0 \leq (L - cD)L^v H^{n-v-1} = -c \sum_i D_i L^v H^{n-v-1}.$$

Each number $D_i L^v H^{n-v-1}$ is nonnegative, so all of them vanish. When $v = 0$, ampleness then forces $D = 0$; the signed representative consequently gives $L \sim_{\mathbb{Q}} 0$.

For $v > 0$, semiampleness identifies the numerical dimension of the restriction to each component of D with that component's image dimension under ϕ . The vanishing just proved bounds this dimension by r . Moreover,

$$0 < L^v H^{n-v} = \sum_i a_i D_i L^r H^{n-v},$$

so the bound is attained by a component with positive coefficient.

To control where that positive part meets the rest of the boundary, consider the symmetric matrix

$$Q_{ij} = D_i D_j L^r H^{n-r-2}.$$

Distinct effective $\mathbb{Q}$ -Cartier divisors have effective intersection cycles, so the off-diagonal entries are nonnegative. The ambient intersection form has at most one positive direction by the mixed Hodge index theorem: approximate by ample classes and apply the ordinary Hodge index theorem on complete-intersection surfaces. In this form L is isotropic and orthogonal to every D_i , while its

pairing with H is strictly positive. Indeed, writing Q also for the ambient bilinear intersection form, we have

$$Q(L, L) = L^{r+2}H^{n-r-2} = 0, \quad Q(L, D_i) = D_i L^{r+1}H^{n-r-2} = 0,$$

whereas $Q(L, H) = L^{r+1}H^{n-r-1} > 0$ because $r+1 = v$. The form on the orthogonal space to L is therefore negative semidefinite. In particular this holds on the span of the D_i . Pulling back to a resolution gives the same conclusion, so smoothness of X is unnecessary here.

Write $a = (a_i)$ for the coefficient vector. The signed relation gives the vector equation $Qa = 0$: for every i ,

$$\sum_j Q_{ij}a_j = D_i L^{r+1}H^{n-r-2} = 0.$$

Temporarily write $a = a^+ - a^-$ for its decomposition into positive and negative coefficient vectors. We obtain

$$Q(a^+, a^+) = Q(a^+, a^-) \geq 0.$$

Negative semidefiniteness forces both sides to vanish. A vector on which a negative semidefinite quadratic form vanishes belongs to its kernel, so $Qa^+ = 0$. In a row with $a_i \leq 0$, the diagonal term is absent and every summand $Q_{ij}a_j^+$ is nonnegative. Their sum is zero; hence each intersection with a positive component has zero L^r -degree against H^{n-r-2} . Semiample on D then makes its image dimension less than r . For $r = 0$, any nonzero effective intersection would have positive ample degree, so the intersection is empty. $\square$

For the rest of the construction assume $0 < v < n$, and put $N = n - 1$.

3.2 The divisor and adjunction data for lifting

We will replace the positive boundary by a reduced Cartier divisor whose normal line comes from the semiample image. Its powers must be actual line bundles throughout the construction. For a positive integer ℓ , the root gerbe

$$\mathcal{P} = \sqrt[\ell]{\mathcal{O}_P(1)}$$

parametrizes ℓ -th roots of the indicated line bundle without the choice of a section. Denote its tautological line by C ; thus C^ℓ is the pullback of $\mathcal{O}_P(1)$. Further pullbacks of C will carry the same notation. The use of normalized cyclic covers and their eigenspace decompositions follows the classical covering method of Esnault–Viehweg [20, Section 3.5, Claim 3.10 and Corollary 3.11]. Here we also retain the negative and coefficient-zero boundary components and keep track of the fixed adjunction isomorphism on the quotient charts. We will also place the Stein image in a smooth target: this allows differential operators and projective Hodge-module direct images to be used on ordinary scheme charts. The infinitesimal neighborhoods will later be cut down to a projective fiber before they are algebraized.

Proposition 3.3. *Choose ℓ divisible by m and every nonzero integer $|ma_i|$, and set $a = \ell/m$. There is a normal tame Deligne–Mumford stack $\mathcal{X}$, considered on a neighborhood of a reduced Cartier divisor S of pure dimension $N = n - 1$, together with a morphism to X and a reduced boundary T having no component in common with S , with the following properties.*

- (i) *The pair $(\mathcal{X}, S + T)$ is log canonical, and a fixed isomorphism of reflexive sheaves is given by*

$$\omega_{\mathcal{X}}(S + T) \simeq \mathcal{O}_{\mathcal{X}}(aS). \quad (*_{\text{adj}})$$

The support of T is locally set-theoretically principal. The ambient space and S are Cohen–Macaulay on scheme charts. Off T , the ambient space is Gorenstein and the displayed isomorphism is ordinary Cartier adjunction data.

- (ii) Put $J = (S \cap T)_{\text{red}}$. Both S and J are Du Bois on scheme charts, the ideal $\mathcal{I}_{J/S}$ is maximal Cohen–Macaulay, and

$$\mathcal{B} := \text{Hom}_S(\mathcal{I}_{J/S}, \omega_S) \simeq \mathcal{O}_S(aS).$$

The identification agrees off J with the residue convention specified by $(*_\text{adj})$.

- (iii) There is a finite representable morphism

$$S \longrightarrow D \times_P \mathcal{P}$$

whose image covers $\text{Supp } G_+$. The line $\mathcal{O}_S(S)$ is the pullback of C . The image of J in P has dimension less than r , whereas the image of S has dimension r .

- (iv) There is a smooth projective bundle $p : \mathcal{Y} \rightarrow \mathcal{P}$ and a representable projective morphism $g : S \rightarrow \mathcal{Y}$. This is a morphism from the reduced divisor S ; no extension to a neighborhood of S in $\mathcal{X}$ is asserted here. Writing $h = p \circ g$ and $\mathcal{T} = \text{Spec}_{\mathcal{P}} h_* \mathcal{O}_S$, the morphism g factors through a closed embedding $\mathcal{T} \hookrightarrow \mathcal{Y}$, and

$$g_* \mathcal{O}_S = \mathcal{O}_{\mathcal{T}}, \quad \mathcal{O}_S(S) = g^* C, \quad \mathcal{B} = g^* C^a.$$

- (v) For a separated quasi-projective scheme chart $U \rightarrow \mathcal{Y}$ that is etale and of finite type, put $S_U = S \times_{\mathcal{Y}} U$. The etale morphism $S_U \rightarrow S$ extends compatibly to every nilpotent thickening qS in $\mathcal{X}$. Only this etale chart is extended; an extension of g to qS is not needed to define the chart. The resulting qS_U are quasi-projective schemes, separated and quasi-finite over X .

All the adjunction and duality identifications are compatible with changes of etale chart.

Proof. We construct the divisor and its adjunction data first, then place its Stein image in a smooth projective bundle. The final step verifies that the infinitesimal neighborhoods have the scheme charts required for formal lifting.

Roots and a fixed adjunction isomorphism. Let $\rho : \mathcal{R} \rightarrow X$ be the normalized simultaneous root stack of the Cartier divisors mG_+ , mG_- , and mD_0 , taking an ℓ -th root of each divisor together with its section; the construction allows empty divisors. Denote the tautological root divisors by S_+, S_-, S_0 on $\mathcal{R}$. Their reducedness can be checked at a generic prime of downstairs multiplicity b . The chosen divisibility conditions give $b \mid \ell$. A normalized chart of $y^\ell = x^b$ has ramification index ℓ/b , and y has order one. Since m divides ℓ , this also applies to D_0 . The divisors are Cartier, so normality and generic reducedness give their reducedness everywhere.

Put $B = S_+ + S_- + S_0$ on $\mathcal{R}$. With this full reduced boundary, log ramification gives a log canonical pair and the relation

$$K_{\mathcal{R}} + B \sim_{\mathbb{Q}} a(S_+ - S_-).$$

The ambient charts are klt. They are klt off the boundary, and, on a log resolution, decreasing every boundary coefficient slightly removes all zero-discrepancy places. The difference between the two sides is a torsion integral Weil class, represented by

$$\mathcal{F} = (\omega_{\mathcal{R}}(B) \otimes \mathcal{O}_{\mathcal{R}}(-a(S_+ - S_-)))^{**}.$$

Choose a periodicity isomorphism $\mathcal{F}^{[q]} \simeq \mathcal{O}_{\mathcal{R}}$, where square brackets denote reflexive powers. It defines multiplication in the finite algebra

$$\mathcal{A} = \bigoplus_{i=0}^{q-1} \mathcal{F}^{[i]}.$$

Let $\tau : \mathcal{R}^\dagger \rightarrow \mathcal{R}$ be the normalization of $\mathrm{Spec}_{\mathcal{R}} \mathcal{A}$. In codimension one this is the ordinary cover of a torsion line bundle and is étale. The cover preserves log canonicity, klt ambient singularities, and the reduced Cartier boundary. From now on S_+, S_-, S_0 denote their pullbacks to $\mathcal{R}^\dagger$.

Tautological evaluation on this cover trivializes the pullback of $\mathcal{F}$ in codimension one and hence fixes the adjoint isomorphism. Because this evaluation is a sheaf morphism on the cover stack itself, the adjoint isomorphism is equivariant on every atlas. This equivariance will be needed for the quotient below.

Separating the positive and negative parts. Let $p_1 : \tilde{\mathcal{R}} \rightarrow \mathcal{R}^\dagger$ be the normalized blowup of the ideal of $S_+ \cap S_-$. The spaces constructed so far are related by

$$\tilde{\mathcal{R}} \xrightarrow{p_1} \mathcal{R}^\dagger \xrightarrow{\tau} \mathcal{R} \xrightarrow{\rho} X.$$

The blown-up ideal is locally generated by two elements, so the ordinary blowup embeds in a relative projective line. Its fibers have dimension at most one, as do the fibers after finite normalization. Hence each exceptional divisor lies over a codimension-two component of the intersection. Downstairs such a component is a dlt stratum, generically SNC. Separate normalized Kummer charts and purity for the index cover at the smooth generic locus give the same description upstairs.

The tautological exceptional divisor E is consequently reduced Cartier, and the divisors

$$S'_+ = p_1^* S_+ - E, \quad S'_- = p_1^* S_- - E$$

are reduced Cartier and disjoint on $\tilde{\mathcal{R}}$. Since no codimension-two two-branch stratum is contained in S_0 , its pullback $p_1^* S_0$ is reduced Cartier and has no exceptional component. The total boundary

$$S'_+ + S'_- + E + p_1^* S_0$$

is crepant and log canonical. To check the ambient singularities as well, denote this boundary on a chart V' by B' . The crepant equality gives log canonicity for every valuation. Outside B' , the blowup is an isomorphism to the old klt boundary complement. A divisor F with $a(F; V', B') = 0$ therefore has center in $\mathrm{Supp} B'$. Since B' is effective Cartier, $\mathrm{ord}_F(B') > 0$, whence

$$a(F; V', 0) = a(F; V', B') + \mathrm{ord}_F(B') > 0.$$

Positive pair discrepancies also remain positive after the boundary is dropped. Thus V' is klt even at its higher-codimension singular loci. On $\tilde{\mathcal{R}}$, put $S = S'_+$ and $T = S'_- + E + p_1^* S_0$. Near S , the section of the disjoint divisor S'_- is a unit, and the fixed linear equivalence reads

$$\omega(S + T) \simeq \mathcal{O}(aS).$$

Klt singularities make the ambient charts Cohen–Macaulay. All the boundary divisors in this display are Cartier, so the isomorphism also makes these charts Gorenstein.

We also need finiteness of $S \rightarrow S_+$ on $\mathcal{R}^\dagger$. Locally write $S_+ = (u = 0)$ and $S_- = (v = 0)$. Before normalization, the strict transform of S_+ lies in the blowup chart with ratio u/v , where its equation is $u/v = 0$. Thus each fiber of this strict transform over S_+ has at most one point. Properness and

finite normalization prove finiteness of $S \rightarrow S_+$. A codimension-one point of $S \cap T$ therefore lies over a codimension-two intersection downstairs. The generic SNC description shows that $T|_S$ is generically reduced. Being Cartier on the Cohen–Macaulay scheme S , it is reduced everywhere.

Retaining the required root line. The construction so far has supplied reduced Cartier divisors and fixed adjunction data. We now remove the root characters that are unnecessary, while preserving the line attached to S . On $\tilde{\mathcal{R}}$ the line $\mathcal{O}(S - S'_-)$ has ℓ -th power equal to the pullback of N_0 and hence defines a morphism to the gerbe of ℓ -th roots of N_0 on X . Denote this gerbe by $\mathcal{G}$, and let $\kappa : \tilde{\mathcal{R}} \rightarrow \mathcal{X}$ be the relative coarse-space morphism over $\mathcal{G}$. On a trivializing chart, we quotient by the kernel of the finite-group character on the indicated root line. The quotients are ordinary quasi-projective schemes: the covers used above are finite, the blowup is projective, and finite quotients preserve quasi-projectivity. These chartwise quotients patch.

Before taking the quotient, the section of the disjoint divisor S'_- is a unit near S . There the retained line $\mathcal{O}(S - S'_-)$ identifies with $\mathcal{O}(S)$, so the line of S and its section both descend through the kernel quotient. Continue to write S, T for their reduced images under κ ; they now lie on $\mathcal{X}$. In particular S remains Cartier. The finite-group norm of a local equation for T upstairs makes its image set-theoretically principal; Cartierness of the reduced image T is not needed. Divisorial ramification occurs only on the boundary. Log ramification thus preserves log canonicity and identifies the descended adjoint isomorphism with $(*_\text{adj})$. The ambient chart and S are Cohen–Macaulay because their structure sheaves are invariant direct summands of the finite CM covers.

More explicitly, the kernel acts trivially on the retained root line and hence on $\mathcal{O}(aS)$. Equivariance of the fixed isomorphism gives the same kernel character on $\omega(S + T)$. Taking invariants descends the isomorphism, and log ramification identifies the invariant log dualizing sheaf. Off T , this isomorphism makes the canonical sheaf invertible, giving the asserted Gorenstein property.

Duality on the boundary. The reduced supports S and J are unions of log canonical centers, since intersections of lc centers are again unions of lc centers. Both are therefore Du Bois by [43, Theorems 1.4 and 1.7]. To identify the ideal and its dual, work on a quotient chart and denote the finite cover above by $f : S^\text{up} \rightarrow S$. Reducedness of $T^\text{up}|_{S^\text{up}}$ gives

$$\mathcal{I}_{J/S} = (f_* \mathcal{O}_{S^\text{up}}(-T^\text{up}|_{S^\text{up}}))^{\text{inv}}.$$

An invariant function vanishes on the reduced quotient image exactly when its pullback vanishes on this reduced preimage. The displayed invariant sheaf is maximal Cohen–Macaulay: the upstairs sheaf is invertible on a CM scheme, finite pushforward preserves its depth over the quotient, and in characteristic zero invariants form a direct summand.

Apply finite-map duality with trace and then take invariants:

$$\mathcal{H}om_S(\mathcal{I}_{J/S}, \omega_S) = (f_* \omega_{S^\text{up}}(T^\text{up}|_{S^\text{up}}))^{\text{inv}} \simeq \mathcal{O}_S(aS).$$

The last isomorphism comes from Cartier adjunction upstairs and descent of the line of S . Finite-map duality applies in the presence of ramification; it requires no invertibility of ω_S itself. Off J , the identification is the ordinary residue fixed by the ambient isomorphism. Normalized trace preserves this residue convention, so the identifications agree on overlaps.

The target of the lifting argument. On S , the root gerbe of N_0 is $D \times_P \mathcal{P}$. Finiteness of the strict divisor, proved above, and removal of the relative inertia kernel give a finite representable map $S \rightarrow D \times_P \mathcal{P}$. Its image covers the positive boundary, and the root line is $\mathcal{O}_S(S) = C$. Lemma 3.2 therefore gives image dimension r for S in P . To check the smaller image of J , recall that before the quotient S'_- is disjoint from S , the exceptional divisor E maps into $\text{Supp } G_+ \cap \text{Supp } G_-$, and

$p_1^*S_0$ maps into D_0 . Taking the quotient changes none of these images in X . The image of J in D is consequently contained in

$$\text{Supp } G_+ \cap (\text{Supp } G_- \cup D_0).$$

The same lemma gives image dimension less than r for this locus in P . We have thus obtained a representable projective morphism $h : S \rightarrow \mathcal{P}$; it remains to embed its Stein image in the required smooth bundle.

The Stein algebra defines a finite stack $\mathcal{T} = \text{Spec}_{\mathcal{P}} h_* \mathcal{O}_S$ over $\mathcal{P}$. This coherent algebra is generated as a module by vector bundles on $\mathcal{P}$. Indeed, decompose it by the finitely many inertia characters, twist each summand by a power of C so that it descends to P , and use Serre generation there. The resulting vector-bundle surjection embeds $\mathcal{T}$ in a vector bundle over $\mathcal{P}$. Since $\mathcal{T}$ is proper over the base, it remains closed in the projective completion. This completion is $\mathcal{Y}$ and gives the map $g : S \rightarrow \mathcal{Y}$ from the reduced divisor. The Stein construction yields $g_* \mathcal{O}_S = \mathcal{O}_{\mathcal{T}}$, and the line identities follow from those already established.

Finally, start with the etale chart $S_U \rightarrow S$ obtained from g . Invariance of the etale site under nilpotent thickening extends this chart uniquely to $qS_U \rightarrow qS$, compatibly as q varies. This construction does not use a map $qS \rightarrow \mathcal{Y}$; lifting the map from S is the task of the next section. The reduction S_U is a scheme; its extension qS_U is an algebraic space with trivial inertia. The chosen chart is separated, the map to the root gerbe is finite, and that gerbe has finite diagonal, so qS_U is separated over X . The finite representable map $S \rightarrow D \times_P \mathcal{P}$ and the etale map $S_U \rightarrow S$ show that the finite-type geometric fibers of $S_U \rightarrow X$ are zero-dimensional. Nilpotent thickening changes neither geometric points nor fiber dimensions. The morphism $qS_U \rightarrow X$ is therefore quasi-finite as well. Zariski's main theorem embeds qS_U in a scheme finite over X , making it a quasi-projective scheme as required. $\square$

4 Hodge theory and formal lifting

Our goal is to lift functions and a transverse parameter through every infinitesimal neighborhood of the divisor constructed in Proposition 3.3. We recall its data. $\mathcal{P} = \sqrt[\ell]{\mathcal{O}_{\mathbb{P}^s}(1)}$ is a root gerbe, C is its tautological line bundle, and

$$S \xrightarrow{g} \mathcal{Y} \xrightarrow{p} \mathcal{P}, \quad h = p \circ g.$$

Here g is representable and projective, and p is a smooth projective bundle. The Stein factor of h is finite over $\mathcal{P}$; it has been embedded as $\mathcal{T} \subset \mathcal{Y}$ so that $g_* \mathcal{O}_S = \mathcal{O}_{\mathcal{T}}$. The reduced Cartier divisor $S \subset \mathcal{X}$ has dimension $N = n - 1$. If $J = (S \cap T)_{\text{red}}$, then S, J are Du Bois, $\mathcal{I}_{J/S}$ is maximal Cohen–Macaulay, and

$$\mathcal{B} := \text{Hom}_S(\mathcal{I}_{J/S}, \omega_S) \simeq \mathcal{O}_S(aS) = g^* C^a, \quad \mathcal{O}_S(S) = g^* C, \quad (2)$$

for an integer $a > 0$. Off T , the fixed ambient isomorphism $\omega_{\mathcal{X}}(S) \simeq \mathcal{O}_{\mathcal{X}}(aS)$ supplies these identifications by adjunction.

Put $I = \mathcal{O}_{\mathcal{X}}(-S)$. For a separated quasi-projective scheme etale chart $U \rightarrow \mathcal{Y}$ of finite type, write $S_U = S \times_{\mathcal{Y}} U$ and again $g : S_U \rightarrow U$ for the induced projective morphism. Denote by qS_U the unique extension of $S_U \rightarrow S$ to the nilpotent thickening qS . These extensions are compatible in q . Proposition 3.3 makes them quasi-projective schemes: they are separated and quasi-finite over X , and hence open in schemes finite over the projective variety X . Powers and quotients of I below are pulled back to the appropriate thickenings. On these quotients, g_* denotes direct image for the common underlying topological map; a ringed-space map from a thickening to U has not yet been constructed.

Proposition 4.1. *For every such chart U , every $j \geq 0$, and every $k \geq 1$, the transition*

$$g_*(I^j/I^{j+k+1}) \longrightarrow g_*(I^j/I^{j+k})$$

is surjective as a map of sheaves of abelian groups. Its kernel is $\mathcal{O}_{\mathcal{T}_U} \otimes C^{-j-k}$, where $\mathcal{T}_U = \mathcal{T} \times_{\mathcal{Y}} U$. The statements are compatible with further etale changes of charts. On an affine chart, these transitions are also surjective on global sections.

In their isolated-component case with numerical dimension one, Liu–Xu use finite covers and Du Bois cohomology to obtain infinitesimal motion [47, Theorem 3.1]. Here the semiample boundary image may have positive dimension, so its local functions and the transverse parameter must be lifted together.

For $j \geq 0$ and $k \geq 1$, the kernel of the transition from length $k+1$ to length k in I^j/I^{j+k+1} is $\mathcal{O}_S \otimes C^{-j-k}$. The corresponding connecting homomorphism therefore takes values in

$$R^1 g_* \mathcal{O}_S \otimes C^{-j-k}.$$

We will identify this sheaf with a negative twist of the lowest filtered piece $F_0 M$ of a Hodge-module direct image. Lifting local functions gives Čech cocycles in this coherent sheaf. A calculation with their graphs shows that the coordinate cocycle lies in the kernel of the differential-operator symbol map. Negative-twist vanishing will make that kernel have no global sections. The same graph calculation then kills the error in lifting the transverse parameter.

4.1 The lowest Hodge piece

We work with graded-polarizable mixed Hodge modules on ordinary complex algebraic varieties. The inputs are projective strictness and the usual functorial operations [56, Theorem 2.14 and Section 4], the comparison with the Du Bois complex and its coherent dual [57, Theorem 0.2, Corollary 0.3 and Proposition 5.3], and Kodaira–Saito vanishing [56, Proposition 2.33]. On a smooth stack, we use compatible systems of these objects on scheme etale charts and descend their filtered differential modules. Thus the argument requires no Hodge-module theory on stacks.

All differential modules in this section are right modules with an increasing filtration. In the Spencer de Rham complex the module itself occupies degree zero. Accordingly, on a smooth chart V , the term in degree $-i$ of $\mathrm{Gr}_k^F \mathrm{DR}_V(M)$ is

$$\mathrm{Gr}_{k-i}^F M \otimes_{\mathcal{O}_V} \bigwedge^i T_V.$$

When $F_{<0} M = 0$, the lowest graded de Rham complex therefore consists of the sheaf $F_0 M$ in degree zero.

We now identify a Hodge module whose lowest piece contains the coherent lifting obstructions. On each chart U , write $J_U = J \times_{\mathcal{Y}} U$. For $j : S_U \setminus J_U \hookrightarrow S_U$, put

$$A_U^\bullet = \mathbf{D} \left(j! \mathbb{Q}_{S_U \setminus J_U}^H[N] \right), \quad A_{0,U} = {}^p H^0 A_U^\bullet, \quad M_U = {}^p H^1 g_* A_{0,U}.$$

Here $\mathbf{D}$ is Hodge-module duality, ${}^p H^i$ denotes perverse cohomology, and g_* in the definition of M_U is the derived Hodge-module direct image. Later, g_+ will denote the corresponding direct image of differential modules. These constructions commute with etale restriction. Thus the M_U form a system M supported on $\mathcal{T}$.

Lemma 4.2. *The following identifications hold compatibly on the charts:*

$$F_{<0}A_{0,U} = 0, \quad F_0A_{0,U} = \mathcal{B}|_{S_U}, \quad F_{<0}M_U = 0, \quad F_0M_U = R^1g_*(\mathcal{B}|_{S_U}).$$

In a smooth ambient embedding $S_U \hookrightarrow V$ of codimension c , the underlying module of $A_{0,U}$ is $\mathcal{H}_{S_U}^c(\omega_V)$, localized off J_U . The lowest-piece inclusion is the adjunction, or Ext-to-support, inclusion off J_U , extended by meromorphic localization.

Proof. The proof must identify not just a coherent sheaf but its actual inclusion into the differential module: the graph calculation will differentiate classes in that inclusion. We first compute the sheaf by duality. Du Bois comparison applied to the difference triangle for the constant objects of S_U and J_U identifies its degree-zero graded de Rham complex with $\mathcal{I}_{J_U/S_U}$. Coherent duality and the maximal-Cohen–Macaulay property give

$$\mathrm{Gr}_k^F \mathrm{DR}(A_U^\bullet) = 0 \quad (k < 0), \quad \mathrm{Gr}_0^F \mathrm{DR}(A_U^\bullet) = \mathcal{B}|_{S_U}[0]. \quad (3)$$

Here $[N]$ cancels the dimension shift of the dualizing complex. In the smooth case, our convention gives $\mathbf{D}(\mathbb{Q}^H[N]) = \mathbb{Q}^HN$; its right differential module has ω as its lowest piece, at index zero.

To pass to perverse cohomology, work locally on a fixed chart and let q be the smallest filtration index occurring in any perverse cohomology object of $A_U^\bullet$. Boundedness of the complex and goodness of the filtrations provide such a lower bound. At index q , each of these objects has a graded Spencer complex consisting only of its degree-zero term. The spectral sequence for perverse truncation has a single nonzero row at that index, and therefore gives

$$\mathcal{H}^i \mathrm{Gr}_q^F \mathrm{DR}(A_U^\bullet) = F_q {}^p H^i A_U^\bullet.$$

If $q < 0$, this contradicts Equation (3). It follows that all these objects have $F_{<0} = 0$. The same argument at index zero yields $F_0A_{0,U} = \mathcal{B}|_{S_U}$, while the lowest pieces of the other perverse cohomology objects vanish. This passage does not require the shifted constant complex to be perverse.

We next identify the underlying unfiltered module. Ambient duality in V identifies the dual of the reduced constant complex with

$$R\Gamma_{[S_U]}(\omega_V)[c].$$

The degree-zero module is the first nonzero support cohomology, $\mathcal{H}_{S_U}^c(\omega_V)$. The support of J_U is locally set-theoretically principal in S_U by the geometric construction. Lift a local defining function to V and invert it. This exact meromorphic localization realizes j_* on the underlying module and gives the description of $A_{0,U}$ in the statement.

The two descriptions must agree as inclusions, with the fixed residue normalization. On the smooth locus of $S_U \setminus J_U$, filtered duality and smooth graph pushforward give the ordinary dualizing sheaf and its residue inclusion. This is precisely Ext-to-support with the normalization specified by ambient adjunction. Naturality in smooth etale coordinates gives the same scalar on each chart. The agreement extends throughout $S_U \setminus J_U$, because the first support-cohomology module has no sections supported on a smaller-dimensional subset. One can see this from the Cousin resolution of the smooth dualizing sheaf: its first term supported on S_U is a sum of injective modules at the codimension- c generic points. Localization extends the agreement across J_U . We have therefore identified the actual lowest-piece inclusion, including its residue convention.

We have now identified $F_0A_{0,U}$ and its inclusion before direct image. To identify F_0M_U , compute g_+ by a graph embedding followed by a smooth projection. At filtration zero, only the top term

of the relative Spencer complex survives: it is the direct image of $\mathcal{B}|_{S_U}$ on the graph. Projective strictness yields

$$F_{<0}M_U = 0, \quad F_0M_U = R^1g_*(\mathcal{B}|_{S_U}),$$

and identifies this sheaf as a coherent subsheaf of the unfiltered degree-one direct image. All constructions in this identification are canonical under etale changes of charts. $\square$

The obstruction sheaf for the transition indexed by j, k is thus $F_0M \otimes C^{-(a+j+k)}$, by Equation (2) and projection formula. We do not need all its global sections to vanish. The lifting proof will show that the coordinate obstruction lies in the kernel of a symbol map, and will apply the following vanishing to that kernel.

4.2 Negative-twist vanishing

The support of the system M just constructed is finite over $\mathcal{P}$. We prove the required vanishing for any system with this support property.

Lemma 4.3. *Let M be a compatible system of mixed Hodge modules in perverse degree zero on the scheme etale charts of $\mathcal{Y}$. Suppose its support is finite over $\mathcal{P}$. Then, for every integer k , every positive integer b , and every $j < 0$,*

$$\mathbb{H}^j(\mathcal{Y}, \mathrm{Gr}_k^F \mathrm{DR}_{\mathcal{Y}}(M) \otimes p^*C^{-b}) = 0.$$

Here the graded de Rham complex is the descended coherent complex.

Proof. The line C lives on the root gerbe, so we first push the complex to $\mathcal{P}$. We will then pull back to projective space, where C becomes $\mathcal{O}(1)$ and ordinary Kodaira–Saito vanishing applies. Since p is finite on the support of M , its direct image is concentrated in perverse degree zero; call the resulting system on $\mathcal{P}$ M' . On ordinary scheme charts, projective strictness gives

$$Rp_* \mathrm{Gr}_k^F \mathrm{DR}_{\mathcal{Y}}(M) \simeq \mathrm{Gr}_k^F \mathrm{DR}_{\mathcal{P}}(M'). \quad (4)$$

For the desired global hypercohomology vanishing, this identity must hold for the descended complexes. We verify that compatibility before passing to a finite cover.

For right differential modules, use the transfer bimodule

$$\mathcal{D}_{\mathcal{Y} \rightarrow \mathcal{P}} = \mathcal{O}_{\mathcal{Y}} \otimes_{p^{-1}\mathcal{O}_{\mathcal{P}}} p^{-1}\mathcal{D}_{\mathcal{P}},$$

with its order filtration. Derived tensor over $\mathcal{D}_{\mathcal{Y}}$, followed by derived direct image, forms the filtered direct image; equivalently, these operations can be carried out on Rees modules. De Rham on the base is a further derived tensor with $\mathcal{O}_{\mathcal{P}}$, filtered from degree zero. The filtered Spencer resolution, associativity of derived tensor, and projection formula identify this with direct image of the de Rham complex upstairs. Indeed, tensoring the transfer module over the base differential operators with the base structure sheaf gives $\mathcal{O}_{\mathcal{Y}}$.

These sheaf constructions are canonical on the etale site, and the Spencer resolutions are locally free over differential operators. They commute with chart changes already before taking associated gradeds. The ordinary projective direct-image theorem on the base charts supplies strictness and concentration in perverse degree zero. Passing to the etale site does not change coherent cohomology on those charts. This proves Equation (4) globally, as an identity computing coherent hypercohomology. The same verification will apply to the finite representable map below.

To pass from the gerbe to a scheme, use the representable finite surjective morphism

$$v : \mathbb{P}_z^s \longrightarrow \mathcal{P}$$

defined by the power map $[z_0 : \cdots : z_s] \mapsto [z_0^\ell : \cdots : z_s^\ell]$ and the root $\mathcal{O}_{\mathbb{P}_z^s}(1)$; thus $v^*C = \mathcal{O}_{\mathbb{P}_z^s}(1)$. Trivial roots on the standard affine opens of $\mathbb{P}^s$ give scheme étale charts $\check{U}_i \rightarrow \mathcal{P}$. The restriction of v to $z_i \neq 0$ factors through $\check{U}_i$. On each such chart take the derived Hodge-module inverse image of M' and then its degree-zero perverse cohomology. Thus the object being constructed chartwise is

$$H = {}^p H^0(v^* M').$$

The chosen root and functoriality identify these objects on overlaps, producing a system on the Zariski opens of $\mathbb{P}_z^s$.

We check that H is a global graded-polarizable mixed Hodge module, so that ordinary vanishing applies to it. The perverse, filtered, and weight data glue, as do the strict-support decompositions of the pure weight gradeds, by uniqueness. On a smooth connected dense stratum of an irreducible support, a polarization on a nonempty Zariski open extends to a flat pairing: the fundamental group of that open surjects onto the fundamental group of the stratum. Hodge compatibility extends by continuity, and the extended flat pairing remains nondegenerate and positive. The local Hodge modules already give boundary quasi-unipotence. Saito's extension theorem for polarizable variations [56, Section 3.b] and uniqueness of strict-support extension identify the global pure object with the glued object. Projectivity of $\mathbb{P}_z^s$ leaves no additional boundary at infinity. The weight extensions give the required mixed object.

Vanishing for H will imply vanishing for M' once we realize M' as a retract of $v_+ H$. On a chart one must use the *whole* base-changed finite cover. Over $\check{U}_i$, this is a disjoint union of copies of the corresponding affine coordinate chart of $\mathbb{P}_z^s$. The unit on unshifted constants and its dual trace, formed using smooth duality in equal dimensions, compose to multiplication by the degree. This can first be checked on the finite étale locus, where the trace sums over the sheets. On each connected smooth base chart, the shifted constant Hodge module is the rank-one intersection complex, with endomorphism ring $\mathbb{Q}$, and its endomorphisms are determined on a dense open. The composition therefore equals the degree globally, with no contribution supported on the branch locus. This argument permits ramification.

Tensor the unit and trace with M' , apply proper projection formula, and take perverse cohomology in degree zero. Finite direct image is perverse exact, so it commutes with this degree-zero cohomology. We obtain maps

$$M' \longrightarrow v_+ H \longrightarrow M'$$

whose composite is multiplication by the degree of the whole base-changed cover. Dividing the second map by that degree gives the retraction. Each map is canonical under étale base change; the retraction consequently descends on filtered differential modules as well.

Applying the finite version of Equation (4) and projection formula now exhibits the hypercohomology in the statement as a direct summand of

$$\mathbb{H}^j(\mathbb{P}_z^s, \mathrm{Gr}_k^F \mathrm{DR}(H) \otimes \mathcal{O}_{\mathbb{P}_z^s}(-b)).$$

Ordinary negative-ample Kodaira–Saito vanishing makes this group zero for $j < 0$. For a mixed object, apply the vanishing to the pure weight gradeds and then to the weight-filtration exact sequences, whose Hodge filtrations are strict. $\square$

4.3 Lifting through the infinitesimal neighborhoods

We now combine the lowest-piece identification with negative-twist vanishing. The remaining step is to identify the relation satisfied by the coordinate and conormal obstructions.

Proof of Proposition 4.1. We induct on k , simultaneously for all powers I^j . The connecting homomorphisms will factor through the first graded layers and together form a derivation. We must show that this derivation vanishes both on functions from U and on a local generator of I/I^2 . The graph calculation relates these two errors in the differential module M_U . Its order-one symbol, together with negative-twist vanishing, first kills the function error. The original relation then kills the conormal error.

Define

$$E_{j,k} = g_*(I^j/I^{j+k}), \quad j \geq 0, \quad k \geq 1.$$

The first graded layer is

$$E_{j,1} = \mathcal{O}_{\mathcal{T}_U} \otimes C^{-j}.$$

For adjacent lengths, the exact sequence has kernel $g_*\mathcal{O}_{S_U} \otimes C^{-j-k}$, and its connecting homomorphism takes values in

$$R^1g_*\mathcal{O}_{S_U} \otimes C^{-j-k}.$$

The obstruction derivation. Assume inductively that all shorter transitions are surjective. A section of $E_{j,k}$ with zero length-one term comes from $E_{j+1,k-1}$ when $k > 1$. Induction lifts it locally from $E_{j+1,k}$, so its connecting class vanishes. Induction also makes $E_{j,k} \rightarrow E_{j,1}$ surjective. The obstruction thus factors through the length-one layer; for $k = 1$ this factorization is immediate. More precisely, these factorizations give additive maps

$$D_{k,j} : \mathcal{O}_{\mathcal{T}_U} \otimes C^{-j} \longrightarrow R^1g_*\mathcal{O}_{S_U} \otimes C^{-j-k}.$$

Together they define D_k on the graded algebra $\bigoplus_{j \geq 0} \mathcal{O}_{\mathcal{T}_U} \otimes C^{-j}$, with graded shift k ; no $\mathcal{O}_U$ -linearity is asserted. Its values are computed by choosing local lifts and taking Čech differences. For a product, this difference is the sum of the two first-order differences: the product of two errors vanishes in the layer under consideration. Thus D_k is a $\mathbb{C}$ -derivation, with values in the graded module whose degree- j term is $R^1g_*\mathcal{O}_{S_U} \otimes C^{-j}$.

Restrict its degree-zero part along $\mathcal{O}_U \rightarrow \mathcal{O}_{\mathcal{T}_U}$. This derivation factors through Ω_U^1 , defining a section of $T_U \otimes R^1g_*\mathcal{O}_{S_U} \otimes C^{-k}$. As the chart varies, these sections descend to

$$e \in H^0(\mathcal{Y}, T_{\mathcal{Y}} \otimes R^1g_*\mathcal{O}_S \otimes C^{-k}) = H^0(\mathcal{Y}, T_{\mathcal{Y}} \otimes F_0M \otimes C^{-(a+k)}). \quad (5)$$

The equality follows from Equation (2), Lemma 4.2, and projection formula. For the descent assertion, function lifts pull back along the unique étale thickenings. Ordinary flat base change on the reductions pulls back their Čech classes in the coherent obstruction sheaves. Differentials also pull back and generate under an étale map, giving the required compatibility.

Coordinate errors and adjunction. We now compute this derivation in local coordinates. After shrinking U , choose étale coordinates $t_1, \dots, t_d$, with $d = \dim U$, and a frame of C^{-1} . Write y for its pullback, a conormal frame. Induction lifts the coordinates along g modulo I^k and lifts y to I/I^{k+1} . Lift them one step further on an open cover of S_U , obtaining functions $h_{i,\lambda}$ modulo I^{k+1} and generators y_i modulo I^{k+2} . The conormal generator must be lifted one order farther than the coordinates: its next error lies in I^{k+1}/I^{k+2} , whereas a coordinate error lies in I^k/I^{k+1} . By formal

etaleanness of the coordinate map, each h_i defines a local map from $(k+1)S_U$ to U . Write the overlap errors as

$$h_{j,\lambda} - h_{i,\lambda} = e_{ij,\lambda} y_i^k, \quad y_j - y_i = \eta_{ij} y_i^{k+1}. \quad (6)$$

Thus $D_k(t_\lambda)$ is represented by $(e_{ij,\lambda} y_i^k)$, and $D_k(y)$ by $(\eta_{ij} y_i^{k+1})$. The powers of y record the target line bundles; $e_{ij,\lambda}$ and η_{ij} are their coefficients on the reduction. Denote the induced frame of $\mathcal{B}|_{S_U}$ by $\sigma = y^{-a}$.

Off J_U , the fixed adjunction isomorphism gives

$$\omega_{(k+1)S_U} \simeq \mathcal{O}_{\mathcal{X}}((a+k)S)|_{(k+1)S_U}.$$

Via this isomorphism, the local equation y_i defines a dualizing generator, denoted $b_i = y_i^{-(a+k)}$. This notation denotes a frame of the line bundle $\mathcal{O}_{\mathcal{X}}((a+k)S)$ restricted to the thickening; it does not invert the nilpotent function y_i in its structure sheaf. Realize the dualizing sheaves in support cohomology. Under the trace inclusion $\omega_{S_U} \hookrightarrow \omega_{(k+1)S_U}$, we have

$$b_j - b_i = -(a+k)\eta_{ij}\sigma, \quad y_i^k b_i = \sigma, \quad y_i^{k+1} b_i = 0. \quad (7)$$

The first identity is the binomial expansion of $(1 + \eta_{ij} y_i^k)^{-(a+k)}$: all quadratic terms vanish because $2k \geq k+1$. In the last two identities, σ is viewed in the dualizing sheaf of the thickening through the trace inclusion. They express Cartier trace and the annihilator of the thickening.

The graph calculation. We next put both errors in the same differential module. Embed $(k+1)S_U$ as a closed subscheme of a smooth open $Z \subset \bar{Z} = \mathbb{P}^m$. The graph of the reduced map g is closed in $\bar{Z} \times U$, since it is proper over U . Let $\iota : S_U \hookrightarrow \bar{Z} \times U$ be this reduced graph embedding, and let $\mathcal{N}_\Gamma$ be the underlying unfiltered differential module of $\iota_* A_{0,U}$. This module is distinct from the direct-image module M_U on U . For each local graph lift h_i , Ext-to-support cohomology sends the dualizing section b_i to a section $\delta_i(b_i)$ of $\mathcal{N}_\Gamma$. All the lifted graphs have the same reduced support, namely the graph of g , so these sections lie in one support-cohomology module rather than in different modules for the different lifts. These sections need not lie in its lowest filtered piece. We will prove the identity

$$\delta_j(b_j) - \delta_i(b_i) = -(a+k)\eta_{ij}\sigma + \sum_{\lambda=1}^d (e_{ij,\lambda}\sigma) \cdot \partial_{t_\lambda}. \quad (8)$$

On the right, each reduced dualizing section σ is included in $\mathcal{N}_\Gamma$ by the lowest-piece map of Lemma 4.2. The right differential-operator action is taken in this graph module.

The calculation also applies when S_U is singular. Put $c_0 = \dim Z - N$. Ext-to-support identifies the dualizing sheaf of the thickening with its annihilator submodule in $\mathcal{H}_{S_U}^{c_0}(\omega_Z)$. To see this, use the support-cohomology spectral sequence. Support cohomology vanishes below c_0 , so in total degree c_0 the Ext group consists of homomorphisms from the structure sheaf of the thickening to the first support-cohomology module. In particular, no local complete-intersection hypothesis on $S_U \subset Z$ is needed.

Lift the $h_{i,\lambda}$ locally to functions on Z . For a dualizing section b of the thickening, its graph inclusion in $Z \times U$ is the generalized fraction

$$\frac{b dt_1 \wedge \cdots \wedge dt_d}{\prod_{\lambda=1}^d (t_\lambda - h_{i,\lambda})}.$$

It represents successive support cohomology in the additional coordinate equations, equivalently the Koszul residue for the graph, and can be computed etale-locally near the graph branch. After

taking support cohomology from Z , the translated new coordinates form a regular sequence; their support cohomology is consequently concentrated in the degree equal to their number. Either set of translated coordinates gives the same localization of this support-torsion module, because the two translations differ nilpotently on every section.

The change from h_i to h_j thus has a finite Taylor expansion on b_j . Again using $2k \geq k+1$, products of two differences from Equation (6) annihilate b_j . The linear terms are determined by

$$(h_{j,\lambda} - h_{i,\lambda})b_j = e_{ij,\lambda}\sigma.$$

The right action on a top differential form satisfies

$$\left(\frac{dt_\lambda}{t_\lambda - h}\right) \cdot \partial_{t_\lambda} = \frac{dt_\lambda}{(t_\lambda - h)^2}.$$

The doubled pole therefore contributes the positive sign in Equation (8). Combining these terms with Equation (7) proves the identity. So far the calculation is off J_U . Both its sections and its identities extend meromorphically across J_U in the *unfiltered* localized module, where arbitrary finite pole orders are allowed.

Vanishing of the obstruction. Push the graph identity along the projection $\pi : \bar{Z} \times U \rightarrow U$, using the relative Spencer complex. Closed direct image is perverse exact, and $\pi_{+\iota+}A_{0,U} = g_+A_{0,U}$. Thus degree-one holonomic differential-module cohomology of this Spencer direct image is the underlying module of $M_U = {}^pH^1g_*A_{0,U}$. The cochain $(\delta_i(b_i))$ belongs to the top, degree-zero term, from which there is no outgoing relative Spencer differential. Its Čech difference is a boundary in total degree one. A sufficiently refined ambient cover around the closed graph computes this boundary; all the sheaves in question are supported there. By Lemma 4.2, the reduced cocycles in Equation (8) represent their $R^1g_*\mathcal{B}$ classes in F_0M . Right differentiation in the U -coordinates acts on the pushforward complex. Consequently the Čech classes satisfy

$$-(a+k)[\eta_{ij}\sigma] + \sum_{\lambda=1}^d [e_{ij,\lambda}\sigma] \cdot \partial_{t_\lambda} = 0 \quad \text{in } M_U.$$

Here each bracketed class belongs to $R^1g_*\mathcal{B} = F_0M_U$; after differentiation, its image belongs to F_1M_U . Thus every term of the relation lies in F_1M , and the η -term lies in F_0M . The equality itself was proved in the unfiltered differential module. The inclusion $F_1M \subset M$ is an inclusion of actual sheaves, so vanishing in M implies vanishing in F_1M . Taking the order-one symbol shows that the global section e from Equation (5) is killed by

$$T_{\mathcal{Y}} \otimes F_0M \longrightarrow \mathrm{Gr}_1^F M, \tag{9}$$

after twisting by $C^{-(a+k)}$. Only the relation is used in this filtered step; no bound on the Hodge filtration of the auxiliary cochain $(\delta_i(b_i))$ has been imposed.

Because $F_{<0}M = 0$, the entire complex at filtration index one is

$$\mathrm{Gr}_1^F \mathrm{DR}(M) = [T_{\mathcal{Y}} \otimes F_0M \longrightarrow \mathrm{Gr}_1^F M],$$

in degrees -1 and 0 , with differential the symbol map in Equation (9). After the indicated twist, its degree- -1 hypercohomology is therefore exactly the space of global sections of the kernel of that map. The twist is negative, since $a+k > 0$. Lemma 4.3 annihilates this space, and hence $e = 0$.

It remains to kill the conormal part of the derivation. With coordinates and a line trivialization fixed locally, $e = 0$ says that every class $[e_{ij,\lambda}\sigma]$ vanishes in F_0M , and thus in M . Its actual

differential-operator image vanishes as well, before passing to symbols. Returning to the unfiltered relation gives

$$(a + k)[\eta_{ij}\sigma] = 0.$$

Injectivity of the lowest-piece inclusion and $a + k > 0$ then give $D_k(y) = 0$. The degree-zero derivation also vanishes: its restriction to $\mathcal{O}_U$ is the coordinate error e , and $\mathcal{O}_U \rightarrow \mathcal{O}_{\mathcal{T}_U}$ is surjective because $\mathcal{T}_U$ is a closed subscheme of U . The coefficient algebra and the conormal frame y generate the full graded length-one algebra locally. Thus $D_k = 0$ in every degree, completing the induction and proving surjectivity of all the sheaf transitions.

The kernels are the stated coherent sheaves $\mathcal{O}_{\mathcal{T}_U} \otimes C^{-j-k}$. If U is affine, their first cohomology vanishes, so the exact sequences of sheaves of abelian groups give surjectivity on global sections as well. Successive choices of lifts then give compatible formal lifts of any chosen functions on $\mathcal{T}_U$ and, after trivializing C , of the transverse conormal frame. $\square$

5 Compact null families and descent

We now complete Theorem 3.1. Retain $0 < v < n$, $r = v - 1$, $d = n - 1 - r = n - v$, and the construction of Proposition 3.3. The formal lifts first produce compact subvarieties outside the entire boundary, of dimension $n - v$, on which L is numerically trivial. Their existence bounds the dimension of the nef reduction; descent along that reduction then proves abundance.

The immediate geometric problem is to move a projective fiber off the boundary while keeping it projective. A normal coordinate supplies the direction of motion; parameters on the image of the boundary family keep the fiber dimension fixed. The global root and Hodge constructions have already supplied these functions formally. Their lifts concern the common underlying topological map of the thickenings; a map of ringed spaces to the base has not been assumed. The construction below obtains a deformation directly from the lifted functions, without requiring one ambient scheme neighborhood for all orders.

Proposition 5.1. *Assume the transition surjectivity of Proposition 4.1. There is a dominating algebraic family of integral projective subvarieties of X of dimension*

$$d = n - 1 - r = n - v$$

whose general members avoid D and on which L is numerically trivial.

Proof. We construct a formal deformation of a general boundary fiber, algebraize it over $\mathbb{C}[[s]]$, and show that the resulting subvarieties belong to a family dominating X .

Deforming a boundary fiber. Choose a separated affine etale chart $U \rightarrow \mathcal{Y}$ near a general point of a top-dimensional component of its Stein image $\mathcal{T}_U$. After shrinking, C is trivial, that component is smooth of pure dimension r , and S_U is flat over it. Since the image of J in P has dimension less than r , we may also arrange that S_U is disjoint from J_U . Choose a closed point t in this open set. After a further shrinking, regular parameters $t_1, \dots, t_r$ on $\mathcal{T}_U$ cut out only t . The fiber

$$F = S_{U,t}$$

is projective and has pure dimension d . Along F , the pullbacks of the parameters form a regular sequence. The ideal sheaf $\mathcal{I}_{J/S}$ is maximal Cohen–Macaulay by Proposition 3.3(ii). Since S_U is disjoint from J_U , the etale pullback of this ideal sheaf is $\mathcal{I}_{J_U/S_U} = \mathcal{O}_{S_U}$; hence S_U is Cohen–Macaulay. The quotient F by the regular sequence is Cohen–Macaulay as well. We will use this property to control the dimensions of the components after algebraization.

Write $I = \mathcal{O}_{\mathcal{X}}(-S)$. By Proposition 4.1, the transitions between the sheaves $g_*(I^j/I^{j+k})$ are surjective as the length k increases. Their kernels are the coherent sheaves $g_*\mathcal{O}_{S_U} \otimes C^{-j-k}$. Affineness of U also gives surjectivity on global sections. We can therefore lift the parameters compatibly through all thickenings, and lift the chosen conormal frame to a compatible element $\tilde{y}$ that formally generates I .

Let Z_q be the closed subscheme of qS_U cut out by the r lifted parameters. Regard it as a scheme over

$$R_q = \mathbb{C}[s]/(s^q), \quad s \mapsto \tilde{y}.$$

The construction has the diagram

$$\begin{array}{ccccc} F & \hookrightarrow & Z_q & \longrightarrow & X \times \operatorname{Spec} R_q \\ \downarrow & & \downarrow & & \downarrow \\ \operatorname{Spec} \mathbb{C} & \hookrightarrow & \operatorname{Spec} R_q & \xlongequal{\quad} & \operatorname{Spec} R_q. \end{array}$$

The middle vertical map is defined by the lifted normal parameter. The lifted base parameters cut out Z_q ; they are not provided by a preexisting morphism of the ambient thickening to the base.

The schemes Z_q are compatible under reduction, have closed fiber F , and are flat over R_q . For flatness, the powers of the Cartier generator first identify the successive s -layers of qS_U with $\mathcal{O}_{S_U}$. This proves flatness before imposing the parameters. The local flatness criterion then preserves flatness when we quotient by lifts of the regular sequence on the closed fiber.

Algebraization and boundary avoidance. The map $Z_q \rightarrow X \times \operatorname{Spec} R_q$ is quasi-finite. It is also proper: its reduction is the map from the projective scheme F , and nilpotent thickenings do not change properness of a finite-type morphism. Thus it is finite. Put $R = \mathbb{C}[[s]]$. Projective Grothendieck existence algebraizes the compatible finite algebra sheaves on $X \times \operatorname{Spec} R_q$ to a finite algebra on $X_R = X \times \operatorname{Spec} R$. Full faithfulness algebraizes their multiplication and unit [62, Lemmas 30.24.1 and 30.24.3]. The relative spectrum gives a finite morphism

$$Z \longrightarrow X_R$$

with exactly the prescribed reductions. In particular, Z is projective over R . Formal flatness gives flatness along the closed fiber; on the generic fiber over $\mathbb{C}((s))$ it is automatic. Hence Z is flat over R .

The images of the pole and coefficient-zero boundary parts miss F . Their inverse images in Z are closed and proper over R , and must therefore be empty: any nonempty closed subset of a proper R -scheme specializes to the closed fiber. Recall also the exceptional divisor E of the normalized blowup separating S_+ and S_- in Proposition 3.3. Its image in X lies in $\operatorname{Supp} G_+ \cap \operatorname{Supp} G_-$, so it is already excluded by avoidance of the pole part.

On the formal neighborhood from which the Z_q were cut out, the root construction therefore identifies the divisor of the pulled-back section s_0 with ℓS . Comparing its root-line frame with the formal Cartier generator $\tilde{y}$ gives a frame in which this section is $\tilde{y}^\ell$. Restricting to the Z_q gives compatible trivializations

$$N_0|_{\widehat{Z}} \simeq \mathcal{O}_{\widehat{Z}}, \quad s_0|_{\widehat{Z}} = s^\ell.$$

Full faithfulness algebraizes this line-bundle trivialization and its section identity on Z ; no algebraic map from Z to the root stack is needed. The generic fiber of Z therefore misses the positive part of D as well, and hence avoids all of D .

We next check the dimension assertion needed for the family. At a closed point z of F , flatness makes s a nonzerodivisor in $\mathcal{O}_{Z,z}$, and its quotient is the Cohen–Macaulay ring $\mathcal{O}_{F,z}$ of dimension d .

Thus $\mathcal{O}_{Z,z}$ is Cohen–Macaulay of dimension $d + 1$. These local rings are also equidimensional: a Cohen–Macaulay local ring essentially of finite type over the excellent discrete valuation ring R has this property. Every component of Z meets F by properness, and no component is contained in F by flatness. The dimension formula now gives dimension d for every component of the generic fiber. Its finite image in $X_{\mathbb{C}((s))}$ therefore has pure dimension d . After a suitable finite extension of $\mathbb{C}((s))$, choose a geometrically integral component of the reduced base change of this image.

A family dominating X . We have constructed compact subvarieties outside D . To obtain a dominating family, choose a positive component D_i with image dimension r , together with a component of S covering it. As the general chart and t vary, the images of points of F contain a dense open subset of D_i . We will show that each point in this open subset lies in the closed evaluation image of an algebraic family of d -dimensional subvarieties disjoint from D .

Now consider all Hilbert schemes of d -dimensional subvarieties of X . The geometrically integral loci parametrizing members disjoint from D have a countable stratification by integral parameter spaces with irreducible universal families. Denote the irreducible closures of their evaluation images in X by W_α . Fix a point x in the dense open subset of D_i , and a point of a boundary fiber F mapping to it. Since Z has no vertical component, this point lies in the closure of a generic component. That component has dimension d by the preceding argument. Pass to a finite extension of the discrete valuation field over which the reduced generic components are geometrically integral, and let R' be the extended discrete valuation ring. Choose a generic component after this base change whose closure contains a point over the chosen point of F . Flatness still excludes vertical components, so such a choice is possible. Its reduced finite image is geometrically integral. Take the schematic closure of that image in $X_{R'}$. The structure sheaf of the closure has no uniformizer torsion, so the closure is flat over the discrete valuation ring and defines a morphism to the Hilbert scheme. Its generic member lies in one of the strata above. The corresponding W_α is closed and contains the generic image, hence also its special-fiber support and the point x . Thus every point of the chosen open subset of D_i belongs to some W_α . The generic component and the index α may depend on x .

An irreducible variety over the uncountable field $\mathbb{C}$ cannot have a dense open covered by countably many proper closed subsets. Some W_α therefore contains D_i . By definition it also contains points outside D . Since D_i is a prime divisor in the integral variety X , the only proper irreducible closed subset containing it is D_i itself. Hence $W_\alpha = X$, and the family dominates X . On each member, avoidance of D makes the rational section s_0 regular and nowhere zero. It trivializes N_0 , proving that L is numerically trivial on that member. $\square$

5.1 Descent along the nef reduction

The compact null subvarieties will bound the dimension of the nef-reduction base. To descend the signed representative to that base, we first prove a statement for vertical divisors using relative nefness and a fiber-dimension bound.

Lemma 5.2. *Let $f : Y \rightarrow B$ be a surjective projective morphism from a normal integral variety to a smooth projective variety, with geometrically connected integral generic fiber. Suppose all fibers have dimension at most $\dim Y - \dim B$. Let G_Y be a vertical $\mathbb{Q}$ -Cartier divisor that is relatively nef. Then*

$$G_Y = f^*G_B$$

for a $\mathbb{Q}$ -divisor G_B on B .

Proof. When B is a point, every vertical divisor is zero. When the relative dimension is zero, the fiber bound makes f finite. Its geometrically integral connected generic fiber makes it birational,

and normality of B then makes it an isomorphism. The assertion follows in both cases. We may therefore assume that the base dimension and the relative dimension are positive.

The fiber bound forces every vertical prime divisor to map onto a prime divisor of B . For each such prime P , write its full pullback as $f^*P = \sum_j m_j E_j$. Let b_j be the coefficient of G_Y along E_j , with zero coefficients included, and assign coefficient $\min_j(b_j/m_j)$ to P in G_B . Only finitely many of these coefficients are nonzero. The residual divisor

$$R' = G_Y - f^*G_B$$

is effective, vertical, $\mathbb{Q}$ -Cartier, and relatively nef; over each base prime, at least one component is absent from its support. It remains to prove that $R' = 0$.

Suppose $R' \neq 0$, and choose a base prime under its support. Intersect B with general hyperplanes to obtain a complete-intersection curve meeting that prime at a general point; if $\dim B = 1$, use B itself. Normal Bertini makes its inverse image in Y normal. This inverse image is integral: generic geometric integrality supplies the dominating component, and the fiber bound rules out further vertical components. Intersect upstairs with $\dim Y - \dim B - 1$ general very ample hyperplanes. The result is a normal integral surface over the base curve, with geometrically connected generic fiber. Over the chosen general base point, these cuts retain curves both in the residual support and in a component missing from it. The curves remain distinct because the original components were distinct at the generic point of the base prime.

Resolve this surface. The pullback of the residual divisor stays effective, vertical, and relatively nef. It has zero intersection with the full fiber and nonnegative intersection with every fiber component. All those component intersections must therefore be zero. The fiber is connected, by generic connectedness and Stein factorization over the normal base curve. Its intersection matrix is negative semidefinite, with kernel generated by the full fiber. The residual part over the chosen point is thus a multiple of the full fiber. Its missing component forces that multiple to be zero, contradicting the retained component of its support. $\square$

Completion of the proof of Theorem 3.1. The cases $v = 0$ and $v = n$ are covered by Lemma 3.2. For $0 < v < n$, Propositions 3.3, 4.1, and 5.1 give a dominating family of compact L -trivial subvarieties of dimension $d = n - v$.

Apply the nef-reduction theorem to a Cartier multiple of L [2, Theorem 2.1]. The resulting almost holomorphic rational map has connected fibers. The divisor L is numerically trivial on its compact general fibers and has positive degree on every noncontracted curve through a very general point of X . Write b for the base dimension. A compact L -trivial d -fold through such a point must be contracted: otherwise, general ample slices through the point would produce a noncontracted curve of L -degree zero. Consequently

$$b \leq n - d = v.$$

To apply vertical descent, resolve the graph, flatten the main component over a modification of the base, resolve the base, and normalize the main transform. This gives projective morphisms

$$\begin{array}{ccc} Y & \xrightarrow{\pi} & X \\ f \downarrow & & \\ B & & \end{array}$$

with π birational, B smooth, Y normal, and geometrically connected integral generic fiber of f . Every fiber has dimension at most $n - b$. Indeed, flatness and generic integrality keep the flat main

transform integral after the base modification, and finite normalization preserves the bound on fiber dimension. This bound is the needed hypothesis; normalization need not preserve flatness.

We claim that π^*D has no horizontal component. Restrict to a very general fiber of f . There, π^*L is numerically trivial and $\pi^*(L - cD)$ restricts to a pseudo-effective class. To justify the second assertion, choose a sequence of effective approximants with ample errors tending to zero. Shrink to a flat open of the base with geometrically integral fibers. This is possible because the generic fiber is geometrically integral. On this common open, the locus over which a given support contains the whole fiber is proper closed, by upper semicontinuity of fiber dimension. Avoid the countably many such bad closed loci in the base. Every approximant then restricts effectively to a very general fiber, and passage to the limit gives pseudo-effectivity of the restriction. The resulting class is $-c\pi^*D$. A negative nonzero effective divisor cannot be pseudo-effective on a projective variety, as its intersection with an ample power shows. The restriction of π^*D must therefore be zero. This also excludes $b = 0$, which would force $D = 0$ and contradict $v > 0$.

It follows that $G_Y = \pi^*G$ is vertical. It is also relatively nef because $G_Y \sim_{\mathbb{Q}} \pi^*L$. Lemma 5.2 now gives

$$\pi^*L \sim_{\mathbb{Q}} f^*G_B$$

for a $\mathbb{Q}$ -divisor G_B on the smooth base. Every curve in B is dominated by a curve in Y , where the pullback has nonnegative degree; hence G_B is nef. Numerical dimension is unchanged by these pullbacks; the intersection formula gives

$$v = \nu(X, L) = \nu(Y, \pi^*L) = \nu(B, G_B) \leq b.$$

Together with $b \leq v$, this yields $b = v = \nu(B, G_B)$. Since G_B is nef on the b -dimensional base, it follows that $G_B^b > 0$, so G_B is big. Pullback of sections then gives $\kappa(X, L) \geq b = v$. The inequality $\kappa(X, L) \leq \nu(X, L)$ for nef divisors provides the reverse bound and proves abundance. $\square$

6 Geometric exclusions for a nonvanishing counterexample

We now prepare the smooth nonvanishing step. Assume only the lower-dimensional good-model hypothesis of Assumption 2.1, and suppose that a smooth projective n -fold X , with $n > 0$, satisfies

$$K_X \text{ is pseudo-effective,} \quad \kappa(X, K_X) = -\infty. \quad (10)$$

We first restrict to moving proper subvarieties and positive canonical currents, and then exclude covering curves of bounded genus and normalized degree. Two further conclusions will handle the remaining jet configurations: the reduced-boundary adjoint of a signed canonical representative is big, and fixed finite covers admit no family of birational maps whose graphs dominate their product. The argument takes place over $\mathbb{C}$, and the two displayed properties persist on smooth projective birational models. Numerical dimension for a pseudo-effective divisor means Nakayama's numerical dimension κ_{σ} ; for a nef divisor it agrees with the intersection definition used in Theorem 3.1.

6.1 The Albanese reduction and algebraic webs

Subadditivity first shows that the Albanese map is constant. This will let us pass from an effective numerical representative of a rational divisor to an effective rational-linear representative. We then construct the rational quotient generated by a moving family of subvarieties.

Lemma 6.1. *Every smooth projective birational model of X has irregularity zero. Consequently, on such a model, or on a projective $\mathbb{Q}$ -factorial terminal birational model, numerical equivalence of rational divisors implies their $\mathbb{Q}$ -linear equivalence.*

Proof. If $q(X) > 0$, resolve the Stein factorization of the Albanese map to obtain a morphism $f : W \rightarrow B$ with connected fibers, with W, B smooth projective and $\dim B > 0$. The map from B to a subvariety of $\text{Alb}(X)$ is generically finite. Wedges of invariant one-forms on the abelian variety therefore give a nonzero section of K_B : at the generic point, choose $\dim B$ independent pulled-back one-forms. Thus $\kappa(B, K_B) \geq 0$.

For a very general smooth fiber F , the restriction of the pseudo-effective class K_W is pseudo-effective. For example, restrict a positive current representing it to almost every fiber, or use effective approximations with arbitrarily small ample error. Adjunction identifies this restriction with K_F . If F has positive dimension, the induction hypothesis gives $\kappa(F, K_F) \geq 0$; for a point the same assertion holds by convention. Applying Theorem 1.2 with both boundaries empty gives $\kappa(W, K_W) \geq 0$, contradicting (10).

This is the only use of Theorem 1.2 in the proof. Irregularity is a smooth birational invariant. Finally, when $\text{Pic}^0 = 0$, a numerically trivial line bundle is torsion, since the group of numerically trivial line bundles modulo Pic^0 is finite. Clear denominators for rational divisors. On a terminal $\mathbb{Q}$ -factorial model, pull back to a smooth resolution and then descend the rational linear equivalence. $\square$

To handle subvarieties through very general points, parameterize them in countably many Hilbert families, then resolve and stratify. On a suitable open parameter space the domains form a smooth projective family with integral fibers, and evaluation is dominant. When each fiber maps generically finitely onto its image, general ample cuts of the parameter space make total evaluation generically finite while preserving dominance. We resolve and compactify this total evaluation before using its ramification divisor. Invariance of plurigena for smooth projective families [55, Theorem 1] permits the same construction for the Iitaka fibers of general members: choose a divisible pluricanonical system on the general member, spread it by base change, and resolve its relative rational map. When the member has nonnegative Kodaira dimension but is not of general type, the smooth general fibers of this map have positive dimension. They form the new family used below.

Lemma 6.2 (Algebraic web quotient). *Let a smooth projective variety be dominated generically finitely by the total space of a family with smooth integral projective general fibers. On a dense regular open, form the distribution generated by the tangent spaces of the fiber images, taking spans, saturation, and Lie brackets. Its general leaves are dense opens of algebraic subvarieties. Their closures are the general fibers of a rational map, which has connected general fiber after Stein factorization.*

Proof. Shrink to an open U where the evaluation has finitely many etale sheets, the parameter map is smooth, and the generated involutive distribution $\mathcal{F}$ has constant rank. Its construction is algebraic: spans and brackets stabilize at the generic point after finitely many operations, and descend from the etale sheets. The nonempty open parts of the integral parameter fibers are connected. A chain step moves between two points in the image of one such open fiber. We use only steps lying in U , so each step remains in one analytic leaf of $\mathcal{F}$.

For $x \in U$, endpoints of chains of at most k steps form a constructible set, by the algebraic fiber-product construction and Chevalley's theorem. Every endpoint lies in the analytic leaf through x . A locally closed smooth variety contained in an analytic leaf has dimension at most $\text{rk } \mathcal{F}$. Indeed, in a Frobenius chart the leaf meets at most countably many plaques: use finite plaque chains in a countable foliation atlas. The transverse coordinate map on any connected smooth piece then has countable image and is constant.

There is therefore a maximal dimension among the closures of all such endpoint loci; it is attained for some finite k . The loci are nested because we allow at most k steps, including the identity step. Choose an irreducible component V of maximal dimension and a dense open $O \subset V$ of reachable

points. On each etale sheet, the parameter map defines the algebraic relation consisting of pairs in the same fiber. Restrict its first coordinate to the inverse image of O , take the component through the diagonal, and map the second coordinate to U . The closure of this image contains O , hence V , while its points are reachable in at most $k + 1$ steps. Maximality forces that closure to equal V . At a general diagonal point, smoothness of the parameter map identifies its vertical tangent with the corresponding web tangent. Every web tangent is consequently tangent to V , and so is every bracket. Hence $\dim V \geq \operatorname{rk} \mathcal{F}$, while the reverse inequality was proved above. A plaque is thus open in V . The equations of V , pulled back to the connected immersed leaf, vanish on a nonempty open and hence on the entire leaf. Its closure is exactly V .

These invariant subvarieties of dimension $\operatorname{rk} \mathcal{F}$ have countably many Hilbert or Chow parameter spaces. Tangency on the regular locus is an algebraic condition after stratification, so one such family dominates. An invariant irreducible variety meeting U contains the leaf through any of its points in U : the tangent vector fields preserve its reduced ideal, first on its smooth locus and then everywhere by closure. A leaf closure of dimension $\operatorname{rk} \mathcal{F}$ is therefore unique through a general point. Assigning that closure gives the desired rational map. Resolve its graph and take the Stein factorization. $\square$

For positive closed $(1, 1)$ -currents we shall use this quotient in the following way. If a current vanishes on almost every parameter fiber away from a fixed removed divisor, it annihilates the corresponding fiber tangents on a dense open. In submersion coordinates, choose locally integrable plurisubharmonic potentials. Fubini's theorem makes the pure fiber Hessian zero as a distribution; positivity then makes the mixed coefficients in fiber directions zero as well. A generically etale evaluation transfers this assertion to the target. Annihilation passes to brackets by the identities

$$\mathcal{L}_v T = d(\iota_v T) + \iota_v(dT), \quad \iota_{[v,w]} T = \mathcal{L}_v(\iota_w T) - \iota_w(\mathcal{L}_v T).$$

Pullbacks by dominant morphisms and restrictions to almost every smooth parameter fiber are defined by the same local potentials.

6.2 The general-type exclusion

The next proposition turns the lower-dimensional good-model hypothesis into a restriction on every moving proper subvariety. Its proof passes from Iitaka fibers to the algebraic quotient of their tangent directions and then back to nonvanishing.

Proposition 6.3. *Every positive-dimensional proper subvariety through a very general point of X is of general type on resolution. The assertion holds also on every projective birational model.*

Proof. Otherwise take a covering family of nongeneral-type subvarieties, with smooth domains V , and slice its parameters as above. Restricting the ramification formula of its generically finite total evaluation shows that K_V dominates the restriction of the pulled-back K_X by an effective divisor. Thus K_V is pseudo-effective. The induction hypothesis gives a good minimal model of V , so $\kappa(V) \geq 0$. Since V is not of general type, its general Iitaka fibers have positive dimension and Kodaira dimension zero. Spreading these fibers gives another covering family with smooth domains G , of dimension less than n , and $\kappa(G) = 0$. Each G has a good minimal model and $\kappa_\sigma(K_G) = 0$.

To exploit this zero-Iitaka-dimension family, fix a positive current T representing K_X . After slicing and resolving the total evaluation of the G 's, its restriction to almost every parameter fiber, plus the effective restricted ramification divisor, is a positive current in K_G . Every positive current

in that class is the current of the fixed canonical divisor of G . To see this, let R be such a current and take a common resolution $u : H \rightarrow G$, $v : H \rightarrow G_{\min}$ with the good minimal model. The current

$$u^*R + [K_H - u^*K_G]$$

is positive and represents K_H . Since $K_{G_{\min}}$ is torsion, this class is represented by an effective v -exceptional divisor. Its intersection with the $(\dim G - 1)$ -st power of an ample pullback from $G_{\min}$ is zero. That form is strictly positive off the exceptional locus, so the current is supported there. The support theorem makes it divisorial, and independence of exceptional divisor classes, by negativity, determines its coefficients. Pushing forward by u determines R itself. After shrinking the parameter space, these fixed canonical divisors are the fixed divisor of a sufficiently divisible relative pluricanonical system. Remove this divisor and the excluded loci of the construction from the total family. Their images under generically finite evaluation are proper algebraic subsets of X .

It follows that T annihilates the web distribution of the G 's on a dense open. If that distribution has full rank, T is supported on a proper algebraic set. The support theorem for positive closed $(1, 1)$ -currents expresses it as a finite nonnegative real combination of prime divisors [58, 17]. Its rational cohomology class then has a nonnegative rational representative: solve the finite rational linear system for the coefficients on the same face of the nonnegative orthant. Lemma 6.1 converts numerical effectivity to $\mathbb{Q}$ -linear effectivity, a contradiction.

Otherwise Lemma 6.2 gives a rational quotient with general smooth fiber F on a smooth resolved graph, where $0 < \dim F < n$. Its canonical divisor is pseudo-effective: restrict the pseudo-effective canonical class of the smooth graph to a very general fiber, as in Lemma 6.1. The moving G 's still generate its tangent space at general points. Indeed the quotient is constant on every general web member, hence induces a rational map on their parameter space; restricting to a general quotient fiber preserves dominance of evaluation. A pluricanonical rational map of F is constant along each such G . Slice the parameters inside F to make the total evaluation generically finite: ramification injects the restricted pluricanonical sections into sections of a multiple of K_G , and these span a space of dimension at most one. Fix a nonzero denominator section and work where it does not vanish. Its restriction is nonzero on a general member of the covering family, so all ratios are constant on that member. Their differentials vanish on the web and its brackets, so the pluricanonical map is constant on F . Lower-dimensional nonvanishing therefore gives $\kappa(F) = 0$, and its good minimal model gives $\kappa_\sigma(K_F) = 0$.

The quotient has now reached the exact setting of the numerical-zero-fiber reduction of Gongyo–Lehmann [31, Theorem 1.3 and Corollary 4.5]. For a projective $\mathbb{Q}$ -factorial klt rational pair with a connected-fiber morphism and numerical dimension zero on the general fiber, that theorem produces a klt pair on a smooth birational base whose good-minimal-model existence is equivalent. Here the total space is the smooth graph, the boundary is zero, and the base has dimension less than n . The required numerical dimension is κ_σ , which is zero by the good model of F ; thus the numerical-dimension qualification, with the corrected convention discussed in [23, Section 3 and Theorem 3.2], causes no issue. The induction hypothesis supplies the base good model, hence nonvanishing upstairs, contradicting (10). $\square$

Corollary 6.4. *Let Y be a projective $\mathbb{Q}$ -factorial terminal birational model of X . If a rational divisor P has $\kappa(Y, P) \geq 1$, then $K_Y + cP$ is big for every rational $c > 0$. Consequently, if $\alpha \neq 0$ is a supporting functional of the pseudo-effective cone with $\alpha(K_Y) = 0$, then $\alpha(P) > 0$.*

Proof. Resolve a moving subsystem of a multiple of P . If its map is generically finite, P is big and the assertion follows from pseudoeffectivity of K_Y . Otherwise its general fiber is a proper positive-dimensional subvariety through very general points and is of general type by Proposition 6.3.

The canonical divisor of the smooth resolution is relatively big, so adding a sufficiently large ample pullback from the image makes it big. Since that pullback is bounded by a multiple of the resolved moving subsystem, pushforward gives $K_Y + aP$ big for some $a > 0$. Convexity with the pseudo-effective K_Y , and then addition of the pseudo-effective P , gives the assertion for every $c > 0$. A nonzero nonnegative functional on a closed convex cone is strictly positive on its interior. Applying it to $K_Y + cP$ proves the last assertion. $\square$

6.3 Valuations and positive currents

The next two exclusions involve currents whose numerical intersection gaps tend to zero. A limiting inequality alone is insufficient: we need one family on which the gap is exactly zero. The ACC statement below supplies that step for a fixed current and bounded discrepancies.

For a positive closed $(1,1)$ -current T on a smooth projective variety V , let $\nu_E(T)$ denote the generic Lelong number of its pullback at a divisorial valuation E . The normalization is such that a reduced smooth divisor has generic number one. We write $A_V(E) = a(E; V, 0)$ for log discrepancy.

Lemma 6.5. *For a fixed T and a fixed M , the set*

$$\{\nu_E(T) : A_V(E) \leq M\}$$

satisfies the ascending chain condition.

Proof. We induct on the integer part of M ; discrepancies over a smooth variety are positive integers, so the assertion is empty when $M < 1$. Suppose that a strictly increasing sequence exists, and discard initial terms so its members exceed a fixed positive number c . Skoda integrability and change of variables give

$$\nu_E(T) \leq A_V(E)\nu_x(T) \tag{11}$$

at a general point x of the center. Indeed every exponent smaller than $1/\nu_x(T)$ is locally integrable, whereas integrability after pullback imposes the corresponding discrepancy bound. Hence all centers lie in the fixed proper Siu locus $\{\nu_x(T) \geq c/M\}$, which is algebraic by Siu analyticity and projectivity [58]. Passing to a subsequence, all centers lie in one irreducible component of this locus.

If, after passage to a subsequence, the centers lie in a fixed codimension-at-least-two subvariety, principalize its ideal. All lifted centers lie in the exceptional locus, where the relative canonical divisor has positive integral order. The discrepancy bound for the new smooth ambient space has therefore decreased by at least one. Pull back T and apply induction.

Otherwise the centers lie in a fixed prime divisor D of the Siu locus. Write $T = c_D[D] + T'$ with T' positive and with zero generic Lelong number along D . The integers $\text{ord}_E(D)$ are bounded: the fixed effective divisor D has positive log canonical threshold, and $\text{lct}(D)\text{ord}_E(D) \leq A_V(E)$. Pass to a constant order. The residual numbers $\nu_E(T')$ are still strictly increasing, and after discarding a term have a positive lower bound. Equation (11) places their centers in a fixed Siu locus of T' , which does not contain D . Their intersection with D has codimension at least two, so the preceding case applies. $\square$

To compare those Lelong numbers with movable algebraic systems, use the following multiplier-ideal approximation [17, Theorems 5.11, 6.27, and 14.2]. On a smooth projective variety, if $\{T\}$ is a real algebraic class, one can choose integral line bundles L_k such that $c_1(L_k) - k\{T\}$ stays in a bounded, uniformly sufficiently ample set and $L_k \otimes \mathcal{J}(kT)$ is globally generated. Round the coefficients of $k\{T\}$ in a fixed integral basis and add a fixed sufficiently positive integral class. Nadel vanishing and Castelnuovo–Mumford regularity give generation. For every divisorial valuation,

$$\text{ord}_E \mathcal{J}(kT) \leq k\nu_E(T). \tag{12}$$

The local Bergman weight is bounded below by the original weight up to a constant; this inequality persists after pullback and gives (12). These are statements for a fixed current, not uniform assertions over all currents.

Proposition 6.6. *Let Y be a fixed projective $\mathbb{Q}$ -factorial terminal birational model of X , with K_Y nonbig. A positive current in the pullback class of K_Y on a smooth resolution cannot, on a dense regular Zariski open, annihilate the tangent spaces of the general fibers of a rational fibration of positive relative dimension.*

Proof. Fix a current T as in the statement and suppose such a fibration exists. Consider all dominant rational maps from Y with connected general fiber of positive dimension for which the pulled-back current annihilates the fiber tangents on a dense regular open of a resolved graph. Choose one with smallest base dimension b . We will descend the current to this base and find a covering family of curves on which the descended current vanishes. Their quotient will give a member of the same collection with smaller base dimension. If $b = 0$, the current vanishes on a dense open; the divisorial-current argument in Proposition 6.3 contradicts (10). Thus $0 < b < n$. Resolve the graph and the space carrying the current:

$$\begin{array}{ccc} W & \xrightarrow{p} & Y \\ h \downarrow & & \\ B & & \end{array}$$

Here W, B are smooth projective and h has connected general fiber. All further modifications replace this graph and base by birational models; they do not change the chosen rational map or the function field of its base. Flatten the main component over a modification of the base, resolve that base, normalize the main transform, and then resolve upstairs. Every vertical prime on the flat transform maps to a divisor of the base: a prime over base codimension c would have generic fiber dimension at least $n - b + c - 1$, whereas all fibers have dimension at most $n - b$. A prime of W either maps onto a prime of the flat transform, where this dimension count applies, or is exceptional over that transform. In the latter case it is also exceptional over Y , because the transform maps birationally to Y . Thus every vertical divisor on W over base codimension at least two is exceptional over Y . This remains true after later resolutions and base modifications: a prime dominating a divisor of Y already has divisorial center on the old base, and its center on a higher base model is the strict transform of that divisor.

For divisors and numerical classes put

$$T_B = p_* h^*.$$

This strict pullback is well defined on numerical classes because Y is $\mathbb{Q}$ -factorial: numerical classes upstairs decompose into pulled-back classes from Y and exceptional classes. It preserves pseudoeffectivity. For a higher base model $r : B' \rightarrow B$, its analogue satisfies $T_{B'} r^* = T_B$, by computing on a common graph. The operator $T_{B'}$ kills divisors exceptional over B : otherwise a prime in such a pullback that dominates a divisor of Y would have center of codimension at least two on B , contrary to the property just established.

Descent of the current. Let T also denote the pulled-back current on W . Shrink to a smooth open B^0 so that h is smooth proper, the removed set contains no vertical divisor over B^0 , and the remaining open in every fiber is nonempty and connected. Away from that fixed removed algebraic set upstairs, T descends to a positive current S^0 . Indeed in submersion coordinates the horizontal coefficient distributions are independent of fiber variables by closedness, and the open parts of general fibers are connected. They therefore glue to S^0 on B^0 . The difference $T - h^* S^0$ is closed

of order zero and supported on the removed algebraic set. The support theorem expresses it as a signed divisor sum; a closed order-zero $(1, 1)$ -current supported in codimension at least two is zero. Its horizontal coefficients are nonnegative, since a pullback has zero generic Lelong number along a horizontal divisor.

Subtract those finitely many global Siu components from T , obtaining a positive current T' that equals h^*S^0 on all of $h^{-1}(B^0)$. Choose a Kähler form ω representing an ample algebraic class and put $d = n - b$. Fiber integration gives $h_*(\omega^d) = c > 0$, constant on B^0 , by closedness. The projection formula therefore gives

$$c^{-1}h_*(T' \wedge \omega^d)|_{B^0} = S^0.$$

The left side defines a global positive closed extension. Its class is real algebraic, since $\{T'\}$ is the algebraic class p^*K_Y minus real divisor classes and algebraic intersection and pushforward preserve such classes. Remove its divisorial parts along $B \setminus B^0$, and call the result S . On a dense open $T = h^*S$. Their difference is again closed of order zero, so the same support theorem leaves only a signed divisor sum. Over B^0 , this difference consists only of the horizontal components removed above. Its coefficient at a prime dominating a base prime that meets B^0 is therefore zero. Along a base prime contained in $B \setminus B^0$, the current S has zero generic Lelong number by construction. At a prime upstairs dominating it, the generic local map is a transverse power map times a submersion. Local target balls of a radius equal to a fixed power of the source radius give the Lelong comparison, so h^*S also has zero generic number there. Positivity of T then makes the coefficient of $T - h^*S$ nonnegative. Thus all coefficients at primes dominating base primes are nonnegative. The only possible negative coefficients lie over base codimension at least two and are exceptional over Y . We have therefore obtained a positive closed current S in a real algebraic class on B , with $T - h^*S$ a divisor current whose only possible negative coefficients are p -exceptional. Push this difference forward by p . The exceptional terms disappear and the remaining terms are effective, so

$$K_Y - T_B\{S\} \text{ is pseudo-effective.} \quad (13)$$

The horizontal components removed during descent contribute effective divisors to the same difference.

A negative canonical direction on the base. Choose a nonzero supporting functional α of the pseudo-effective cone at K_Y , and put $\eta = T_B^*\alpha$. The corresponding functionals η' on higher smooth base models are movable classes by pseudo-effective/movable duality [8, Theorem 0.2]. They annihilate base-exceptional divisors. Since S is positive, $\eta \cdot \{S\} \geq 0$. Applying α to (13) gives the reverse inequality, because $\alpha(K_Y) = 0$. Hence $\eta \cdot \{S\} = 0$. If $c_D[D]$ is a positive Siu component of S , both $S - c_D[D]$ and $[D]$ are positive. It follows that $\eta \cdot D = 0$, or equivalently $\alpha(T_B D) = 0$.

Before applying multiplier approximation, we need to control the divisorial part of this descended current. Only finitely many prime divisors on B have positive divisorial Lelong coefficient for S . Otherwise their nonzero strict pullbacks are effective divisors on Y killed by α , with no common prime components. Indeed, a nonexceptional prime of Y cannot map into the intersection of two base primes. Only finitely many base primes have zero strict pullback, since their total pullbacks are supported in the finite exceptional locus of p . In the finite-dimensional rational space of divisor classes, infinitely many remaining strict pullbacks have a rational relation. This relation must have both positive and negative coefficients: a nonzero effective sum has positive intersection with an ample $(n - 1)$ -fold product on Y , so cannot be numerically zero. The two sides are thus nonzero effective rational divisors with no common prime component. Lemma 6.1 makes them $\mathbb{Q}$ -linearly equivalent. Clearing denominators gives two distinct linearly equivalent effective integral divisors and hence a pencil, whose class is killed by α . This contradicts Corollary 6.4.

We claim that

$$K_B \cdot \eta < 0. \quad (14)$$

The smooth general fiber F of h is of general type by Proposition 6.3. The relative-positivity theorem of Kovács–Patakfalvi [44, Theorem 9.9] applies to a log canonical fiber space with smooth base, SNC total pair, and log-general-type geometric generic fiber; for a rational base divisor M with $\kappa(M) \geq 0$, it gives relative subadditivity with the term h^*M . Here both spaces are smooth projective, the boundary is zero, and we take $M = 0$. Invariance of plurigenera on the smooth locus identifies the generic-fiber Kodaira dimension with that of F . Thus

$$\kappa(W, K_W - h^*K_B) \geq \kappa(F, K_F) = n - b > 0. \quad (15)$$

This is an established general-type-fiber theorem, not another use or strengthening of Theorem 1.2. Choose compatible canonical divisors and set $P = K_Y - T_B(K_B) = p_*(K_W - h^*K_B)$. For every sufficiently divisible m , pushing forward an effective divisor $\text{div}_W(\varphi) + m(K_W - h^*K_B)$ gives $\text{div}_Y(\varphi) + mP \geq 0$. The resulting injection of sections preserves their ratios, so $\kappa(Y, P) \geq 1$. Corollary 6.4 now gives $0 < \alpha(P) = -K_B \cdot \eta$, proving (14).

Rational curves and an exact zero gap. Apply the multiplier approximation to S on B , and principalize $\mathcal{J}(kS)$ by $r_k : B_k \rightarrow B$. If F_k is its divisor, the class

$$P_k = \frac{r_k^*L_k - F_k}{k}$$

is nef. Put $\eta_k = T_{B_k}^* \alpha$. The class η_k is movable and F_k is effective, so $0 \leq P_k \cdot \eta_k \leq (L_k \cdot \eta)/k$. Since $\eta \cdot \{S\} = 0$ and $L_k - k\{S\}$ stays in a bounded set, this gives the first estimate below. The second uses the fact that η_k kills exceptional divisors:

$$0 \leq P_k \cdot \eta_k = O(k^{-1}), \quad K_{B_k} \cdot \eta_k = K_B \cdot \eta < 0.$$

Fix an ample divisor H on B , and write $\delta = -K_B \cdot \eta > 0$. On B_k , choose an ample divisor H_k and a sufficiently small rational $\epsilon_k > 0$ so that the ample rational class

$$Q_k = P_k + k^{-1}r_k^*H + \epsilon_k H_k$$

satisfies $Q_k \cdot \eta_k = O(k^{-1})$. This is possible because $\eta_k \cdot r_k^*H = \eta \cdot H$ is fixed. Approximate η_k by a positive sum $\gamma_k = \sum_{i=1}^{N_k} a_i [C_i]$, with $a_i > 0$, of covering-curve classes, closely enough that

$$Q_k \cdot \gamma_k \leq C/k, \quad -K_{B_k} \cdot \gamma_k \geq \delta/2$$

for a constant C independent of k . Keep only terms with $K_{B_k} \cdot C_i < 0$. Their total anticanonical degree is at least $\delta/2$, while their total Q_k -degree is at most C/k . Taking the weighted average of the ratios therefore gives one covering curve Γ_k with

$$\frac{Q_k \cdot \Gamma_k}{-K_{B_k} \cdot \Gamma_k} \leq \frac{2C}{\delta k}.$$

The quantitative bend-and-break estimate [51, Theorem 5] gives a rational curve through a general point of Γ_k with Q_k -degree at most $2 \dim B$ times this ratio. Parameterization and uncountability give a covering family of such rational curves, denoted R_k . Since $Q_k - P_k - k^{-1}r_k^*H$ is ample and both P_k and r_k^*H are nef, we obtain

$$P_k \cdot R_k = O(k^{-1}), \quad H \cdot r_{k*}R_k = O(1). \quad (16)$$

Shrink each parameter space so that the curve class and its contacts with the exceptional divisors and the strict transforms of the Siu divisors of S are constant. The general parametrized member is free in characteristic zero. Consequently $K_{B_k} \cdot R_k \leq -2$. Put $\bar{R}_k = r_{k*} R_k$. Its bounded H -degree bounds $K_B \cdot \bar{R}_k$, and effectivity of the relative canonical divisor gives

$$0 \leq K_{B_k/B} \cdot R_k = K_{B_k} \cdot R_k - K_B \cdot r_{k*} R_k = O(1). \quad (17)$$

In particular, the upper bound is independent of k . If an exceptional prime E has coefficient $a_E = A_B(E) - 1$ in $K_{B_k/B}$, a positive contact with it contributes $a_E(E \cdot R_k)$ to this bounded sum. Both factors are positive integers. Thus the number of exceptional primes met by R_k , their log discrepancies $A_B(E) = a_E + 1$, and their contact degrees are uniformly bounded. Contacts with the finitely many strict transforms of divisorial Lelong components of S are bounded by their fixed degrees against $\bar{R}_k$. Subtracting all these divisorial parts from $r_k^* S$ leaves a positive current. Its restriction to almost every member of the family is defined and positive, by the local-potential and Fubini argument following Lemma 6.2. Its degree is the first difference below; this proves that difference is nonnegative. Using (12) and (16) gives

$$\begin{aligned} 0 &\leq \{S\} \cdot \bar{R}_k - \sum_E \nu_E(S) E \cdot R_k \\ &\leq \{S\} \cdot \bar{R}_k - (F_k/k) \cdot R_k \leq O(k^{-1}). \end{aligned} \quad (18)$$

The sum includes exceptional primes and those strict divisorial components; terms with zero contact are irrelevant. For the last bound, the expression involving F_k equals $P_k \cdot R_k + (\{S\} - L_k/k) \cdot \bar{R}_k$. The first term is $O(k^{-1})$ by (16); the second has the same bound because $L_k - k\{S\}$ stays bounded and the H -degrees of $\bar{R}_k$ are bounded.

Bounded ample degree gives only finitely many numerical classes of the integral cycles $\bar{R}_k$. Pass to one class, so $\{S\} \cdot \bar{R}_k$ is now one fixed real number. The current S on B also remains fixed. By Lemma 6.5, the sums in (18) belong to an ACC set: there are a bounded number of terms, their nonnegative integral multiplicities are bounded, and all relevant discrepancies are bounded. Finite sums of this kind preserve ACC, by successively taking nonincreasing subsequences of their entries. We can now replace the limiting gap by an exact equality: the nonnegative gaps in (18) must equal zero for some covering families. Otherwise a sequence tending to zero has a strictly decreasing positive subsequence, giving a strictly increasing sequence of the complementary sums.

For a family with zero gap, restrict the residual current to almost every parameter fiber as above. Its integral on that complete curve is zero. A positive current on a curve is a positive measure, so zero integral forces it to vanish. Off the removed divisors the residual current agrees with $r_k^* S$. The discussion following Lemma 6.2 and that lemma itself yield a positive-relative-dimensional rational quotient of B whose vertical directions are annihilated by S on an open. Composing with h and taking the Stein factorization gives a quotient of smaller base dimension whose vertical directions are annihilated by T . It belongs to the collection minimized at the start, contradicting the choice of b . $\square$

6.4 Excluding normalized bounded curves

We apply the same exact-zero mechanism to curves on the small modifications of a fixed late model. The normalization by volume is chosen so that its scale tends to zero, while the current and the discrepancy lattice stay fixed.

Run a canonical LMMP on X with scaling of a fixed very ample rational divisor A . For rational $t > 0$, its positive-threshold stages give terminal $\mathbb{Q}$ -factorial models Y_t on which $K_t + tA_t$ is nef, big,

and semiample [6]. There are only finitely many divisorial contractions, since each lowers Picard number. Fix a late model Y after the last such contraction. All subsequent maps $Y \dashrightarrow Y_t$ are small, and are canonical nonpositive birational maps. If the program terminates, use the last model repeatedly. Define

$$\mu_t = \text{vol}(K_X + tA)^{1/n}. \quad (19)$$

Since K_X is not big, $\mu_t \rightarrow 0$. In particular K_Y is nonbig. The transform A_t is effective up to rational linear equivalence; all intersections below use general covering curves, which are not contained in a chosen such representative.

Lemma 6.7 (Exceptional current comparison). *Fix a positive current T on a smooth resolution in the pullback class of K_Y . On a common smooth resolution of $Y \dashrightarrow Y_t$, write*

$$p^*K_Y = p_t^*K_t + J_t, \quad J_t \geq 0,$$

where J_t is exceptional over Y_t . Then $\nu_E(T) \geq \text{coeff}_E J_t$ for every prime on that resolution.

Proof. Pass to a common smooth model V also dominating the fixed resolution carrying T . Choose on the fixed resolution a sufficiently positive line bundle H . For sufficiently divisible k , multiplier approximation gives a globally generated $\mathcal{O}(kp^*K_Y + H) \otimes \mathcal{J}(kT)$. Pull this system to V . For the finite collection of primes on the common resolution whose coefficients we are comparing, choose a general member D_k of the underlying effective divisor system. At every such prime E , it has multiplicity $\text{ord}_E \mathcal{J}(kT)$; global generation after removal of the ideal ensures no additional generic vanishing. Here the notation H also denotes its pullback to V .

Put $D'_k = p_{t*}D_k$ and $H_t = p_{t*}H$. The first is effective, and both are $\mathbb{Q}$ -Cartier because Y_t is $\mathbb{Q}$ -factorial. Pushing the rational linear relation gives

$$D'_k \sim_{\mathbb{Q}} kK_t + H_t.$$

Thus the exceptional divisor $D_k - p_t^*D'_k$ is rationally equivalent to

$$kJ_t + H - p_t^*H_t.$$

The two exceptional divisors are equal: their difference is numerically trivial and exceptional, so the negativity lemma applied with both signs makes it zero. Since $p_t^*D'_k$ is effective,

$$\text{coeff}_E D_k \geq k \text{coeff}_E J_t + \text{coeff}_E (H - p_t^*H_t).$$

The last coefficient is fixed for this model and t . Combining with (12), dividing by k , and letting $k \rightarrow \infty$ proves the assertion. The same proof can be made on each common model, so the comparison applies to all divisorial valuations needed later. No uniformity in t of the fixed error is required. $\square$

Proposition 6.8. *There are no sequences $t_j \downarrow 0$ and covering families of curves on Y_{t_j} , with smooth projective domains C_j , for which both*

$$g(C_j), \quad \frac{(K_{t_j} + t_j A_{t_j}) \cdot C_j}{\mu_{t_j}}$$

are bounded by a fixed constant. Degrees mean degrees on the domains, or equivalently intersection with their pushforward cycles.

Proof. Suppose such families exist. Fix once and for all a smooth resolution $p : W \rightarrow Y$ and a positive current T in the class p^*K_Y ; pseudoeffectivity supplies this current. Neither W nor T will depend on t . Let g_0 be a common upper bound for the genera, and choose an integer $m > 0$ such that mK_Y is Cartier.

We first compare the curves on these fixed models without changing their domains. Slice the parameter space so that the smooth total family has dimension n , the parameter space has dimension $n - 1$, and evaluation is generically finite. It has rational maps to Y , to W , and to the common models comparing Y and Y_t . Each target is proper, so each map extends at codimension-one points by the valuative criterion. Its indeterminacy locus has dimension at most $n - 2$, and its image cannot dominate the parameter space. Remove these finitely many images from that space, then resolve and compactify away from the remaining complete fibers. Denote the resulting maps by $\rho_t : U_t \rightarrow W$ and $q_t = p \circ \rho_t : U_t \rightarrow Y$. The smooth general parameter fiber is still the original C_t , with the same genus. The current we use on this varying space is always ρ_t^*T .

For a prime Z of U_t exceptional over Y , the restricted valuation of its function field is $r_Z \text{ord}_E$ for a divisorial valuation E over Y . The coefficient of Z in the ramification difference is

$$\text{coeff}_Z(K_{U_t} - q_t^*K_Y) = r_Z a(E; Y, 0) - 1. \quad (20)$$

This follows by factoring through a model extracting E and calculating the tame ramification of DVRs. It is positive and bounded away from zero: Y is terminal, so $a(E; Y, 0) > 1$, and the index m puts these log discrepancies in $m^{-1}\mathbb{Z}$. In particular $a(E; Y, 0) \geq 1 + 1/m$. The remaining ramification coefficients are nonnegative. On the general parameter fiber,

$$K_{U_t} \cdot C_t = 2g(C_t) - 2, \quad q_t^*K_Y \cdot C_t \geq 0.$$

The latter inequality follows from pseudoeffectivity and the covering property. Thus

$$0 \leq (K_{U_t} - q_t^*K_Y) \cdot C_t \leq 2g_0 - 2.$$

Each exceptional contact contributes $(r_Z a(E; Y, 0) - 1)(Z \cdot C_t)$ to this bounded sum. The first factor is at least $1/m$, and the second is a positive integer. Formula (20) therefore bounds the number of contacted exceptional primes, their log discrepancies, their ramification indices, and their contact degrees. In particular the integers $r_Z(Z \cdot C_t)$ are uniformly bounded. Also

$$q_t^*K_Y \cdot C_t \in m^{-1}\mathbb{Z} \cap [0, 2g_0 - 2],$$

so these degrees take only finitely many values.

The ramification estimate has bounded the possible contacts. We next compare their Lelong contributions with the canonical differences. The generic Lelong number of ρ_t^*T at Z is $r_Z \nu_E(T)$. Extract E , remove its generic divisorial part, and use the transverse-power-map comparison for the zero-generic-number remainder. Lemma 6.7 shows that these coefficients dominate the coefficients of the pulled-back canonical difference J_t . Its support is exceptional over Y as well as Y_t , since the map between these models is small. Subtracting all exceptional divisorial parts of ρ_t^*T leaves a positive current. Shrink the parameter space so that the relevant divisor contacts are constant, and restrict this residual current to almost every fiber. As in (18), its degree is the first difference below and is nonnegative. Consequently

$$\begin{aligned} 0 &\leq q_t^*K_Y \cdot C_t - \sum_{Z \text{ exceptional}} r_Z \nu_E(T) Z \cdot C_t \\ &\leq K_t \cdot C_t \leq (K_t + tA_t) \cdot C_t = O(\mu_t). \end{aligned} \quad (21)$$

All divisor contacts used here are nonnegative, since the curves cover the total space.

On the fixed smooth resolution $p : W \rightarrow Y$, one has

$$a(E; W, 0) = a(E; Y, 0) - \text{ord}_E(K_W - p^*K_Y) \leq a(E; Y, 0),$$

because the relative canonical divisor is effective. Thus Lemma 6.5 applies to the one fixed current T on W , with a fixed discrepancy bound. The sums in (21) form an ACC set: their Lelong numbers belong to this ACC set, and both the number of terms and their integral multiplicities $r_Z(Z \cdot C_t)$ are bounded. Pass to a fixed value of $q_t^*K_Y \cdot C_t$. Since $\mu_t \rightarrow 0$, the exact-zero argument of (18) gives a family for which the left gap is zero.

For that family, the restriction of the residual positive current to almost every parameter fiber is a positive measure of total mass zero, so it vanishes. On the dense open away from the removed exceptional divisors it agrees with ρ_t^*T . Apply Lemma 6.2 to the family evaluated by $\rho_t : U_t \rightarrow W$. Its curve tangents generate a rational fibration of positive relative dimension on the smooth projective variety W , and T annihilates the fiber tangents on a dense open. Transport this rational fibration through $p : W \rightarrow Y$ on their common isomorphism open. This contradicts Proposition 6.6. $\square$

6.5 Special termination for signed representatives

To turn a signed canonical representative into a big logarithmic adjoint, we will run a program whose negative rays meet the reduced boundary. The following special-termination statement uses only the lower-dimensional hypothesis; its proof keeps the programs on the boundary separate from the ambient one.

Proposition 6.9 (Special termination over a point). *Assume Assumption 2.1. Let (V, Δ) be a projective $\mathbb{Q}$ -factorial dlt n -fold with rational boundary and pseudo-effective adjoint. Choose an effective ample rational divisor A such that $(V, \Delta + A)$ is dlt and $K_V + \Delta + A$ is nef. The LMMP for $K_V + \Delta$ with scaling of A has special termination: if the program is infinite, its flipping loci are eventually disjoint from the round-down of the boundary.*

In particular, if at every stage the adjoint is rationally linearly equivalent to a signed divisor supported on that round-down, the program terminates at a log minimal model.

Proof. In an infinite ample-scaling program the scaling limit is zero: a positive limit is covered by the termination theorem for a dlt pair with an ample summand in the scaling divisor [5, Theorem 1.9(ii), equivalently Theorem 4.1(ii)]. Discard the finitely many divisorial contractions and write the flips as $V_i \dashrightarrow V_{i+1}/Z_i$, with scaling numbers $\lambda_i > 0$ tending to zero. Fix a component S_1 of $\lfloor \Delta_1 \rfloor$, write S_i for its normal strict transform, and let T_i be the normalization of its image in Z_i . Since the ambient contraction is small, $S_i \rightarrow T_i$ is projective birational. The standard discrepancy argument makes $S_i \dashrightarrow S_{i+1}$ an isomorphism in codimension one after finitely many steps [3, proof of Lemma 3.6].

To identify the lower-dimensional input, consider the relative program induced on this one component of the floor. Take a small projective $\mathbb{Q}$ -factorialization $p_i : S'_i \rightarrow S_i$, and put

$$D_i = K_{S'_i} + B_i = p_i^*((K_{V_i} + \Delta_i)|_{S_i}), \quad C_i = p_i^*(A_i|_{S_i}).$$

Adjunction gives a dlt pair (S'_i, B_i) , with $B_i, C_i \geq 0$, and $(S'_i, B_i + \lambda_i C_i)$ is lc. The scaling divisor has no floor component, so its restriction is effective. The number λ_i is rational, being the ratio of intersections of rational divisors on the contracted rational curve. The globally nef divisor $L_i = K_{V_i} + \Delta_i + \lambda_i A_i$ descends rationally linearly across $V_i \rightarrow Z_i$. Indeed, for sufficiently small rational $\delta > 0$, the pair $(V_i, (1 - \delta)\Delta_i)$ is klt and its negative adjoint is ample over Z_i . Relative

basepoint freeness applies to a Cartier multiple of L_i ; its numerical triviality and the connected fibers give descent. Restricting and pulling back yields

$$D_i + \lambda_i C_i \sim_{\mathbb{Q}} 0/T_i.$$

The descended divisor on T_i is nef: its pullback to S'_i is the pullback of the restriction of the globally nef L_i .

By [5, Theorem 1.1(3)], a D_i -LMMP over T_i with scaling of a fresh ample divisor terminates. The theorem applies to the effective rational lc boundary $B_i + \lambda_i C_i$, with $\lambda_i C_i$ rational Cartier and the driving pair $\mathbb{Q}$ -factorial dlt. Since the base map is birational, the endpoint is a log minimal model, not a Mori fiber space. Comparison with the relatively ample model S_{i+1} identifies this endpoint with a small $\mathbb{Q}$ -factorialization S'_{i+1} , as in [5, Remarks 2.9–2.10].

The next point is that these finite relative programs concatenate into an absolute program with scaling of the transforms of C_1 . Throughout the i -th piece, $D_i + \lambda_i C_i$ remains the pullback of a nef divisor on T_i , hence is globally nef. Here D_i and C_i also denote their transforms along this relative program. A contracted negative ray R satisfies

$$(D_i + \lambda_i C_i) \cdot R = 0, \quad D_i \cdot R < 0.$$

It follows that $C_i \cdot R > 0$. Every smaller coefficient of C_i leaves this ray negative, while $D_i + \lambda_i C_i$ is globally nef. Thus the absolute scaling threshold at that step is exactly λ_i . The relative cone is a face of the absolute cone, cut out by the pullback of an ample divisor on the projective base T_i ; its extremal rays are therefore absolute extremal rays. Thus the pieces concatenate as asserted in the cited remarks. Rescaling the initial scaling divisor by λ_1 , if necessary, makes its sum with the initial boundary lc and nef.

If the concatenation were infinite, its positive scaling numbers would tend to zero. Fix a late scaling value λ_j . On every earlier ray R contracted at threshold λ_i , the calculation above gives

$$(D_i + \lambda_j C_i) \cdot R = (\lambda_j - \lambda_i) C_i \cdot R \leq 0.$$

Thus all preceding steps are nonpositive for the adjoint with this fixed coefficient λ_j . On a common resolution of the initial and the j -th models, the pullback of $D_1 + \lambda_j C_1$ equals the pullback of the late nef divisor $D_j + \lambda_j C_j$ plus an effective comparison divisor. Push this equality down to the initial model. The pushforward of the nef pullback is pseudo-effective, as is the effective comparison term, so $D_1 + \lambda_j C_1$ is pseudo-effective. Closedness of the pseudo-effective cone and $\lambda_j \rightarrow 0$ show that D_1 is pseudo-effective. Assumption 2.1 therefore gives an absolute log minimal model for that initial pair. The concatenation terminates by [5, Theorem 1.9(iii)], since its zero limit is never attained. The floor-component argument of [3, Lemma 3.6] now gives special termination. The only good-model input in this argument concerns the absolute pair on the lower-dimensional floor component. In particular it does not require an effective representative in dimension n .

For the final assertion, a curve disjoint from the round-down has degree zero against any signed divisor supported there. It therefore cannot generate an adjoint-negative flipping ray. Special termination rules out all sufficiently late flips. There are only finitely many divisorial contractions, since each lowers the Picard number, and pseudo-effectivity excludes a Mori fiber space as the endpoint. Thus the endpoint is nef. $\square$

6.6 The signed alternative

The jet construction in Section 8 will produce signed rational representatives of K_t . The following result turns such a representative into a big logarithmic adjoint on a resolution.

Proposition 6.10. *On any model Y_t , let a signed rational divisor represent K_t up to $\mathbb{Q}$ -linear equivalence. On a log resolution $p : W \rightarrow Y_t$, let D be the reduced SNC divisor consisting of its strict support and all exceptional divisors. Then $K_W + D$ is big.*

Proof. Put $P = K_W + D$, and suppose it is not big. It cannot have Iitaka dimension at least one: by Corollary 6.4, $K_W + cP$ would be big, and $(1 + c)P = (K_W + cP) + D$ would then be big as well. Hence $\kappa(W, P) \leq 0$.

Run the dlt LMMP for P with ample scaling. The divisor P has a signed rational representative supported on the floor D , and its transforms retain that property. Every negative flip must meet the floor: on its complement the representing rational section is nowhere vanishing, so the adjoint has zero degree on every complete curve there. By Proposition 6.9, an infinite sequence would eventually have all flips disjoint from the floor, a contradiction. Divisorial contractions are finite in number. Thus the program terminates at a $\mathbb{Q}$ -factorial dlt log minimal model (Z, D_Z) .

The boundary is reduced, K_Z is pseudo-effective as the birational pushforward of K_W , and the nef adjoint has a signed rational representative supported on D_Z . Its restriction to D_Z is semiample by Proposition 2.3. Apply Theorem 3.1 with $L = K_Z + D_Z$ and $c = 1$, so that the required pseudo-effective class $L - cD_Z$ is K_Z . It gives $\kappa(Z, L) = \nu(Z, L) \geq 0$. The Iitaka dimension is at most zero, as proved above, so both dimensions are zero. Intersecting $K_Z + D_Z \equiv 0$ with an ample $(n - 1)$ -fold product, and using pseudoeffectivity of K_Z , forces $D_Z = 0$. The signed relation then gives $K_Z \sim_{\mathbb{Q}} 0$.

The signed representative of K_W is supported on D : pull back the given representative of K_t and add the relative canonical divisor. Since $D_Z = 0$, every component of this support is contracted over Z . On a common resolution $a : V \rightarrow W$, $b : V \rightarrow Z$, we therefore have $a^*K_W \sim_{\mathbb{Q}} E$ for a signed b -exceptional divisor E . The class a^*K_W is pseudo-effective. A positive current in this class has zero intersection with the $(n - 1)$ -st power of an ample pullback from Z , so is supported on the exceptional locus. The support theorem makes it effective divisorial. Independence of exceptional divisor classes, by negativity, identifies its coefficients with those of E ; hence $E \geq 0$. Pushing forward by a shows that K_W is $\mathbb{Q}$ -linearly effective, contradicting (10). $\square$

6.7 Birational families between fixed covers

The last obstruction concerns fixed finite covers of a birational model of the counterexample. If birational maps between two such covers form an algebraic family whose graphs dominate their product, then one fixed cover is birational to an abelian variety. The positive canonical current rules this out.

Proposition 6.11 (Fixed-cover obstruction). *Let Y be a projective $\mathbb{Q}$ -factorial terminal birational model of the counterexample X in Equation (10), and let $T_i \rightarrow Y$, $i = 1, 2$, be fixed finite surjective morphisms from normal integral projective varieties. There is no integral parameter variety A_0 and closed subvariety*

$$\Gamma \subset A_0 \times T_1 \times T_2$$

flat over A_0 , with each fiber the integral graph of a birational map $T_1 \dashrightarrow T_2$, for which the projection $\Gamma \rightarrow T_1 \times T_2$ is dominant.

Proof. Suppose such a family exists, and denote its birational maps by $\phi_a : T_1 \dashrightarrow T_2$. Choose one map ϕ_{a_0} in this family and use it to identify T_2 birationally with T_1 . The maps $g_a = \phi_{a_0}^{-1} \circ \phi_a$ are then birational selfmaps of T_1 . The evaluation $(a, u) \mapsto (u, g_a(u))$ is dominant onto $T_1 \times T_1$; in particular, the images of a general point under these selfmaps are dense.

We use the birational automorphism group to identify this fixed cover. The cover is non-uniruled, being generically finite over the non-uniruled Y . Hanamura’s non-uniruled birational-group theorem gives a smooth projective birational model for which the reduced birational group is a group scheme, locally of finite type, and its identity component is an abelian variety of regular automorphisms [35, Theorems 2.1–2.2]; see also [7, Proposition 3.7 and Theorems 3.8–3.9]. After conjugating the maps to that model, shrink the irreducible parameter variety anew so that their graph closures are flat with integral fibers and both graph projections remain birational on every fiber. This is a family in the flat graph functor of [7, Definition 3.4], which is represented by the birational scheme [7, Proposition 3.7]. It therefore defines a morphism to that scheme; because the parameter variety is reduced, this morphism factors through its reduction. Its connected image lies in a single component, hence in a translate of the identity component. Translate the family by one member so that it lies in that identity component; this preserves dominance of evaluation. Thus the identity component, an abelian variety, has a dense orbit. That orbit is the quotient by the stabilizer of a general point. A quotient of an abelian variety is again an abelian variety, so the fixed cover is birational to an abelian variety.

It remains to see why this abelian cover contradicts the original counterexample. Resolve the generically finite rational map from that abelian variety to X : on a smooth model U , ramification gives

$$K_U = f^*K_X + R, \quad R \geq 0.$$

The canonical class of U also has the effective representative E_A exceptional over the abelian variety: choosing a nowhere-zero top form on that variety gives $K_U \sim E_A$. This second choice of representative need not be the canonical divisor compatible with the displayed ramification formula. Let T be a positive current in K_X . The current $f^*T + [R]$ is positive and represents K_U . Let α be an ample class on the abelian variety. The intersection of K_U with the $(n-1)$ -st power of its pullback is zero, since $K_U \sim E_A$ and E_A is exceptional. The positive current $f^*T + [R]$ therefore has zero mass against that power. On the open where the birational map to the abelian variety is an isomorphism, the pulled-back ample form is strictly positive, so the current vanishes there. Its support is consequently exceptional over the abelian variety. The positive summand f^*T is supported there as well, and the support theorem makes it divisorial. Push forward this summand alone: $f_*(f^*T)$ is a positive divisorial current whose class is $\deg(f)c_1(K_X)$ by the projection formula. Rationality of the numerical class supplies a rational effective representative, and irregularity zero turns numerical effectivity into nonvanishing, as in Lemma 6.1. This contradicts $\kappa(X, K_X) = -\infty$. $\square$

7 Tools for moving base components

This section follows the base components of a complete kernel of jets as the marked point moves. Their multiplicities give degree and canonical-intersection bounds. A separate finiteness lemma treats the covers that arise when a component projects generically finitely onto a fixed base. These arguments use neither nonvanishing, abundance, minimal models, nor logarithmic subadditivity. Section 8 applies them to moving components produced by failure of the scalar or two-slot jet estimates.

All varieties in the moving-center arguments are over $\mathbb{C}$. A rational Cartier divisor is a rational divisor some positive integral multiple of which is Cartier. A requirement that kP be Cartier means that kP is an actual integral Cartier divisor. For a nef class on an integral subvariety, intersections are computed after pullback to its normalization and a resolution. For a linear subseries of a line bundle, its base ideal is the image of its evaluation map after tensoring by the inverse line bundle.

“Isolated at the generic point of V ” means that this ideal is primary for the maximal ideal of $\mathcal{O}_{Z,\eta_V}$. All powers of ideals below are ordinary powers.

7.1 Numerical subadjunction on the center

The first result converts a generic log canonical center into a canonical-intersection inequality. Its testing class is allowed to live on the center itself. The proof uses dlt adjunction and the klt-trivial canonical bundle formula; it does not require the auxiliary pair to be log canonical away from the generic center.

Lemma 7.1 (Numerical generic subadjunction). *Let Z be a projective $\mathbb{Q}$ -factorial klt variety over $\mathbb{C}$, let $\Theta \geq 0$ be a rational divisor, and let $V \subset Z$ be an integral positive-dimensional subvariety. Suppose that (Z, Θ) is lc at the generic point of V and that a divisor of log discrepancy zero has center V . Let $f = \dim V$, let P be a nef rational Cartier divisor on V , and let $\rho : V^* \rightarrow V$ be a smooth projective resolution, factoring through the normalization. Then*

$$K_{V^*} \cdot (\rho^* P)^{f-1} \leq (K_Z + \Theta)|_V \cdot P^{f-1}. \quad (22)$$

The right side is the intersection on V of the restricted rational Cartier class with P^{f-1} . No $\mathbb{Q}$ -Gorenstein hypothesis on the normalization of V is required.

Proof. Apply the dlt blowup theorem for an effective rational divisor [29, Theorem 2.10]. It gives a projective birational morphism $p : Q \rightarrow Z$ with Q $\mathbb{Q}$ -factorial and

$$p^*(K_Z + \Theta) = K_Q + B_Q + E_Q, \quad (23)$$

where (Q, B_Q) is dlt and $E_Q \geq 0$ is the surplus over the non-lc locus. In this construction the coefficients of the strict boundary are truncated at one and the relevant extracted divisors are reduced; the coefficients left over form E_Q . The surplus is absent over the open where (Z, Θ) is lc. In particular it is absent over a neighborhood of the generic point of V .

There are lc strata of (Q, B_Q) mapping onto V . Indeed over that lc neighborhood the construction is crepant, and the center of a log-discrepancy-zero divisor on the dlt model is an lc stratum. Choose a stratum S minimal among those mapping onto V . Iterated dlt adjunction makes S normal and gives an effective rational dlt different on S ; its lc centers are precisely the smaller ambient strata in S [42]. The divisor E_Q does not contain S . Since Q is $\mathbb{Q}$ -factorial, a Cartier multiple of E_Q restricts to an effective Cartier divisor on S . Adding that restriction to the different gives an effective rational divisor B_S such that

$$K_S + B_S \sim_{\mathbb{Q}} (p|_S)^*((K_Z + \Theta)|_V). \quad (24)$$

Over the generic point of V this pair is klt: the surplus is absent there, and a non-klt center of the different dominating V would be a smaller stratum, contradicting the choice of S .

Factor the map through the normalization V^ν and its Stein factorization:

$$S \xrightarrow{q} V' \xrightarrow{\eta} V^\nu \xrightarrow{\nu} V.$$

Here q has connected fibers and η is finite. Put $D_V = \nu^*((K_Z + \Theta)|_V)$ and $D' = \eta^*D_V$. We apply the canonical bundle formula on the connected-fiber base V' , then push the resulting intersection inequality through the finite map η . Its ramification will contribute an effective term. The final comparison is between the canonical cycle on V^ν and the canonical divisor on V^* .

We verify the klt-trivial-fibration hypotheses for $q : (S, B_S) \rightarrow V'$. The generic pair is sub-klt, and Equation (24) is the required rational-linear pullback expression. For the rank condition, take

a log resolution over the generic point of V' . Since the boundary on S is effective and the generic pair is klt, the round-up of its discrepancy divisor has coefficient zero along every strict boundary component and has nonnegative coefficients only on exceptional divisors. Over this dense klt open, its pushforward to the normal S is $\mathcal{O}_S$: a rational function allowed poles only over exceptional centers extends across codimension at least two. Connectedness in the Stein factorization therefore gives generic rank one for the required discrepancy round-up pushforward. This verifies the rank condition, rather than deducing it from connected fibers alone.

The canonical bundle formula, with Ambro's moduli positivity in the form recalled by Fujino–Gongyo, now gives

$$D' \sim_{\mathbb{Q}} K_{V'} + B_{V'} + M_{V'}, \quad (25)$$

where M is a b-nef moduli divisor [27, Definition 3.1 and Theorem 3.5]. The definition requires sub-klt over the generic point; it does not require the pair to be globally sub-lc. Thus the possible non-lc fibers caused by $E_Q|_S$ are permitted.

The discriminant $B_{V'}$ on this initial base is effective. To check this at a prime divisor $F \subset V'$, work over its generic point, where the base is regular. A component of its inverse image on S has multiplicity $m_F \geq 1$ and nonnegative boundary coefficient b_F . Its discrepancy bounds the generic lc threshold t_F from above by

$$t_F \leq \frac{1 - b_F}{m_F} \leq 1.$$

Thus the discriminant coefficient $1 - t_F$ is nonnegative. If the fiber is already non-lc, t_F can be negative; this only increases that coefficient.

Let $P' = \eta^* \nu^* P$. On a higher projective base $\mu : \widehat{V}' \rightarrow V'$ where the moduli divisor is nef, its trace satisfies $M_{V'} = \mu_* M_{\widehat{V}'}$. The projection formula gives

$$M_{V'} \cdot (P')^{f-1} = M_{\widehat{V}'} \cdot (\mu^* P')^{f-1} \geq 0.$$

Only this intersection is asserted to be nonnegative; the trace $M_{V'}$ need not itself be nef. Likewise $B_{V'} \cdot (P')^{f-1} \geq 0$ by effectivity and nefness. All intersections with Weil divisors here mean their cycle intersections with rational Cartier powers, so they are defined even when the separate canonical or discriminant divisor is not $\mathbb{Q}$ -Cartier.

Choose compatible canonical Weil divisors. Since η is finite and separable, the codimension-one ramification formula, followed by pushforward, is

$$\eta_* K_{V'} = (\deg \eta) K_{V^\nu} + \eta_* R_\eta, \quad R_\eta \geq 0.$$

This identity of Weil cycles uses the discrete valuation rings at generic prime divisors and does not require K_{V^ν} to be $\mathbb{Q}$ -Cartier. Intersect Equation (25) with $(P')^{f-1}$ and discard the nonnegative discriminant and moduli intersections. Pushforward through η and the nonnegative ramification intersection then give

$$(\deg \eta) D_V \cdot (\nu^* P)^{f-1} \geq (\deg \eta) K_{V^\nu} \cdot (\nu^* P)^{f-1}.$$

Finally the pushforward of K_{V^*} to V^ν is its compatible canonical cycle. All additional divisors on the resolution have image of codimension at least two, so their intersection with $f - 1$ pulled-back Cartier powers is zero. Dividing by $\deg \eta$ proves Equation (22). When $f = 1$ there are no birational exceptional divisors on the normal curve, and the same calculation applies directly. $\square$

7.2 Degree and canonical bounds for a base component

We next apply numerical subadjunction to a boundary built from an actual linear system. Its local multiplicity controls both the degree of the center and the coefficient of that boundary.

Lemma 7.2 (A base component of high multiplicity). *Let Z be a projective $\mathbb{Q}$ -factorial klt variety of dimension d , let P be a nef rational Cartier divisor, and let $k > 0$ be an integer for which kP is Cartier. Let $0 \neq \mathcal{H} \subset H^0(Z, \mathcal{O}_Z(kP))$ be a linear subspace with proper base locus and base ideal $\mathfrak{b}$. Suppose that V is an integral base-locus component of dimension $f < d$, isolated at its generic point η_V , and that $\eta_V \in Z_{\text{sm}}$. Write $a = d - f$ and $\mathfrak{m} = \mathfrak{m}_{\mathcal{O}_{Z, \eta_V}}$. If*

$$\mathfrak{b}\mathcal{O}_{Z, \eta_V} \subset \mathfrak{m}^h \quad (h \in \mathbb{Z}_{>0}),$$

then

$$P^f \cdot V \leq (k/h)^a P^d. \quad (26)$$

There are a rational number c with $0 < c \leq ak/h$ and an effective rational divisor $\Theta \sim_{\mathbb{Q}} cP$ such that (Z, Θ) is lc at η_V and has an lc place with center V . If $f > 0$, on a smooth resolution V^* one consequently has

$$K_{V^*} \cdot P^{f-1} \leq (K_Z + cP)|_V \cdot P^{f-1}. \quad (27)$$

The degree estimate includes $f = 0$; the canonical estimate is asserted only for positive-dimensional V .

Proof. The local ring $R = \mathcal{O}_{Z, \eta_V}$ is regular of dimension a . Isolation means that $\mathfrak{b}R$ is $\mathfrak{m}$ -primary. Choose a general sections of $\mathcal{H}$. Their local equations form a system of parameters in R , since an $\mathfrak{m}$ -primary ideal is contained in no smaller prime. Let $\mathfrak{q}$ be the parameter ideal they generate. It is contained in $\mathfrak{m}^h$. Regularity and the multiplicity comparison give

$$\text{length}(R/\mathfrak{q}) = e(\mathfrak{q}, R) \geq e(\mathfrak{m}^h, R) = h^a.$$

This is the generic intersection multiplicity along V .

The global intersection can be performed without assuming that the series has no other base components. Before each cut, discard the components of the current effective cycle that lie in the base locus. Before the final cut their dimensions are greater than f , so none can contain V : otherwise V would not be a base-locus component. Discarding them therefore leaves the calculation at η_V unchanged. A general next section meets the remaining components properly. Nefness of P shows that discarding effective components can only decrease their total P -degree. After a cuts the surviving effective cycle has P -degree at most $k^a P^d$, and contains V with multiplicity at least h^a . This proves Equation (26).

Let $\lambda = \text{lct}_{\eta_V}(\mathfrak{b})$ be the generic ideal threshold. It is positive and rational. The valuation from blowing up the smooth generic center has log discrepancy a and ideal order at least h , so

$$0 < \lambda \leq a/h.$$

Because $\mathfrak{b}R$ is primary, a divisor computing this threshold has center V . We realize the ideal threshold by a divisor rather than apply subadjunction to an ideal formally. Take a log resolution of Z and $\mathfrak{b}$, so that the pulled-back system has fixed divisor F and a basepoint-free moving part. Choose sufficiently many general members $D_1, \dots, D_N$ of $\mathcal{H}$ and put

$$\Theta = \frac{\lambda}{N}(D_1 + \dots + D_N) \sim_{\mathbb{Q}} cP, \quad c = k\lambda.$$

On the resolution the fixed contribution is λF . The moving contributions have coefficients $\lambda/N < 1$ and, by general choice, meet the fixed resolution divisor with simple normal crossings on the open in question. Thus (Z, Θ) is lc at the generic point of V and has the same lc place there as the ideal threshold. Global log canonicity is unnecessary. For $f > 0$, Lemma 7.1 gives Equation (27), and $c = k\lambda \leq ak/h$. $\square$

7.3 Differentiating the full moving jet kernel

Fixing a high order at a varying point produces a family of linear subspaces of one fixed section space. A component common to two successive base loci acquires high multiplicity, because parameter derivatives of the higher kernel still belong to the lower kernel. We first state the parameter-family conclusion. Local restriction to a fiber is treated separately below. The differentiation argument is related to the method of Ein–Küchle–Lazarsfeld [19, Proposition 2.3]; the ordinary-power and primary-ideal details needed here are included.

Lemma 7.3 (Moving multiplicity). *Let Z be an integral projective variety of dimension d , and let P be a nef, big, semiample rational Cartier divisor. Fix an integer k with kP Cartier and positive real numbers*

$$\tau_0 < \tau_1 < \cdots < \tau_d, \quad \tau_{i+1} - \tau_i = \delta > 0.$$

For a smooth point x , let $\mathcal{H}_i(x)$ be the entire subspace of sections of kP vanishing at x to order at least $\lceil k\tau_i \rceil$. Suppose that for very general x all these subspaces are nonzero and the base locus of $\mathcal{H}_0(x)$ has a positive-dimensional component through x . If $k\delta \geq 4$, then, after an algebraic parameter extension and restriction to nonempty opens, there are an irreducible parameter space T , a fixed adjacent pair of levels, and a family of integral subvarieties V_t common to the two base loci such that the incidence

$$\Gamma = \{(t, z) : z \in V_t\} \subset T \times Z$$

dominates Z . Each general V_t has dimension in $[1, d-1]$, is an isolated component of the higher base locus at its generic point, and its entire higher-series base ideal is contained there in

$$\mathcal{I}_{V_t}^h, \quad h = \lceil k\delta/2 \rceil, \quad k/h \leq 2/\delta. \quad (28)$$

These are ordinary local ideal powers in the smooth ambient open.

The incidence base ideal lies in $\mathcal{I}_\Gamma^h$ on a dense open of Γ . If, in addition, Z is $\mathbb{Q}$ -factorial and klt, then for a general member, with $f = \dim V_t$, one has

$$\begin{aligned} P^f \cdot V_t &\leq (2/\delta)^{d-f} P^d, \\ K_{V_t}^* P^{f-1} &\leq (K_Z + cP)|_{V_t} P^{f-1}, \quad 0 < c \leq 2(d-f)/\delta. \end{aligned} \quad (29)$$

In particular the weaker bound $c \leq 2d/\delta$ also holds. Positivity of $P^f \cdot V_t$ follows if V_t meets the open where the big semiample contraction is an isomorphism. No such positivity is asserted for every exceptional subvariety.

Proof. On the smooth marked-point open, evaluation into the finite jet bundle is a morphism from the constant bundle $H^0(Z, kP) \otimes \mathcal{O}$ to a locally free sheaf. Remove the rank-jump loci for the finitely many levels. Its kernels are then vector subbundles whose fibers are exactly the full spaces $\mathcal{H}_i(x)$. Their base loci are nested in increasing order as i increases. Starting with the stipulated component through the mark, follow a component containing it at each later level. Every such component has dimension between 1 and $d-1$: the kernel is nonzero, so its base locus is proper.

Among the $d + 1$ nested components two successive ones have the same dimension and hence are equal.

This tracking can be made algebraic for the fixed k . Follow the finitely many components of the geometric generic base loci, their inclusions, and their incidences with the mark. They are defined after a finite extension of the generic parameter field. Spread over a corresponding irreducible parameter variety and shrink for the required flatness, geometric integrality, and constant-rank conditions. There are only finitely many choices of adjacent step; one of them gives a family dominant over the marked-point open. It gives the asserted T and Γ . The evaluation $\Gamma \rightarrow Z$ is dominant because the moving mark belongs to each V_t and its map to Z is dominant.

We now prove ordinary high-order vanishing. Work on a smooth parameter open, and denote the higher-kernel bundle there by $\mathcal{K}$. Evaluation on $T \times Z$ gives a map from the pullback of $\mathcal{K}$ to the pullback of $\mathcal{O}_Z(kP)$. Let $\mathfrak{b}_T$ be its base ideal. After trivializing the target line bundle, a local frame of $\mathcal{K}$ generates $\mathfrak{b}_T$. We show that every frame section lies in $\mathcal{I}_\Gamma^h$. Each frame element has the form

$$s(t, z) = \sum_{\alpha} c_{\alpha}(t) s_{\alpha}(z),$$

where the s_{α} are a fixed basis of the global space $H^0(Z, kP)$. Parameter differentiation with z fixed only differentiates the coefficients. Every derivative, at each fixed parameter, is consequently a global section of the same line bundle kP .

Write $x(t)$ for the moving mark. In local coordinates $z - x(t)$, the section starts in order $\lceil k\tau_{i+1} \rceil$. Each parameter derivative lowers this order by at most one. A derivative of order r therefore belongs to the lower kernel if

$$r \leq \lceil k\tau_{i+1} \rceil - \lceil k\tau_i \rceil.$$

The right side is at least $k\delta - 1$. Since $k\delta \geq 4$, every derivative of order less than $h = \lceil k\delta/2 \rceil$ is allowed. It thus vanishes along Γ , which lies in the lower base locus. A change of higher-kernel frame contributes only parameter coefficients and derivatives of orders no larger than r , so the conclusion holds for any local frame.

Here a first-order normal-space assertion must be upgraded to all orders. At a general smooth point of Γ , its projection to Z is submersive. Thus

$$T_{(t,z)}(T \times Z) = T_{(t,z)}\Gamma + (T_t T \times \{0\}).$$

In particular, parameter directions span the normal space to Γ . Choose parameter coordinates $t = (t', t'')$, where t' has length $a = \text{codim}_{T \times Z} \Gamma$, such that the normal equations have an invertible Jacobian minor in the t' directions. The analytic implicit-function theorem writes

$$\Gamma : \quad t' = F(t'', z).$$

Set $u = t' - F(t'', z)$. With t'' and z held fixed, differentiation in t' is exactly differentiation in u , to every order. Use a frame of kP depending only on z . The vanishing on Γ of all parameter derivatives of order below h says that every normal Taylor coefficient of degree below h vanishes. Therefore $s \in \mathcal{I}_\Gamma^h$ in the ordinary analytic local ring. Equivalently, in arbitrary relative normal coordinates a product of r normal vector fields expands into parameter derivatives of orders at most r , so no unaccounted higher-coordinate terms arise.

The sections and ideals are algebraic. Faithful flatness of the analytic local ring over the algebraic local ring gives the same ideal membership algebraically. Applying this to a finite frame and clearing the finitely many denominators gives containment on a dense algebraic open of the incidence. Generic smoothness of $\Gamma \rightarrow T$ identifies its scheme fiber there with the reduced general component

V_t . Restriction therefore gives $\mathcal{I}_t^h \mathcal{O}_{\{t\} \times Z} = \mathcal{I}_{V_t}^h$ locally. The specialized frame is a basis of the entire higher kernel, so the containment concerns its whole base ideal.

If Z is $\mathbb{Q}$ -factorial and klt, apply Lemma 7.2 to the higher kernel. Its generic base ideal is primary because V_t is a base-locus component, and its generic point is smooth by incidence dominance. The bounds follow from $k/h \leq 2/\delta$. On the isomorphism open of the big semiample contraction the image of V_t has dimension f , so the pullback of an ample class has positive top intersection on V_t . This proves the stated positivity qualification. $\square$

7.4 Restriction to a suitable fiber

The restriction step is local and must be separated from the global positivity and singularity assumptions of subadjunction.

Lemma 7.4 (Restriction of an isolated base ideal). *In the setting of Lemma 7.3, let $g : Z \rightarrow B$ be a morphism. There is a dense open of the incidence on which the following holds. Choose (t, z) in that open, let $b = g(z)$, and suppose that V_t is smooth over the smooth open of its actual image at z . Let F be the reduced component through z of its scheme fiber. Locally on Z_b at z , restriction of the higher kernel's base ideal has support exactly F and is contained in $\mathcal{I}_F^h$. If F is proper in an integral ambient fiber component containing it, the restricted series is nonzero there and F is an isolated base component at its generic point. Application of Lemma 7.2 on that ambient component requires all its hypotheses, including projectivity, $\mathbb{Q}$ -factoriality, klt singularities, and smoothness at the generic point of F , separately; they are not conclusions of this restriction lemma.*

Proof. Choose the incidence point after removing the other base components, the locus where the ideal containment is unavailable, and the failures of generic smoothness for the incidence over its parameter space and for the component over its actual image. These are proper closed subsets after shrinking. A prescribed very general condition on the evaluation can also be imposed because the incidence dominates Z . The component is then chosen through this point, rather than by selecting a possibly special fiber in advance.

Locally the original base support is exactly V_t . Smoothness over the actual image makes its scheme fiber reduced at the chosen point. Thus in the ambient fiber

$$\mathcal{I}_{V_t} \mathcal{O}_{Z_b} = \mathcal{I}_F, \quad \mathcal{I}_{V_t}^h \mathcal{O}_{Z_b} = \mathcal{I}_F^h$$

near that point. Restricting the actual sections extends their base ideal, which proves both asserted local properties. No other base component contains the chosen point, so none can contain the component F through it. If this component is proper in the chosen integral ambient component, the base support is proper there, and the restricted series cannot vanish identically. At its generic point the resulting ideal is primary. This is a statement about this reduced general fiber and does not assert transversality or nonzero restriction for arbitrary special fibers. $\square$

7.5 A fixed branch complement in a finite-type family

For later applications, the local moving-center estimates are followed by a global finiteness step. A degree bound alone does not make a list of branched covers finite. The following statement records exactly the additional family information that is needed. In the correspondence case of Proposition 8.3, it will turn bounded degree and branch divisors that do not sweep the base into finitely many normal covers.

Lemma 7.5 (Finite covers after excluding sweeping branch divisors). *Let Y be a normal projective complex variety. Let $\mathcal{V} \rightarrow T$ be a smooth projective family with integral fibers, over an integral parameter variety, and let $g : \mathcal{V} \rightarrow Y$ be a morphism whose general fiber maps $g_t : V_t \rightarrow Y$ are dominant and generically finite of degree at most a fixed integer e . After a finite parameter extension and shrinking, spread the geometric generic components of the zero divisor of the relative Jacobian over Y_{sm} to divisors $\mathcal{R}_1, \dots, \mathcal{R}_N$ in $\mathcal{V}$. Retain the notation $\mathcal{V}, T, g$ after this base change. Let I be the set of indices j for which the image closure $\overline{g_t((\mathcal{R}_j)_t)}$ is a divisor in Y for general t . Assume*

$$\overline{g(\mathcal{R}_j)} \neq Y \quad (j \in I).$$

Then, after shrinking again, one smooth nonempty open $Y^\circ \subset Y$ makes every general finite normal Stein cover of g_t finite étale over Y° . There are only finitely many such covers of Y , up to isomorphism over Y . The same conclusion holds simultaneously for finitely many projections. The open and finite lists may depend on all fixed parameters of the family.

Proof. The relative determinant is nonzero on a dense open because the general maps are generically finite in characteristic zero. Its zero divisor has finitely many irreducible components. Shrink away vertical components and make a finite parameter extension to define the geometric generic components individually. Their closures give the divisors $\mathcal{R}_j$ in the statement. Shrink further to keep their fiber-image dimensions constant. These components account for every branch prime of the general finite Stein cover: a ramification prime there has a strict transform on V_t , because a proper birational morphism to a normal variety is an isomorphism at each generic divisorial point. A divisor exceptional over the Stein cover has image of codimension at least two on that cover and hence on Y , so it cannot hide a branch prime. Components confined to special parameters disappear upon shrinking.

Put

$$B = Y_{\text{sing}} \cup \bigcup_{j \in I} \overline{g(\mathcal{R}_j)}.$$

Each closed set in this finite union is proper by hypothesis, so $B \neq Y$. The finite normal covers are unramified in codimension one over the smooth open $Y^\circ = Y \setminus B$. Purity of the branch locus makes them finite étale there [32, Exposé X, Theorem 3.1]. The topological fundamental group of Y° is finitely generated. Indeed, triangulate the compact semialgebraic space Y compatibly with its closed subset B [15, Theorem 1.10 in the author's notes]. After barycentric subdivision the subcomplex for B is full. On its complement, normalize the sum of the barycentric coordinates at vertices outside that subcomplex; decreasing the other coordinates to zero gives a deformation retraction onto a finite subcomplex. Its fundamental group is finitely generated. A finitely generated group has only finitely many permutation representations in each symmetric group of degree at most e . Riemann existence [32, Exposé XII, Theorem 5.1] therefore gives finitely many finite étale covers of these degrees. A finite normal cover of Y is the normalization in the function field of its restriction to Y° , and is determined by that restriction. Taking the union of the proper image closures for finitely many projections proves the simultaneous assertion. $\square$

8 Very-general jet estimates

This section derives two contrasting consequences of the assumed counterexample: an upper bound for the vanishing of scalar sections, and strong jet separation on a projective bundle over the product. Both arguments use the exclusion of covering curves of bounded genus and normalized degree.

We retain the counterexample in Equation (10) and the induction hypothesis of Assumption 2.1. Thus X is smooth projective, K_X is pseudo-effective, and $\kappa(X, K_X) = -\infty$, whereas good minimal

models exist in smaller dimensions. We use the geometric exclusions of Section 6. Dimension one is impossible, since a smooth curve with pseudo-effective canonical divisor has genus at least one. Hence $n = \dim X \geq 2$.

Throughout the section, fix the late terminal $\mathbb{Q}$ -factorial model Y from Proposition 6.8. Write $K = K_Y$ and $A = A_Y$. The subsequent scaling models Y_t are small modifications of Y ; the transforms A_t are effective up to $\mathbb{Q}$ -linear equivalence, and $K_t + tA_t$ is nef, big, and semiample. Put

$$\mu_t = \text{vol}(X, K_X + tA)^{1/n}.$$

Here, in the volume on X , A denotes the original ample divisor. We have $\mu_t \rightarrow 0$. Choose an integer $m > 0$ with mK Cartier.

8.1 Normalization and two local tools

We first normalize the small adjoint divisors so that their volumes stay bounded away from zero. Fix a small rational number $\epsilon > 0$. An integer $q > 0$ will be chosen after ϵ and before t . As rational $t \downarrow 0$, choose positive integers $r = r(t)$ with

$$r\mu_t \rightarrow \epsilon, \quad L = r(K + tA), \quad L_b = L + bmK \quad (0 \leq b \leq q). \quad (30)$$

Only rational weights b are used for model constructions; volume functions in integrals are extended continuously to real weights. The positive part of L_b is the nef semiample class

$$N_b = (r + bm)(K_s + sA_s) \quad \text{on } Y_s, \quad s = \frac{rt}{r + bm}.$$

When emphasizing the weight, denote this axis model by $Y(b) = Y_s$ and write A_b for its transform of A . All references to positive parts below mean these classes on their scaling models, or their pullbacks to common resolutions. Monotonicity of volume in pseudo-effective order gives

$$\text{vol}(L) \leq \text{vol}(L_b) \leq \left(1 + \frac{qm}{r}\right)^n \text{vol}(L). \quad (31)$$

Indeed $L_b - L = bmK$ is pseudo-effective, and $(1 + bm/r)L - L_b = bmtA$ is effective up to equivalence. Consequently N_b^n is bounded above and bounded away from zero, uniformly for $b \in [0, q]$, and tends uniformly to ϵ^n . More explicitly, for every fixed q , once t is sufficiently small depending on q ,

$$\frac{\epsilon^n}{2} \leq N_b^n \leq 2\epsilon^n \quad (0 \leq b \leq q). \quad (32)$$

The constants in this display are independent of q ; only the threshold for t depends on the fixed choice of q . All unspecified constants in this section may depend on n, ϵ, q, Y, A, m , but not on sufficiently small t or on b .

This normalization gives a convenient form of the curve exclusion. A family of curves with uniformly bounded geometric genus and N_b -degree cannot cover the corresponding Y_s for arbitrarily small t . In fact $s/t \rightarrow 1$ uniformly in b , and

$$(r + bm)\mu_s = \text{vol}(L_b)^{1/n}$$

is bounded above and away from zero. Thus such curves would have uniformly bounded $(K_s + sA_s)$ -degree divided by μ_s , contrary to Proposition 6.8. We call this consequence the *normalized curve exclusion*.

We will apply the local estimates of Section 7 on several different models. In each application, let Z denote the ambient model of dimension d , let P be its nef, big, semiample class, and let V be a moving base component of dimension f . Lemma 7.3 applies to the *entire* kernels of jets in one Cartier degree kP , at levels separated by a gap δ . It gives vanishing along V in the ordinary ideal power of order

$$h = \lceil k\delta/2 \rceil, \quad k\delta \geq 4,$$

and, when Z is $\mathbb{Q}$ -factorial and klt, the estimates

$$P^f \cdot V \leq (2/\delta)^{d-f} P^d, \quad K_{V^*} P^{f-1} \leq (K_Z + cP)|_V P^{f-1}, \quad 0 < c \leq 2(d-f)/\delta \leq 2d/\delta. \quad (33)$$

Here V^* is a smooth resolution of V . The canonical estimate uses Lemma 7.1, which allows the nef class on the center itself and requires log canonicity only near its generic point. The last, weaker coefficient is sufficient here. At every application we choose the incidence point in the isomorphism open of the big semiample contraction. Dominance of the incidence permits this choice; it ensures that $P|_V$ is big and hence $P^f \cdot V > 0$. Exceptional components not meeting that open are not assigned this positivity. The separate fiber-restriction Lemma 7.4 will be used only at a general incidence point where the scheme fiber is reduced and the selected base component is isolated.

To apply normalized curve exclusion, we need curves with bounded genus as well as bounded degree. The preceding intersection estimates provide them on components of general type. Suppose a moving f -fold V is of general type, $P|_V$ is nef and big, and

$$0 < P^f \cdot V \leq C_0, \quad K_{V^*} P^{f-1} \leq C_1 P^f \cdot V. \quad (34)$$

Effective birationality gives a uniform pluricanonical degree whose moving part, on a further resolution, is a big basepoint-free Cartier divisor H defining a birational morphism [33, Theorem 4.0.1(3)]. Thus $HP^{f-1} \leq C_2 P^f \cdot V$. Mixed Hodge index gives

$$H^j P^{f-j} \leq C_2^j P^f \cdot V \quad (0 \leq j \leq f);$$

no lower bound on $P^f \cdot V$ is needed here. For $f > 1$, cut by $f-1$ general members of $|H|$. Their P -degree is $H^{f-1}P \leq C_2^{f-1}C_0$, and their images under the birational morphism defined by H have degree $H^f \leq C_2^f C_0$. A general plane projection is birational on such an image curve; the arithmetic genus of the plane image therefore bounds its geometric genus in terms of this degree. For $f = 1$, the canonical-degree bound already bounds the genus. We will also use this construction for a log smooth pair of log general type with reduced boundary, using coefficients in the fixed set $\{1\}$ in effective birationality.

If such components sweep an ambient space mapping to Y , and their curves project nonconstantly with N_b -degree bounded by their P -degree, we obtain forbidden covering curves on Y_s . When the component has positive-dimensional image in the base, general complete intersections from the big birational system project nonconstantly. The components sweep, and their chosen incidence points can lie outside any prescribed countable union of proper closed subsets. Hilbert schemes and spaces of maps have only countably many components, so these curves can be parameterized in covering algebraic families. Thus, whenever Equation (34) is established below, it remains to check general type and the stated covering and projection conditions to invoke normalized curve exclusion.

We first apply the two local tools to scalar sections. Excessive vanishing produces a proper moving component, whose general type turns the numerical estimates into the forbidden curves.

Proposition 8.1 (Scalar jet bound). *For sufficiently small t , let P be the positive part of $2r(K+2tA)$, and put $\rho = (P^n)^{1/n}$. At a very general point, every nonzero section of every Cartier multiple kP*

has order at most $4k\rho$. The same statement holds on a common resolution after adding an effective exceptional divisor that does not change the section spaces. Moreover

$$2r\mu_t \leq \rho \leq 4r\mu_t,$$

so in particular $\rho \leq 8\epsilon$ for small t .

Proof. First, pseudo-effectivity of K gives the volume bounds:

$$\mathrm{vol}(K + tA) \leq \mathrm{vol}(K + 2tA) \leq 2^n \mathrm{vol}(K + tA).$$

Suppose the asserted order bound fails for arbitrarily small t . There are countably many Cartier degrees, and nonzero jet kernels in each degree are detected by algebraic rank loci. Thus failure at a very general point gives a degree in which a violating section exists on a common constant-rank marked-point open. Taking powers makes this degree k arbitrarily large and divisible. Choose $n + 1$ levels strictly between ρ and 4ρ , with equal gaps comparable to ρ . The full jet kernels at these levels are nonzero. Write τ_0 for the lowest level. If the marked point were isolated in its base locus, local Bezout for n general kernel sections would contribute at least $(k\tau_0)^n > k^n P^n$, exceeding the total intersection. Lemma 7.3 therefore supplies a moving proper positive-dimensional component V .

We can now apply the curve construction. The component is of general type by Proposition 6.3. Here let Z be the scaling model for $K + 2tA$, so that $P = 2r(K_Z + 2tA_Z)$. The effective transform A_Z does not contain a general such component, and $K_Z|_V \leq (2r)^{-1}P|_V$ in effective order. Equations (33) consequently imply Equation (34) with constants uniform in t ; the gap is bounded above and away from zero because ρ is comparable to the fixed ϵ . The resulting bounded-genus, bounded-normalized-degree covering curves contradict Proposition 6.8. The argument applies to every Cartier multiple, not only the divisible degrees used for tracking: a section violating the bound in any Cartier degree could be raised to a power in an arbitrarily divisible degree. Its order and its degree increase by the same factor. Finally, exceptional additions preserve the section spaces and the orders at general points. This proves the assertion on resolutions. $\square$

8.2 The two-slot bundle

We next seek large jet separation by placing two copies of the normalized adjoint divisor on a projective bundle. The weight decomposition of its sections will let us compare moving components with either copy of the base. On $Y \times Y$, let

$$Z_0 = \mathbb{P}(mK_1 \oplus mK_2), \quad \xi = \mathcal{O}_{Z_0}(1), \quad M = L_1 + L_2 + q\xi. \quad (35)$$

We use the convention that sections of $k\xi$ are symmetric powers of the indicated direct sum. Fixing one base slot gives the one-slot bundle $\mathbb{P}(\mathcal{O}_Y \oplus mK)$, with class $L + q\xi$, up to a constant line from the fixed slot. We call it a *slice*. The torus open in either bundle is the complement of its two axes.

Lemma 8.2 (Models, volume, and weights). *Both the total class M and the slice class have big semiample positive parts P on $\mathbb{Q}$ -factorial klt models Z , with*

$$K_Z + \Delta \sim_{\mathbb{Q}} c_t P, \quad \Delta \geq 0,$$

where c_t is bounded independently of small t . Their volumes satisfy, respectively,

$$\mathrm{vol}(M) = \frac{(2n+1)!}{(n!)^2} \int_0^q \mathrm{vol}(L_b) \mathrm{vol}(L_{q-b}) db, \quad (36)$$

$$\mathrm{vol}(L + q\xi) = (n+1) \int_0^q \mathrm{vol}(L_b) db. \quad (37)$$

In particular both volumes are bounded above and below by positive constants times q . For each rational weight, on common resolutions,

$$P \sim_{\mathbb{Q}} N_{b,1} + N_{q-b,2} + E_b, \quad E_b \geq 0, \quad (38)$$

with the analogous one-term formula on slices. At a generic point of a base divisor of a slice, and on a general torus fiber, the full system has no fixed subtraction.

Proof. We construct the semiample models as models of big klt adjoints. The weight decomposition then gives both the volume formulas and the comparison with the axis models.

The product $Y \times Y$ is $\mathbb{Q}$ -factorial. To see this, take a smooth resolution $\tilde{Y} \rightarrow Y$. Its irregularity is zero by Lemma 6.1, so every line bundle on $\tilde{Y} \times \tilde{Y}$ is a tensor product of line bundles from the two factors. The strict transform of a Weil divisor on $Y \times Y$ is therefore linearly equivalent to a sum of two factorwise pullbacks. Push this relation down to $Y \times Y$. The resulting factor divisors are $\mathbb{Q}$ -Cartier because Y is $\mathbb{Q}$ -factorial, and hence so is the original divisor. Products of canonical singularities are canonical, and the projective bundles are klt. The two disjoint Cartier axes form a plt boundary D , and

$$K_{Z_0} + D = K_{\text{sum}}, \quad D \sim 2\xi - mK_{\text{sum}}.$$

The notation K_{sum} means $K_1 + K_2$, or K on a slice.

For $0 < \sigma \leq 1$, put

$$\gamma = 2\sigma + \frac{q(1 + \sigma m)}{r}, \quad \Xi_t \sim_{\mathbb{Q}} \xi + \frac{(1 + \sigma m)t}{\gamma} A_{\text{sum}}.$$

Both endpoint weight systems are big, so choose an effective rational representative of Ξ_t containing neither axis, and denote the representative by Ξ_t as well. To make the adjoint construction uniform, we first obtain a threshold bound independent of σ . Restrict to $0 < t \leq \sigma/(1 + m)$. Since $\gamma \geq 2\sigma$, the coefficient

$$a = \frac{(1 + \sigma m)t}{\gamma}$$

lies in $(0, 1]$. Thus the numerical classes $\xi + aA_{\text{sum}}$, and their restrictions to either fixed axis, range over bounded segments independent of σ, q, r, t . On one fixed smooth resolution the effective pullbacks of these chosen divisors have uniformly bounded degree against a fixed very ample class. That degree bounds their multiplicity at every point, including the coefficients of exceptional components. Rational denominators do not enter this estimate. The smooth lc threshold is at least the reciprocal of maximal multiplicity.

To transfer this estimate to the fixed klt space, write the crepant boundary on its resolution as an SNC divisor with coefficients below one. The minimum of one and the positive numbers one minus its positive coefficients bounds its log discrepancies below by a fixed positive multiple of smooth log discrepancies. Therefore the preceding smooth bound gives a common klt threshold $\eta > 0$ for Ξ_t on the original space. The same reasoning on each axis gives, after decreasing η , a klt threshold bound for the restricted divisor there. The first bound will be used away from the axes; the restricted bounds allow inversion of adjunction along the coefficient-one axes.

Now choose rational $0 < \sigma < \min\{1, \eta/4\}$, independently of q, t , and then take t small enough that $t \leq \sigma/(1 + m)$ and $q(1 + \sigma m)/r < \sigma$. It follows that $\gamma < 3\sigma < \eta$. Inversion of adjunction with the disjoint coefficient-one axes gives plt near them; away from them the threshold bound gives klt. Lowering their coefficients to $1 - \sigma$ consequently shows that

$$(Z_0, (1 - \sigma)D + \gamma\Xi_t)$$

is klt. This choice respects the order: first ϵ , then q , then sufficiently small t with r as in Equation (30). A direct calculation gives its adjoint class

$$K_{Z_0} + (1 - \sigma)D + \gamma\Xi_t \sim_{\mathbb{Q}} \frac{1 + \sigma m}{r} M.$$

The big klt adjoint has a good minimal model by [6]. Pushing the boundary to this model gives the stated Δ and $c_t = (1 + \sigma m)/r$.

We turn to the volume formulas. The weight- l summand in degree k has base classes $kL_{l/k}$ and $kL_{q-l/k}$; on a slice there is one such factor. The section decomposition and asymptotic Riemann–Roch give Equations (36)–(37) as Riemann sums. One can justify passage to the integral without assuming uniform asymptotic Riemann–Roch: partition $[0, q]$ into rational bins, compare weight classes in each bin by adding and subtracting the bin width times a fixed very ample divisor dominating both mK and $-mK$, take divisible-degree limits, and then shrink the bins. Volume continuity gives the formulas. Equation (31) supplies their uniform bounds and, in particular, proves bigness used above. In the range of Equation (32), the bounds needed when choosing q are explicitly

$$\text{vol}(M) \geq \frac{(2n+1)!}{4(n!)^2} q\epsilon^{2n}, \quad \text{vol}(L + q\xi) \leq 2(n+1)q\epsilon^n.$$

Their coefficients are independent of q ; the smallness threshold for t is allowed to depend on q .

It remains to establish the effective weight comparisons. On a common resolution and in compatible divisible degrees, let F_{full} and F_b be the fixed divisors of the full system and its weight- b subsystem, normalized by the degree. Since the second is a subsystem, $F_b \geq F_{\text{full}}$. Its fixed divisor consists of the toric monomial and the base-model fixed differences, and its moving class is the sum of the indicated pullbacks of N_b . Hence the full moving class is that sum plus $E_b = F_b - F_{\text{full}} \geq 0$. This proves Equation (38). The same comparison in individual sufficiently divisible degrees shows that every section at that weight, after the full fixed subtraction, contains the corresponding multiple of E_b .

At a generic base-divisor point the small base-model maps are isomorphisms. The two endpoint systems are free there in divisible degree, so together they have no common zero on the projective-line fiber. The full system therefore has no fixed subtraction there. The same holds on a general fiber. Finally, when restricting Equation (38) to a component sweeping the torus open, choose a general component not contained in $\text{Supp } E_b$. There are only countably many rational weights, so these effective restrictions can be required simultaneously. $\square$

The bundle models are now available, and their volumes can be made large by choosing q . We use this growth to prove a lower bound for local positivity. Recall that the Seshadri constant of a nef $\mathbb{Q}$ -Cartier class P on a projective variety Z at a smooth point $x \in Z$ is

$$\varepsilon(P; x) = \inf_{\substack{C \subset Z \\ \text{integral curve} \\ x \in C}} \frac{P \cdot C}{\text{mult}_x C}.$$

Proposition 8.3 (Large two-slot Seshadri constant). *For each fixed sufficiently small $\epsilon > 0$, there is an integer $q > 0$ such that, for all sufficiently small rational $t > 0$ and r as in Equation (30), the positive part P of Equation (35) has*

$$\varepsilon(P; x) > 2n + 2$$

at a very general smooth point of its torus open. In particular, for some sufficiently divisible integer $k > 0$, the complete system $|kP|$ separates jets of order strictly greater than $(2n + 2)k$ at such a point. The section spaces and these general-point jets agree on the original bundle and on common birational resolutions.

Proof. The tracking lemma reduces failure of the assertion to a moving base component. We will first analyze its fibers over the two base factors, and then treat components that are generically finite over both.

Write $d = 2n + 1$. Choose a gap $\delta > 0$ large enough that

$$2(n + 1)\epsilon^n(2/\delta)^n < 1.$$

This inequality will exclude a component that is a whole projective-line fiber of a slice. Now choose q sufficiently large that

$$\left(\frac{(2n + 1)!}{4(n!)^2} q \epsilon^{2n}\right)^{1/d} > 2n + 3 + (d + 1)\delta.$$

For all sufficiently small t , the $d + 1$ levels $\tau_i = 2n + 3 + (i + 1)\delta$, $0 \leq i \leq d$, then lie below the volume root of the total positive part. Both choices precede the choice of t . Assume that the Seshadri assertion fails for arbitrarily small t . Because the highest level is below the volume root, jet counts show that the full kernels at all these levels are nonzero in sufficiently large divisible degree. A curve through the marked point with degree-to-multiplicity ratio below the lowest level is contained in that system's base locus. Such a curve exists from the assumed Seshadri bound and the choice $\tau_0 > 2n + 3$, whether or not the infimum defining the constant is attained. Thus the lowest kernel has a positive-dimensional base component, and Lemma 7.3 supplies a moving proper component V and both estimates (33).

The numerical estimates already rule out components of general type. For all moving components under consideration, the effective boundary Δ of Lemma 8.2 does not contain the component. Thus $K_Z|_V \leq c_t P|_V$ in effective order. Whenever V is of general type, the curve construction following Equation (34) and weight domination give a contradiction. To treat the remaining components, we begin with their fibers over a base factor.

Restriction to a slice. Resolve the rational projection from the total model to the first copy of Y , and work on the open where this resolution and the original bundle agree. Fixing the first coordinate there gives the opposite slice. Write Z_{sl} for its model from Lemma 8.2 and P_{sl} for its positive part; their dimension and volume are $n + 1$ and Equation (37), respectively. We use these slice objects until returning to the total model below. Choose one sufficiently divisible Cartier degree for the total model, the slice model, and their comparison on a common resolution. Choose a general incidence point on the common isomorphic torus open, away from the other base components and where the ordinary-power containment of Lemma 7.3 holds. Generic smoothness allows us also to require that V be smooth over its actual image in the first slot there. Let F be the reduced component of the scheme fiber through that point, and suppose $0 < \dim F < n + 1$. Lemma 7.4 now shows that, near that point, the restricted higher-series base ideal has support exactly F and lies in the ordinary power $\mathcal{I}_F^h$, with $h = \lceil k\delta/2 \rceil$. Because F is proper in the slice, the restricted series is nonzero and its generic base ideal is primary.

The fixed differences of the full and slice models are units on a further common open. After removing these fixed divisors and trivializing the fixed-slot lines, all restricted higher-kernel sections form a subspace of $H^0(Z_{\text{sl}}, kP_{\text{sl}})$, with the same local base ideal. This subseries need not be the full jet kernel on the slice: the numerical estimates come from Lemma 7.2, applied to the actual restricted subseries. The model Z_{sl} is projective, $\mathbb{Q}$ -factorial and klt, and P_{sl} is nef, big and semiample. Choose the incidence point also in its smooth open. The component F is proper, its generic base ideal is primary, and the preceding restriction gives ordinary multiplicity at least $h \geq k\delta/2$. Writing

$f = \dim F$, the lemma therefore gives

$$\begin{aligned} P_{\text{sl}}^f \cdot F &\leq (2/\delta)^{n+1-f} P_{\text{sl}}^{n+1}, \\ K_{F^*} P_{\text{sl}}^{f-1} &\leq (K_{Z_{\text{sl}}} + c P_{\text{sl}})|_F P_{\text{sl}}^{f-1}, \quad 0 < c \leq 2(n+1-f)/\delta. \end{aligned}$$

The restricted components must also move sufficiently to apply curve exclusion. These F 's may be chosen in families sweeping the opposite slice. Indeed the incidence of the V -family dominates the total torus. Choose a general incidence point, not necessarily the original marked point, and then a general fiber of its first slot evaluation. Dominance gives the required dominant evaluation onto that fixed opposite slice. Choose a component with dominant evaluation there; generic flatness permits the fiber-component choices just made. For each fixed q, t , very general choices also avoid the exceptional sets for all rational weights. Choose this point also in the isomorphism open of the slice's big semiample contraction and outside the relevant fixed differences. The restriction of its positive part to F is then nef and big, so its top intersection on F is positive, as required for the curve construction.

Proper base images on a slice. If F is generically finite over a proper positive-dimensional image W in the remaining base, then W , and hence F , is of general type by Proposition 6.3. The slice estimates and subadjunction give the required bounds on this component, while weight domination bounds the degrees of its projected curves. We therefore obtain forbidden bounded curves.

Over a proper base image, the other possibility is that F is saturated over its base image W : on the original bundle it is the full projective-line bundle over W . Put $w = \dim W = \dim F - 1$. The nef weight comparison gives

$$q N_b^w \cdot W \leq P_{\text{sl}}^{w+1} \cdot F. \quad (39)$$

Indeed $P_{\text{sl}} - \pi^* N_b$ is effective on the resolved graph of the slice projection π to the base, so successive mixed nef intersections give $P_{\text{sl}}^{w+1} \cdot F \geq P_{\text{sl}}(\pi^* N_b)^w \cdot F$. The latter is $q N_b^w \cdot W$, since endpoint freeness gives fiber degree q . If W is a point, the lower bound q contradicts the slice Bezout upper bound $2(n+1)q\epsilon^n(2/\delta)^n$, because our choice of δ makes the latter strictly less than q .

Suppose $w > 0$. The ruled component F itself need not be of general type, so we seek the adjunction estimate on its base W . We obtain it from the coefficients of the slice sections. Use the section-space identification to view the subseries on the original slice, where the toric weight decomposition is defined. The full fixed factors are units at the chosen generic torus point, so this identification preserves the order h . In degree k , expand each of these sections into its toric weights. At the generic point of W , let I_l be the ideal generated by the coefficients of the nonzero weight l over all these sections. Put $h = \lceil k\delta/2 \rceil$. All I_l lie in $\mathfrak{m}_W^h$: vanishing along the generic torus fiber says that the coefficient of every Laurent monomial vanishes to this order. Their sum is $\mathfrak{m}_W$ -primary. Otherwise its larger common zero germ, times the generic torus fiber, would contradict isolation of F .

Our aim is to construct an effective boundary $\Theta \sim_{\mathbb{Q}} c' L_b$, for some rational weight b , such that W is an lc center near its generic point and $c' \leq 2(n-w)/\delta$. Numerical subadjunction will then give a canonical-intersection bound on W , even though F is ruled. Work in the regular local ring at the generic point of W , of dimension $c = n - w$. Resolve the finitely many coefficient ideals simultaneously. The exponents for which their product defines an lc ideal pair form the rational polytope

$$u_l \geq 0, \quad \sum_l u_l \text{ord}_E(I_l) \leq a(E; Y, 0).$$

Here E runs through the divisors needed to test log canonicity on the simultaneous resolution, and $a(E; Y, 0)$ is the log discrepancy. Include the exceptional divisor of the blowup of the closed point. Since every $I_l \subset \mathfrak{m}_W^h$, it gives $\sum_l u_l \leq c/h$. The polytope is compact, and $\sum_l u_l$ has a positive rational maximum on it. Fix a rational maximizing point. For each coordinate l , some equality at this point must involve a valuation with $\text{ord}_E(I_l) > 0$; otherwise u_l could be increased while staying in the polytope. Such a valuation has log discrepancy zero for the ideal pair. Its center C_l is contained in $V(I_l)$ and contains the generic point of W . Their intersection is contained in $V(\sum_l I_l)$, so it is exactly W locally. To apply the intersection property for lc centers, we first realize these ideal lc places by an actual divisor. Choose an integer $M_0 > \max_l u_l$ and M_0 general coefficient divisors $D_{l,j}$ from each system. Set $\Theta = \sum_l (u_l/M_0) \sum_{j=1}^{M_0} D_{l,j}$. On the simultaneous resolution its fixed crepant coefficients are at most one, and its general moving divisors meet transversely with coefficients below one. Thus the pair is lc near the generic point of W , and every active valuation remains an lc place. Each C_l is consequently an lc center of this pair. The intersection property [43, Theorem 1.7], applied to the identity morphism on its lc open, shows that W is an lc center generically. This also applies when some maximizing coordinate u_l is zero: the active valuation blocking that coordinate still has center in $V(I_l)$.

Set

$$b = \frac{\sum_l u_l(l/k)}{\sum_l u_l}, \quad c' = k \sum_l u_l \leq \frac{kc}{h}.$$

Since a weight- l coefficient divisor has class $kL_{l/k}$, the constructed boundary has class $c'L_b$. Also $c' \leq kc/h \leq 2(n-w)/\delta$. Transform it and W to the small scaling model $Y(b)$; our general incidence choice places the generic point of W in their common isomorphism open. On this model the boundary has class $c'N_b$. Numerical subadjunction and $N_b = (r + bm)K_{Y(b)} + rtA_b$ give

$$K_{W^*} N_b^{w-1} \leq \left(\frac{1}{r + bm} + c' \right) N_b^w \cdot W.$$

Indeed a general W is not contained in a fixed effective representative of A_b , so its contribution to this intersection is nonnegative. The slice degree estimate and Equation (39) give, in turn,

$$0 < N_b^w \cdot W \leq \frac{(2/\delta)^{n-w}}{q} P_{\text{sl}}^{n+1} \leq 2(n+1)\epsilon^n (2/\delta)^{n-w}.$$

These are precisely the two bounds required in Equation (34), with constants independent of t . The variety W is of general type by Proposition 6.3; normalized curve exclusion again gives a contradiction.

A slice multisection. Proper base images have now been excluded. The remaining proper positive-dimensional slice case is $\dim F = n$, with F generically finite of degree e over the whole base. Weight comparison and the slice degree bound give

$$eN_b^n \leq P_{\text{sl}}^n \cdot F,$$

so e is bounded uniformly in t . Here the two axes provide the boundary needed to use log general type on the base. The closure of F on the original bundle is neither axis. Intersect this closure with each Cartier axis and push the resulting effective integral cycle to Y ; call the resulting Weil divisors D_1, D_2 . The two axis classes differ by the pullback of mK , and F has generic degree e over Y . The projection formula therefore gives

$$D_1 - D_2 \sim_{\mathbb{Q}} \pm emK. \tag{40}$$

At the appropriate endpoint weight $b \in \{0, q\}$, the difference $E_b|_F$ contains the corresponding toric-axis intersection with coefficient q . There is no full fixed subtraction above generic base-divisor points by Lemma 8.2. Intersecting with N_b^{n-1} and using mixed nef comparison yields

$$qN_b^{n-1}.D_i \leq P_{\text{sl}}^n.F.$$

The other divisor has bounded degree for the same N_b , because of Equation (40) and

$$0 \leq K_{Y(b)}N_b^{n-1} \leq \frac{N_b^n}{r + bm}.$$

We now apply the logarithmic curve construction on the base $Y(b)$, not on the multisection F . The signed divisor $\pm(D_1 - D_2)/(em)$ represents $K_{Y(b)}$ by Equation (40). On a log resolution of this base, add the reduced strict support of $D_1 + D_2$ and all exceptional divisors to the canonical divisor. Proposition 6.10 gives a big adjoint for the reduced strict support of $D_1 - D_2$ and all exceptional divisors. Adding the remaining effective reduced boundary shows that the resulting log canonical divisor is big as well. Its intersection with the pulled-back N_b^{n-1} is bounded: exceptionals vanish under projection, and reduced support has degree no larger than $D_1 + D_2$. Log effective birationality and the curve construction therefore contradict normalized curve exclusion on the base itself.

Reduction to a correspondence. We now return to the total model Z and its positive part P . The slice analysis has excluded fiber dimensions strictly between zero and $n + 1$ over either slot. A full $(n + 1)$ -dimensional fiber over one slot would mean that V contains the whole opposite slice over its first image W . Its fiber dimension over the other slot would then be $\dim W + 1$, strictly between zero and $n + 1$, unless V were the full total space. That is impossible. Hence V projects generically finitely in both slots and $\dim V \leq n$. If either image is proper, general type and the total-space estimates give the previous curve contradiction. The same is true if V is of general type. Thus only the case $\dim V = n$, dominant with bounded degree over both copies of Y , remains. We next show that its branch divisors cannot move; this will reduce the family to finitely many covers.

Moving branch divisors. Consider the dominating algebraic family of these correspondences, resolving maps after shrinking the parameter space. Suppose a divisorial branch component of one projection moves. On a resolution V^* , choose a ramified divisor R above its generic point. There is a rational weight b for which the effective difference $E_b|_{V^*}$ does not contain R . Indeed take a fixed free divisible degree of P ; some section does not vanish generically on R . Monomial-weight sections span that degree, so one of them does not vanish there. Only finitely many weights are tested in this fixed family; their model comparisons can be resolved simultaneously.

Let J be the branch divisor on the corresponding small axis model. Restrict the effective difference to R ; this is permitted because R is not in its support. Mixed nef intersections and projection to J bound its N_b -degree by $P^{n-1}.R$. The ramification formula over the terminal target bounds the latter by $K_{V^*}P^{n-1}$: its other ramification terms are nonnegative, and the pullback of the target canonical class is pseudo-effective. The total-space subadjunction estimate now gives

$$N_b^{n-1}.J \leq P^{n-1}.R \leq K_{V^*}P^{n-1} \leq C_q. \quad (41)$$

The ramification coefficient of R is a positive integer, so is at least one; the other ramification and exceptional terms are nonnegative. The last bound is the total-space subadjunction estimate.

To turn this degree bound into bounded curves, we still need a canonical intersection bound on J . The required adjunction does not assume that $(Y(b), J)$ is globally lc. Terminal $Y(b)$ is smooth in codimension two. Thus normalization and conductor adjunction along J give

$$K_{J^\nu} + C = (K_{Y(b)} + J)|_{J^\nu}, \quad C \geq 0$$

in codimension one. On resolving J^ν , exceptional divisors map to codimension at least two and pair trivially with the pulled-back N_b^{n-2} . Consequently, with $d_J = N_b^{n-1}.J$,

$$\begin{aligned} K_{J^*} N_b^{n-2} &\leq (K_{Y(b)} + J) J N_b^{n-2} \\ &\leq \frac{d_J}{r + bm} + \frac{d_J^2}{N_b^n} \leq C'_q d_J. \end{aligned} \quad (42)$$

For the second line, a general moving J is not a component of a fixed effective representative of A_b , giving the first term; mixed Hodge index gives the second. The lower bound for N_b^n and Equation (41) give the final uniform constant.

A moving J sweeps very general base points and is of general type by Proposition 6.3. Equations (41)–(42) therefore give forbidden bounded curves. Branch components can be named after a finite parameter extension and shrinking. It follows that, for all sufficiently small t , all divisorial branch components in the chosen family are fixed.

Fixed covers. We now fix t . The branch complement is chosen from the one finite-type family of correspondences under consideration, before any cover is classified. After a finite parameter extension and shrinking, resolve the two evaluation maps simultaneously over a smooth parameter open and follow the finitely many irreducible relative ramification divisors. The argument just given excludes each component whose divisorial images sweep the base: such a component would supply the bounded-genus covering curves of Proposition 6.8. For every remaining component the closure of its divisorial evaluation image is a fixed proper closed subset of Y , and the intersection estimates already bound the degrees of both projections. These are precisely the hypotheses of Lemma 7.5 for the smooth projective family of resolved correspondences. It gives a smooth dense open Y^0 and only finitely many finite normal Stein covers in either slot, all étale over Y^0 . Bad parameter fibers are removed by shrinking the parameter space, not by deleting their possibly dominant images in Y . Both Y^0 and the finite cover lists may depend on this fixed t ; no common complement or list as $t \rightarrow 0$ is used.

For a general member V_a^* of the resolved family, factor the two projections through their finite normal Stein covers. The preceding finiteness lets us fix these as $T_1 \rightarrow Y$ and $T_2 \rightarrow Y$ after restricting to a family whose images still dominate $Y \times Y$. Each map $\beta_{i,a} : V_a^* \rightarrow T_i$ is birational, so the two projections determine a birational map

$$\phi_a = \beta_{2,a} \circ \beta_{1,a}^{-1} : T_1 \dashrightarrow T_2.$$

The graphs of these maps form an algebraic family after choosing a Hilbert-scheme component and shrinking for flatness and birationality of both projections. Such a component with dominant evaluation exists: there are only countably many graph Hilbert schemes and finitely many cover choices, whereas the original incidence dominates the base product. Its graph incidence has image of dimension $2n$ in $T_1 \times T_2$, because this product is finite over $Y \times Y$. The product is integral of dimension $2n$, so the graph incidence dominates it. Proposition 6.11 excludes exactly this family of birational maps between the two fixed covers.

Every possible moving component V has now led to a contradiction, so the Seshadri bound holds. To obtain the assertion about jets, observe that at a very general point the big semiample contraction is an isomorphism onto its image near that point. The jet interpretation of the Seshadri constant for the ample class downstairs gives the stated divisible jet-separating multiple. Complete systems agree under the model comparisons, so this conclusion transfers to the original torus open and to common resolutions. $\square$

9 Frobenius comparison and smooth nonvanishing

The scalar and two-slot estimates lead to a contradiction through a comparison in positive characteristic. We first prove that certain fixed divisors, jets, and a movable-curve witness cannot coexist. This theorem and its proof are independent of nonvanishing, abundance, minimal models, and logarithmic subadditivity. We then construct its inputs from the counterexample geometry and obtain smooth canonical nonvanishing.

9.1 The finite-data Frobenius theorem

We now work with arbitrary fixed data, independent of the scaling models and divisors of Section 8. The comparison concerns two orders of vanishing of an actual determinant. Fixed ordinary jets will give a nonzero determinant after reduction. A small polarization bounds the rank on the diagonal and therefore forces a large order there. A fixed movable curve on a blowup bounds every section in each determinant factor, producing the opposite estimate.

For a vector bundle on a smooth projective curve, its slope is degree divided by rank. The least and greatest Harder–Narasimhan slopes are denoted by $\mu_{\min}$ and $\mu_{\max}$. For a line bundle on a smooth variety, to generate jets through order a at a point means surjectivity to its stalk modulo the $(a+1)$ -st power of the maximal ideal.

Theorem 9.1 (Finite-data Frobenius incompatibility). *Let W be a smooth connected projective complex variety of dimension $n \geq 2$. Let L, H be rational Cartier divisors, with H ample, let S be an integral Cartier divisor, and fix an integer $q > 0$ and real numbers $r > 1$, $\epsilon > 0$. Suppose*

$$\begin{aligned} H^n &\leq (2\epsilon)^n, & K_W H^{n-1} &\leq 2H^n/r, & (2L + qS)H^{n-1} &\leq 4H^n, \\ (14\epsilon)^n &< \frac{1}{4}, & 80\epsilon &< \frac{3}{4}. \end{aligned} \tag{43}$$

Assume that the following fixed data exist.

(i) *A smooth integral complete-intersection flag from W to a curve C , whose successive divisor classes are $h_i = l_i H$ for fixed positive integers l_i , with $\mu_{\min}(\Omega_W^1|_C) \geq 0$.*

(ii) *On the projective bundle*

$$Z = \mathbf{P}(\mathcal{O}_W(S)_1 \oplus \mathcal{O}_W(S)_2) \longrightarrow W \times W, \quad \xi = c_1(\mathcal{O}_Z(1)), \quad M = L_1 + L_2 + q\xi,$$

use the convention that the direct image of $\mathcal{O}_Z(a\xi)$ is the a -th symmetric power of the displayed sum. A smooth point z_ in the complement of the two axes and finitely many sections of $k_0 M$, for some integer $k_0 > 0$ with $k_0 L$ Cartier, generate jets through order $(2n+2)k_0$ at z_* .*

(iii) *A point $x \in W$, its blowup $\pi_x : \widehat{W} \rightarrow W$ with exceptional divisor J , and a curve class*

$$\gamma = f_*(A_1 \cdots A_{n-1}),$$

where $f : V \rightarrow \widehat{W}$ is one fixed birational morphism, V is smooth integral projective, and the A_i are ample integral Cartier divisors, such that $J \cdot \gamma > 0$ and

$$\pi_x^*(L + K_W/2 + \lambda S) \cdot \gamma \leq 40\epsilon J \cdot \gamma \quad (0 \leq \lambda \leq q). \tag{44}$$

It suffices to check the two endpoints in this affine inequality.

These data cannot coexist.

There is no relation required between x and the two base coordinates of z_* . The divisors L, S, K_W are not assumed nef, and K_W is not assumed pseudo-effective. The theorem starts with the actual flag and movable witness, rather than with geometric conditions from which one might construct them. All data in its statement are fixed before the residual characteristic tends to infinity.

9.1.1 Spreading the fixed witnesses

We prove the theorem in the next five steps. Put $\Lambda = \prod_{i=1}^{n-1} l_i$. The flag satisfies

$$[C] = \Lambda H^{n-1}, \quad h_i \cdot C = l_i \Lambda H^n. \quad (45)$$

Choose one sufficiently ample integral Cartier divisor T_0 so that every $T_0 + aS$, $0 \leq a \leq (k_0 - 1)q$, has a section nonvanishing at each of the two base coordinates of z_* . Fix these finitely many sections. All models, morphisms, divisors, points, the flag, the complete-intersection witness for γ , and the jet and filler sections spread over an integral finitely generated $\mathbf{Z}$ -algebra of characteristic zero. Shrink its spectrum so that the fibers and flag are smooth projective and geometrically integral, the birational morphisms remain birational, the fixed ample bundles remain ample, and the finite jet surjection and nonvanishings persist. Birationality is preserved by spreading an isomorphism between dense opens. All displayed intersection numbers are constant.

We also preserve $\mu_{\min}(\Omega_W^1|_C) \geq 0$. One way is to spread its characteristic-zero Harder–Narasimhan filtration, make all its factors locally free, and use openness of their semistability on curves; their degrees and ranks are fixed. Equivalently, a single relative ample twist globally generates the curve bundle. Every quotient of rank a then has degree bounded below by $-ad$ for a fixed integer d . A negative-degree quotient has one of finitely many ranks and degrees. The proper relative Quot schemes for those Hilbert polynomials have images missing the generic point: after removal of torsion, a negative-degree coherent quotient there would give a negative-degree vector-bundle quotient. Removing those finitely many images proves the same openness assertion directly.

This base has closed points in arbitrarily large characteristics. Take algebraic closures of their residue fields and exclude denominator primes. Retain the notation for the reductions. The curve γ continues to pair nonnegatively with every effective divisor: pull that divisor back under the fixed birational morphism and intersect with its fixed ample divisors. This tests effective divisors on the reduction itself. No specialization of a characteristic-zero effective cone is used.

Lemma 9.2. *For a sufficiently large residual characteristic p , set*

$$N_p = \lfloor p/k_0 \rfloor, \quad B_p = k_0 N_p L + T_0.$$

The sections of $pq\xi + B_{p,1} + B_{p,2}$ generate jets through order $(2n+2)k_0 N_p$ at z_ . In particular, they surject onto the quotient of its local ring by the p -th powers of all regular parameters.*

Proof. Multiply N_p copies of the fixed jet system. Every monomial of total degree at most $(2n+2)k_0 N_p$ is a product of N_p monomials of degree at most $(2n+2)k_0$. Taking sections with those leading monomials gives a triangular system, ordered by total degree. Starting at degree zero and removing the error in each successive degree proves the required jet surjection. This argument takes place in the local ring and never divides by factorials.

Put $d_p = (p - k_0 N_p)q$, so $0 \leq d_p \leq (k_0 - 1)q$. Among the sections fixed before reduction, choose

$$u_0 \in H^0(W, \mathcal{O}_W(T_0)), \quad v_{d_p} \in H^0(W, \mathcal{O}_W(T_0 + d_p S))$$

nonzero at the first and second base coordinates of z_* , respectively. Their tensor product belongs to the summand of first-slot weight zero in the projective-bundle formula; its fiber monomial is the d_p -th power of the second coordinate. It defines a section of $d_p \xi + T_{0,1} + T_{0,2}$ nonzero at z_* , since z_* lies off both axes. Multiplying the product jet system by this section changes its divisor from $N_p k_0 M$ to $pq\xi + B_{p,1} + B_{p,2}$. Multiplication is invertible on the local jet algebra. Only the finitely many previously fixed values of d_p occur.

Since $\dim Z = 2n + 1$, the quotient by the p -th powers of regular parameters has top total degree $(2n + 1)(p - 1)$. For large p this is at most $(2n + 2)k_0 N_p$. The underlying ordinary/Frobenius ideal comparison is also recorded in [52, proof of Proposition 2.12, Equation (2.8)]. $\square$

9.1.2 Full rank off the diagonal

Let $F : W \rightarrow W'$ be relative Frobenius to the base-field twist. It is finite flat because W is smooth. Write S' for the twist of S , so $F^*S' = pS$, and define

$$\mathcal{E} = F_*\mathcal{O}_W(B_p), \quad s = \operatorname{rk} \mathcal{E} = p^n, \quad R = s^2, \quad U_a = H^0(W, \mathcal{O}_W(B_p + paS)).$$

Projection formula gives an evaluation map on $W' \times W'$:

$$\bigoplus_{\substack{a+b=q \\ a,b \geq 0}} U_a \otimes U_b \otimes \mathcal{O}(-aS'_1 - bS'_2) \longrightarrow \mathcal{E} \boxtimes \mathcal{E}. \quad (46)$$

Lemma 9.3. *The map in Equation (46) has generic rank R .*

Proof. Trivialize the two summands defining Z near the selected base points. Let z be the torus coordinate, whose value at z_* is $z_0 \neq 0$. By the projective-bundle formula, the sections in Lemma 9.2 are weight polynomials in z with weight- j coefficients in

$$H^0(W, B_p + jS) \otimes H^0(W, B_p + (pq - j)S), \quad 0 \leq j \leq pq.$$

Let A_* be the tensor product of the local algebras of the two base Frobenius fibers. The local quotient in Lemma 9.2 is

$$A_*[z]/(z^p - z_0^p),$$

free over A_* with basis $1, z, \dots, z^{p-1}$. Project to the coefficient of 1 in this basis. Exactly the weights $j = pa$ survive, with multipliers z_0^{pa} . Their coefficients are the values of the corresponding summands of Equation (46), in the chosen frames and the induced twisted frames. As the full jet map is surjective, these coefficients span A_* , of dimension $p^{2n} = R$. Thus the evaluation has full rank at this pair of base points, and hence generically. The calculation uses a basis over the possibly nonreduced algebra A_* , so reducedness of the Frobenius fibers is unnecessary. $\square$

The jet calculation at z_* proves generic full rank; the diagonal estimate below will be tested at (x', x') , obtained from the separately fixed point x .

9.1.3 The diagonal filtration

The Frobenius fiber product has the cartesian square

$$\begin{array}{ccc} \Delta_F = W \times_{W'} W & \xrightarrow{\operatorname{pr}_2} & W \\ \operatorname{pr}_1 \downarrow & & \downarrow F \\ W & \xrightarrow{F} & W'. \end{array} \quad (47)$$

Its reduced subscheme is the diagonal W . Put

$$\mathcal{L}_p = \mathcal{O}_{\Delta_F}(B_{p,1} + B_{p,2} + pqS_1).$$

Write $h = F \circ \text{pr}_1 = F \circ \text{pr}_2 : \Delta_F \rightarrow W'$. On this fiber product,

$$\mathcal{O}(pS_1) \simeq h^* \mathcal{O}_{W'}(S') \simeq \mathcal{O}(pS_2).$$

Consequently every pair of weights $a + b = q$ gives the same line bundle:

$$\mathcal{O}_{\Delta_F}(B_{p,1} + B_{p,2} + paS_1 + pbS_2) \simeq \mathcal{L}_p.$$

On the diagonal of $W' \times W'$, every source twist of Equation (46) becomes $-qS'$. Finite base change and projection formula identify the twisted target as

$$(\mathcal{E} \otimes \mathcal{E})(qS') \simeq h_* \mathcal{L}_p.$$

The product sections defining the columns restrict to global sections of this one bundle $\mathcal{L}_p$. Their values at a diagonal point therefore lie in the image of $H^0(\Delta_F, \mathcal{L}_p)$ in the corresponding fiber of $h_* \mathcal{L}_p$. This proves that the rank at every diagonal point is at most

$$h^0(\Delta_F, \mathcal{L}_p). \quad (48)$$

For a rank- n bundle V in characteristic p , denote by $A_j(V)$ the degree- j part of its symmetric algebra modulo the p -th powers of local generators. This construction is independent of frame: the p -th power of a linear combination is the sum of the p -th powers in characteristic p . Each $A_j(V)$ is locally free and a quotient of $V^{\otimes j}$. Set

$$u = n(p-1), \quad e_j = \text{rk } A_j(V), \quad \sum_{j=0}^u e_j = p^n.$$

Filtering $\mathcal{L}_p$ by powers of the ideal of the reduced diagonal in Δ_F gives the graded bundles

$$\mathcal{G}_j = \mathcal{O}_W(2B_p + pqS) \otimes A_j(\Omega_W^1), \quad 0 \leq j \leq u. \quad (49)$$

Indeed, in smooth local coordinates the algebra is generated by parameter differences with their p -th powers zero, and its degree-one conormal is Ω_W^1 . This coordinate calculation can be made étale locally or in completions and identifies the global associated graded. This is the diagonal-ideal form of the canonical Frobenius filtration [41, Corollary 3.5]. The same calculation, using only the bundle from the second factor, filters $F^* \mathcal{E}$ with pieces [59, Theorem 3.7]

$$\mathcal{O}_W(B_p) \otimes A_j(\Omega_W^1). \quad (50)$$

Complementary multiplication is a perfect pairing, giving

$$A_{u-j}(V) \simeq A_j(V)^\vee \otimes (\det V)^{p-1}. \quad (51)$$

On monomials, each exponent vector pairs with its complement to $(p-1, \dots, p-1)$. The top line transforms by the character $(\det V)^{p-1}$: this is immediate on diagonal matrices and hence identifies the character of GL_n on that line.

9.1.4 One slope bound and one flag polynomial

We bound the sections of every graded bundle in Equation (49) by its rank times the same scalar polynomial in p . This uniformity lets us sum the ranks of the graded pieces without an additional factor for their number. We begin with a slope bound on the fixed curve; here slope means ordinary degree divided by rank.

Lemma 9.4. *There is a constant $c \geq 0$, independent of the sufficiently large residual characteristic p and of j , such that for $V = \Omega_W^1|_C$ one has*

$$\mu_{\max}(A_j(V)) \leq (p-1)K_W \cdot C + nc.$$

Proof. Write F_C for absolute Frobenius of C . Langer's instability estimate [45, Corollary 2.5] gives

$$L_{\min}(V) := \lim_{e \rightarrow \infty} p^{-e} \mu_{\min}(F_C^{e*} V) \geq -\frac{c}{p-1}. \quad (52)$$

To check the uniform constant, the discrepancy from $\mu_{\min}(V) \geq 0$ is at most $(\operatorname{rk} V - 1) \deg A_C / (p-1)$, where A_C is a nef line bundle such that $T_C \otimes A_C$ is globally generated. Its degree can be bounded in terms of the fixed genus, using a sufficiently positive line bundle. Here the rank is n , irrespective of the fact that the base is a curve.

We also need, for every integer $i \geq 0$,

$$\mu_{\min}(V^{\otimes i}) \geq i L_{\min}(V). \quad (53)$$

Fix p and i . Choose a line bundle A_e of degree $-\mu_{\min}(F_C^{e*} V) + O(1)$, rounded up with a fixed genus-dependent additive constant, so that $F_C^{e*} V \otimes A_e$ is globally generated. This follows from Serre duality: after subtracting any point, its minimum slope can be made greater than $2g(C) - 2$, which annihilates H^1 and makes evaluation at that point surjective. For every vector-bundle quotient Q of $V^{\otimes i}$, the bundle $F_C^{e*} Q \otimes A_e^{\otimes i}$ is then globally generated. Consequently

$$p^e \mu(Q) + i \deg A_e \geq 0.$$

Divide by p^e and let $e \rightarrow \infty$ to prove Equation (53). This limit is taken for each fixed p, i ; there is no interchange of Frobenius-iteration and characteristic limits.

Since $A_{u-j}(V)$ is a quotient of $V^{\otimes(u-j)}$, the perfect pairing Equation (51) yields

$$\begin{aligned} \mu_{\max}(A_j(V)) &= (p-1) \deg \det V - \mu_{\min}(A_{u-j}(V)) \\ &\leq (p-1)K_W \cdot C + \frac{(u-j)c}{p-1} \\ &\leq (p-1)K_W \cdot C + nc. \end{aligned}$$

□

The divisors $B_p - pL = T_0 - (p - k_0 N_p)L$ range over a fixed finite list. Combining Equation (43), Equation (45), Lemma 9.4 therefore gives, uniformly for $0 \leq j \leq u$,

$$\begin{aligned} \mu_{\max}(\mathcal{G}_j|_C) &\leq p(4 + 2/r)\Lambda H^n + O(1) \\ &\leq M_p := 7p\Lambda H^n + c_0, \end{aligned} \quad (54)$$

where $c_0 \geq 0$ is fixed. The coefficient 7 provides harmless room in this common bound.

Lemma 9.5. *For all sufficiently large p ,*

$$h^0(\Delta_F, \mathcal{L}_p) \leq R(7^n H^n + O(1/p)) < R/4.$$

The error constant is independent of the filtration index j .

Proof. Write the fixed flag as

$$W = W_n \supset W_{n-1} \supset \cdots \supset W_1 = C, \quad W_{n-k} \in |h_k|_{W_{n-k+1}} \quad (1 \leq k \leq n-1).$$

At step k , fix the preceding nonnegative integers $b_1, \dots, b_{k-1}$ and put

$$\mathcal{F}_k = \mathcal{G}_j|_{W_{n-k+1}} \left(- \sum_{i < k} b_i h_i \right).$$

For each $b_k \geq 0$, the divisor restriction sequence is

$$0 \longrightarrow \mathcal{F}_k(-(b_k + 1)h_k) \longrightarrow \mathcal{F}_k(-b_k h_k) \longrightarrow \mathcal{F}_k(-b_k h_k)|_{W_{n-k}} \longrightarrow 0.$$

Iterating it for $b_k = 0, 1, \dots$ and then proceeding to the next member of the flag bounds the section space by

$$h^0(W, \mathcal{G}_j) \leq \sum_{b_1, \dots, b_{n-1} \geq 0} h^0 \left(C, \mathcal{G}_j|_C \left(- \sum_i b_i h_i|_C \right) \right). \quad (55)$$

At each stage the remainder is zero after sufficiently negative ample twisting. Its stopping index need not be uniform in j, p or in the preceding twists: enlarging the nonnegative finite sums to the displayed lattice sum preserves the inequality.

A summand in Equation (55) vanishes when $\sum_i b_i (h_i \cdot C) > M_p$, because then its maximum slope is negative. Each nonzero summand has at most $e_j(M_p + 1)$ sections. Indeed, evaluation at $[M_p] + 1$ distinct points is injective; the kernel has negative maximum slope after subtracting those points. By Equation (45) it follows that

$$\begin{aligned} h^0(W, \mathcal{G}_j) &\leq e_j(M_p + 1) \prod_{i=1}^{n-1} \left(1 + \frac{M_p}{l_i \Lambda H^n} \right) \\ &= e_j p^n (7^n H^n + O(1/p)). \end{aligned} \quad (56)$$

The leading coefficient follows from $\Lambda = \prod_i l_i$. More importantly, the whole scalar polynomial on the first line is the same for every j .

Sum over the filtration in Equation (49). The ranks satisfy $\sum_j e_j = p^n$, so Equation (56) gives the asserted bound with $R = p^{2n}$. There is no extra factor for the number of graded pieces. Finally

$$7^n H^n \leq (14\epsilon)^n < 1/4$$

by Equation (43), proving the strict inequality for all large p . □

9.1.5 The determinant has incompatible orders

Choose R generically independent columns of Equation (46) from bases of its source vector spaces. Their determinant is a nonzero section

$$0 \neq \sigma \in H^0(W' \times W', \mathcal{Q}_1 \boxtimes \mathcal{Q}_2)$$

of an actual external-product line bundle. To specify its factors, let m_i be the sum of the weights of the R chosen columns in slot i . The target determinant is $(\det \mathcal{E})^s \boxtimes (\det \mathcal{E})^s$, and the chosen source determinant is $\mathcal{O}(-m_1 S') \boxtimes \mathcal{O}(-m_2 S')$. Thus

$$\mathcal{Q}_i = (\det \mathcal{E})^{\otimes s} \otimes \mathcal{O}_{W'}(m_i S'), \quad \lambda_i = m_i/R.$$

Every column weight lies between 0 and q , so $0 \leq \lambda_i \leq q$ independently of the columns chosen.

Under the numerical identification with the base-field twist,

$$\frac{c_1(\mathcal{Q}_i)}{R} = \frac{c_1(\mathcal{E})}{s} + \lambda_i S = \frac{B_p}{p} + \frac{p-1}{2p} K_W + \lambda_i S, \quad 0 \leq \lambda_i \leq q. \quad (57)$$

The Frobenius first-Chern-class identity is also [59, Lemma 4.2]; it is an identity of rational divisor classes. For the second equality, use Equation (50). Complementary degrees in Equation (51) have total first Chern class $e_j(p-1)K_W$. Summing gives

$$\sum_j c_1(A_j(\Omega_W^1)) = \frac{s(p-1)}{2} K_W.$$

Thus $c_1(F^* \mathcal{E}) = sB_p + s(p-1)K_W/2$, and Frobenius pullback of twisted divisor classes multiplies by p , as required.

For $0 \leq \lambda \leq q$, the fixed movable inequality gives

$$(L + K_W/2 + \lambda S) \cdot \gamma \leq 40\epsilon J \cdot \gamma,$$

where pullback by π_x is understood. This concerns only intersections of fixed divisors and the fixed complete-intersection witness, so it remains true on the reduction. The class in Equation (57) differs from its left-hand divisor at $\lambda = \lambda_i$ by

$$\frac{1}{p}(B_p - pL - K_W/2).$$

Its numerator belongs to the fixed finite list $T_0 - aL - K_W/2$, $0 \leq a < k_0$. The resulting error after intersection with γ is therefore $O(p^{-1})$, uniformly over the selected determinant columns.

Let x' be the twisted point and let $0 \neq \tau_i \in H^0(W', \mathcal{Q}_i)$. The strict transform of its effective divisor on the blowup at x' pairs nonnegatively with the twist of γ . Since $J \cdot \gamma > 0$, it follows that

$$\frac{\text{ord}_{x'}(\tau_i)}{R} \leq 40\epsilon + O(1/p). \quad (58)$$

The constant is uniform in the columns and in τ_i . This argument tests sections on the reduction itself; it does not require them to lift to characteristic zero.

Künneth identifies the space containing σ with $H^0(W', \mathcal{Q}_1) \otimes H^0(W', \mathcal{Q}_2)$. Choose bases adapted to the orders of vanishing at x' in each factor. In each degree their leading terms are linearly independent. Tensor products of these leading terms remain independent in each bidegree of the two disjoint sets of parameters; terms of different bidegrees cannot cancel. Every nonzero tensor consequently has order at most the sum of the two maximum basis orders. Applying Equation (58), we obtain

$$\text{ord}_{(x', x')} \sigma \leq R(80\epsilon + O(1/p)). \quad (59)$$

On the other hand, Equation (48), Lemma 9.5 bound the matrix rank at (x', x') by a number strictly smaller than $R/4$. Trivialize its line bundles near that point. Invertible constant row

operations make more than $3R/4$ rows vanish at the point. Each entry of those rows belongs to the maximal ideal, so every term of the determinant has order at least $3R/4$. Therefore

$$\text{ord}_{(x',x')} \sigma \geq 3R/4. \quad (60)$$

Equations (59) and (60) contradict $80\epsilon < 3/4$ for sufficiently large p .

This contradiction proves Theorem 9.1. The full-rank calculation used z_* , while the diagonal estimate holds at every point and was tested at the separately chosen x . The errors were controlled by one fixed finite list of divisor classes and one fixed flag polynomial. Thus their constants do not depend on p , the filtration index, or the chosen columns.

9.2 The geometric witnesses and smooth nonvanishing

We return to the notation and the hypothetical counterexample of Section 8. Theorem 9.1 asks for fixed divisors, a small polarization and flag, finite two-slot jets, and one actual strongly movable curve. We construct all these inputs in characteristic zero. The positive parts on the scaling models, rather than the original non-nef classes, provide the small polarization. No minimal-model statement or pseudo-effective cone is specialized to positive characteristic.

Theorem 9.6 (The smooth nonvanishing step). *Assume the lower-dimensional real-boundary good-model hypothesis of Assumption 2.1. If X is a smooth projective complex variety of dimension n and K_X is pseudo-effective, then $\kappa(X, K_X) \geq 0$.*

In dimension zero the assertion is immediate. On a smooth curve, pseudo-effectivity of K_X gives $g(X) \geq 1$, hence $h^0(X, K_X) = g(X) > 0$. We therefore suppose $n \geq 2$ and argue by contradiction. We use the late terminal model Y , the divisors $K = K_Y$ and $A = A_Y$, and the scaling models of the preceding section. Fix $m > 0$ with mK Cartier, and write

$$\mu_t = \text{vol}(K_X + tA_X)^{1/n}, \quad L = r(K + tA).$$

As before, r is a positive integer chosen so that $r\mu_t \rightarrow \epsilon$ as $t \downarrow 0$. Choose the rational number $\epsilon > 0$ sufficiently small for Propositions 8.1 and 8.3, and also so that

$$(14\epsilon)^n < \frac{1}{4}, \quad 80\epsilon < \frac{3}{4}. \quad (61)$$

The strict inequalities leave room for the errors that tend to zero with the characteristic. We next fix a sufficiently large integer q as in Proposition 8.3, and then choose a positive rational t small enough for the estimates below and for $r > qm + 1$. Once those estimates hold, t and r remain fixed through all the subsequent characteristic-zero constructions and reductions.

9.2.1 A small polarization and a cotangent flag

A small ample polarization and a complete-intersection flag will control the number of sections on the Frobenius thickening of the diagonal. We construct them from the positive part on a scaling model.

Choose a smooth common resolution W of Y and the scaling model for $K + tA$. In this application K, A, L also denote their pullbacks to W , and

$$K_W = K + E, \quad E \geq 0.$$

Let H_0 be the pullback of the nef semiample positive part N_0 of L on its scaling model. The comparison of canonical pullbacks and moving parts gives

$$H_0^n \longrightarrow \epsilon^n, \quad LH_0^{n-1} = H_0^n, \quad K_W H_0^{n-1} = K H_0^{n-1} \leq \frac{H_0^n}{r}. \quad (62)$$

To justify these intersections, recall that the maps $Y \dashrightarrow Y_t$ are small by the choice of the late model. Every divisor exceptional over Y is therefore also exceptional over Y_t . The differences discarded when passing to the moving parts are exceptional over the scaling model, and the transform of A is effective up to rational linear equivalence. Intersecting with H_0^{n-1} gives the displayed equalities and inequality. In particular, since $qm/r < 1$,

$$(2L + qmK)H_0^{n-1} \leq (2 + qm/r)H_0^n < 3H_0^n.$$

Choose t small enough that also $H_0^n < (2\epsilon)^n$, and now fix t, r and the resolution W . For a fixed ample divisor H_{amp} , put $H = H_0 + \eta H_{\text{amp}}$, where $\eta > 0$ is rational. The strict margins just obtained, together with $K_W H_0^{n-1} \leq H_0^n/r$, allow us to choose η small enough that continuity gives

$$H^n \leq (2\epsilon)^n, \quad K_W H^{n-1} \leq \frac{2H^n}{r}, \quad (2L + qmK)H^{n-1} \leq 4H^n. \quad (63)$$

We next choose the flag for the section estimates. The canonical divisor of W is pseudo-effective. Generic semipositivity, applied with empty boundary [11, Theorem 2.1], and restriction to a sufficiently general complete-intersection curve give the following fixed data. Apply restriction to the Harder–Narasimhan filtration and its factors [49, Theorem 6.1 and Remark 6.2]. Choose successively large divisible integers l_i such that $h_i = l_i H$ are integral very ample classes, and choose a smooth integral complete-intersection flag

$$W = W_n \supset W_{n-1} \supset \cdots \supset W_1 = C, \quad W_{i-1} \in |h_{n-i+1}|_{W_i},$$

for which

$$\mu_{\min}(\Omega_W^1|_C) \geq 0. \quad (64)$$

Here and below slopes on C mean degree divided by rank. One may obtain the flag by applying restriction to the finitely many Harder–Narasimhan quotients of Ω_W^1 .

9.2.2 A movable witness for all determinant weights

For the vanishing-order estimate, choose a very general point $x \in W$, away from the exceptional loci, and let $\pi_x : \widehat{W} \rightarrow W$ be its blowup, with exceptional divisor J . Set

$$D^+ = 2r(K + 2tA) + 2E.$$

This big divisor has the scalar sections controlled by Proposition 8.1: pushing to Y identifies the section spaces, since the added exceptional part does not change them. If $\rho = \text{vol}(2r(K + 2tA))^{1/n}$, then

$$\rho \leq 4r\mu_t \leq 8\epsilon$$

for our sufficiently small t . Thus every section of a sufficiently divisible multiple of D^+ has normalized order at x at most $4\rho \leq 32\epsilon$.

We convert this order bound into a curve inequality. It follows that $\pi_x^* D^+ - 40\epsilon J$ is not pseudo-effective. Otherwise its convex combination with the big class $\pi_x^* D^+$ would make $\pi_x^* D^+ - cJ$ big

for some rational $32\epsilon < c < 40\epsilon$, giving a section of excessive order. Movable-curve duality [8, Theorem 2.2] therefore supplies an actual covering curve class γ on $\widehat{W}$ such that

$$(\pi_x^* D^+) \cdot \gamma < 40\epsilon J \cdot \gamma, \quad J \cdot \gamma > 0. \quad (65)$$

The strictness lets us fix an algebraic family that will survive reduction. More precisely, the negative pairing can first be detected by a strongly movable curve: take the pushforward of a general complete intersection of very ample divisors on one fixed smooth birational model of $\widehat{W}$. Fix this model, its divisors, and the resulting covering family. Thus the choice consists of finite algebraic data, rather than a limiting real movable class. Positivity of $J \cdot \gamma$ follows from the strict inequality and pseudo-effectivity of $\pi_x^* D^+$.

To turn this curve into the witness required by Theorem 9.1, set $S = mK$ on W ; it is an integral Cartier divisor. We verify all its weight inequalities before proceeding to the jet construction. Put $Q_\lambda = L + K_W/2 + \lambda S$ for $0 \leq \lambda \leq q$. The equality $K_W = K + E$ gives

$$D^+ - Q_\lambda = (r - \frac{1}{2} - \lambda m)K + 3rtA + \frac{3}{2}E. \quad (66)$$

This class is pseudo-effective in characteristic zero: $r > qm + 1$, K is pseudo-effective, and A, E are effective up to the stated equivalences. Pairing with the fixed strongly movable class therefore gives

$$\pi_x^* Q_\lambda \cdot \gamma \leq \pi_x^* D^+ \cdot \gamma < 40\epsilon J \cdot \gamma, \quad J \cdot \gamma > 0.$$

These are the required movable inequalities. Only the two endpoint inequalities need to be retained as finite data, since the pairing is affine in λ . The common theorem spreads these fixed numerical inequalities and the actual birational complete-intersection witness; it does not spread the pseudo-effective assertion used to obtain them.

9.2.3 A finite two-slot jet system

We must similarly realize the two-slot jet estimate by finitely many sections before reducing. Their products will then supply the jet orders needed for arbitrarily large characteristics.

On $W \times W$, let

$$Z = \mathbb{P}(mK_1 \oplus mK_2), \quad \xi = \mathcal{O}_Z(1), \quad M = L_1 + L_2 + q\xi,$$

with the symmetric-power convention for the projective bundle. Proposition 8.3 gives a semiample big model whose Seshadri constant exceeds $2n + 2$ at a very general torus point. On the open set where its birational contraction is an isomorphism, the jet interpretation of the Seshadri constant consequently gives an integer $k_0 > 0$ such that $|k_0 M|$ generates jets of order at least $(2n + 2)k_0$ at a fixed smooth torus point $z_* \in Z$. Indeed, on the ample model choose a rational number c strictly between $2n + 2$ and its Seshadri constant. On the blowup of the selected smooth point, the pullback of the ample class minus c times the exceptional divisor is ample. Serre vanishing and the exceptional-divisor sequence then give jets of order $kc - 1$ for all sufficiently divisible large k . Pulling sections back on the isomorphic open gives the asserted bound. Enlarge k_0 to clear every denominator, in particular that of L , and fix finitely many sections realizing this jet surjection.

9.2.4 Applying the comparison

All the preceding choices are now fixed. With $S = mK$, the dimension, intersection and flag hypotheses of Theorem 9.1 are Equations (61), (63), and (64). The curve construction gives its

movable witness and both endpoint inequalities, and the finite jet system gives the required surjection at z_* . This point need not lie over the separately chosen testing point x . The proof of the common theorem also fixes its auxiliary divisor T_0 and finitely many filler sections before choosing any residual characteristic.

The common theorem now rules out these fixed data. Its contradiction compares two orders of the same nonzero determinant. After reduction, the Frobenius direct image $\mathcal{E}$ used there has rank p^n , so $\mathcal{E} \boxtimes \mathcal{E}$ has rank $R = p^{2n}$. The fixed jets give full evaluation rank R , while the small polarization bounds the rank on the diagonal by $R/4$. Thus a nonzero maximal minor vanishes to order at least $3R/4$ at (x', x') , where x' is the Frobenius image of the testing point x . Its line bundle is an actual external product. Applying the fixed curve inequality to each factor bounds the same order above by $R(80\epsilon + O(p^{-1}))$, a contradiction for large p . The theorem proves these determinant estimates for the fixed witnesses we have now supplied; it requires no additional geometric input here.

All constants are fixed before the residual prime varies. The contradiction excludes the assumed counterexample and proves Theorem 9.6.

10 Completion and change of ground field

Smooth nonvanishing is now available in the order required by the induction. We first finish the complex proof, including descent from the good model to the original normal variety. We then descend the globally generated Cartier multiple to an arbitrary algebraically closed field of characteristic zero.

Proof of Theorem 1.1 over $\mathbb{C}$. Use dimension induction in the real-boundary category of Proposition 2.5. In dimension zero the variety is a point. Suppose the good-model assertion holds in all smaller dimensions. The signed-representative argument (Theorem 3.1), the exclusions, and the jet estimates are then available with exactly that lower-dimensional hypothesis. Theorem 9.6 establishes smooth nonvanishing in the present dimension. Proposition 2.5 therefore proves the real-boundary good-model assertion in the present dimension. This is precisely the hypothesis needed to continue to the next dimension.

Apply this to the rational lc pair (X, B) of Theorem 1.1. Let $f : (X_d, B_d) \rightarrow (X, B)$ be a crepant dlt model and let (V, B_V) be its good log minimal model. Take a common resolution $a : W \rightarrow X_d$, $q : W \rightarrow V$, and put $p = f \circ a$. Then

$$p^*(K_X + B) = a^*(K_{X_d} + B_d) = q^*(K_V + B_V).$$

The first equality is crepancy. The second is the nef comparison in Lemma 2.2, since $K_{X_d} + B_d = f^*(K_X + B)$ is nef. The right side is semiample because (V, B_V) is a good model, so the pullback of $K_X + B$ is semiample. Normality gives $p_*\mathcal{O}_W = \mathcal{O}_X$. Choose a sufficiently divisible Cartier multiple $m(K_X + B)$ whose pullback is globally generated. The projection formula identifies its sections with those of its pullback. Surjectivity of evaluation upstairs implies surjectivity downstairs: otherwise a base point downstairs would make every pulled-back section vanish on its nonempty fiber. Hence $K_X + B$ is semiample. $\square$

Proposition 10.1 (Change of algebraically closed field). *The conclusion of Theorem 1.1 over $\mathbb{C}$ implies its conclusion over every algebraically closed field of characteristic zero.*

Proof. Let (X, B) be defined over such a field k . Choose a finitely generated subfield $k_1 \subset k$ over which the projective variety, the rational boundary, a Cartier multiple of its adjoint, and a log

resolution are all defined. After enlarging k_1 if necessary, these data base change to the given data. Let k_0 be its algebraic closure inside k , and denote the resulting pair by (X_0, B_0) . Choose an embedding $k_0 \hookrightarrow \mathbb{C}$, and write $(X_{\mathbb{C}}, B_{\mathbb{C}})$ for the complex base change.

The chosen log resolution tests log canonicity after either algebraically closed field extension: its boundary remains simple normal crossing and all discrepancy coefficients remain unchanged. The descended pair is consequently lc over both k_0 and $\mathbb{C}$.

Nefness is also invariant under such extensions. Indeed, a curve over an extension is represented by a point of a relative Hilbert scheme after its finite defining data have been spread over a finite-type parameter scheme. The degree of a fixed line bundle is constant in the resulting flat family. Specializing to a closed point over the algebraically closed smaller field preserves a negative degree, if one existed; some irreducible component of the specialized curve would then have negative degree. This contradicts nefness over that field. Conversely, curves over the smaller field remain available after extension. Apply the converse to $k_0 \subset k$: nefness of $K_X + B$ gives nefness of $K_{X_0} + B_0$. Apply the forward direction to $k_0 \hookrightarrow \mathbb{C}$: its complex base change is nef as well.

The complex case supplies an integer $m > 0$, enlarged to a multiple of the Cartier index fixed over k_0 , for which the Cartier line bundle $M = \mathcal{O}_{X_0}(m(K_{X_0} + B_0))$ becomes globally generated over $\mathbb{C}$. Proper flat base change for sections identifies

$$H^0(X_0, M) \otimes_{k_0} \mathbb{C} = H^0(X_{\mathbb{C}}, M_{\mathbb{C}}).$$

Tensoring the evaluation map for M with $\mathbb{C}$ therefore gives the surjective complex evaluation map. Its coherent cokernel is zero by faithful flatness. Extending from k_0 to k preserves this surjectivity, so the same integer m gives a globally generated line bundle on the original X . $\square$

Together with Proposition 10.1, the complex proof establishes Theorem 1.1 in its stated generality.

11 Good models, linear triviality, and boundary consequences

The real-boundary induction and the final line-bundle descent have different natural scopes. We first record good-model existence over $\mathbb{C}$, including the precise smooth canonical comparison used by fiber-space applications. We then deduce actual rational-linear triviality in Iitaka dimension zero over every field covered by Theorem 1.1. Finally, over $\mathbb{C}$, normalization gluing and a section-extension theorem give two consequences for reducible boundaries and nonnormal pairs.

Theorem 11.1 (Real-boundary good models over $\mathbb{C}$). *Every projective log canonical pair (X, B) over $\mathbb{C}$, with effective real boundary and pseudo-effective real Cartier adjoint $K_X + B$, has a good log minimal model. Real semiample has the meaning fixed in Section 2.*

Proof. The simultaneous induction in Section 10 starts with dimension zero and proves the full real-boundary conclusion of Proposition 2.5 at each dimension. In particular, its conclusion applies to the given pair before any rational-boundary or nefness specialization. $\square$

Corollary 11.2 (Smooth canonical good models). *Let X be a smooth projective complex variety with pseudo-effective K_X . There is a normal projective $\mathbb{Q}$ -factorial klt variety V and a K_X -negative birational contraction $X \dashrightarrow V$ such that K_V is $\mathbb{Q}$ -Cartier and semiample. On a common smooth resolution $p : W \rightarrow X$, $q : W \rightarrow V$, compatible canonical divisors satisfy*

$$p^*K_X = q^*K_V + E, \quad E \geq 0 \quad \text{and } E \text{ is } q\text{-exceptional}.$$

In particular $\kappa(X, K_X) = \kappa(V, K_V)$.

Proof. Apply Theorem 11.1 to the klt pair $(X, 0)$. Choose its $\mathbb{Q}$ -factorial log minimal model, as in the construction of Proposition 2.5. The klt clause of Lemma 2.2 shows that it extracts no divisor, remains klt, and has the displayed effective exceptional comparison. Its boundary is zero, being the strict transform of the zero boundary with no extracted divisors. Discrepancies strictly improve at contracted divisors, which is the K_X -negative condition; semiampness is the good-model conclusion. For sufficiently divisible $m > 0$, the effective q -exceptional divisor mE satisfies $q_*\mathcal{O}_W(mE) = \mathcal{O}_V$, since V is normal. Projection formula and the displayed comparison identify the pluricanonical section spaces. Their ratios agree in the common function field, proving the assertion about Iitaka dimension. $\square$

Lemma 11.3. *Let Z be a nonempty integral projective variety over an algebraically closed field, and let A be a semiample $\mathbb{Q}$ -Cartier divisor. If $\kappa(Z, A) = 0$, then $A \sim_{\mathbb{Q}} 0$.*

Proof. Choose a positive integer m such that the line bundle $L = \mathcal{O}_Z(mA)$ is globally generated. Its complete system gives

$$\phi : Z \longrightarrow \mathbb{P}(H^0(Z, L)^*), \quad L \simeq \phi^*\mathcal{O}(1).$$

The image has dimension at most $\kappa(Z, A) = 0$, so it is a point. A linear form nonzero at that point pulls back to a nowhere-vanishing section of L . Thus $L \simeq \mathcal{O}_Z$, an actual line-bundle isomorphism, and $A \sim_{\mathbb{Q}} 0$. $\square$

Corollary 11.4. *Let (X, B) satisfy the hypotheses of Theorem 1.1. If $\kappa(X, K_X + B) = 0$, then $K_X + B \sim_{\mathbb{Q}} 0$. In particular, every nef rational adjoint of Iitaka dimension zero on a projective klt fivefold over an algebraically closed field of characteristic zero is $\mathbb{Q}$ -linearly trivial.*

Proof. Theorem 1.1 gives semiampness on the original normal variety. Lemma 11.3 gives a positive Cartier multiple with trivial associated line bundle. Klt pairs are lc, so the fivefold assertion is a special case. $\square$

Thus the nef adjoint also has numerical dimension zero. This is a consequence of the trivializing Cartier multiple, not an additional hypothesis. The multiple may depend on the pair; no uniform index is asserted.

11.1 Semi-log-canonical abundance and dlt extension

We use the standard semi-log-canonical convention, in which the underlying projective scheme may be reducible or nonnormal. It is reduced, satisfies S_2 , is pure-dimensional, and has ordinary double crossings in codimension one. No component of the effective boundary is contained in the codimension-one singular locus. If $\nu : X^\nu \rightarrow X$ is the normalization, the boundary Θ defined by

$$K_{X^\nu} + \Theta = \nu^*(K_X + \Delta)$$

includes the conductor boundary, and (X^ν, Θ) is componentwise log canonical. These are the conventions of [26, Definition 2.5].

Corollary 11.5 (Semi-log-canonical abundance over $\mathbb{C}$). *Let (X, Δ) be a projective semi-log-canonical pair over $\mathbb{C}$, where Δ is an effective rational divisor and $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. If $K_X + \Delta$ is nef, then it is semiample.*

Proof. Write $X^\nu = \coprod_i X_i^\nu$ for the normalization and put $\Theta_i = \Theta|_{X_i^\nu}$. Each (X_i^ν, Θ_i) is a normal projective log canonical pair with effective rational boundary. The adjoint $K_{X_i^\nu} + \Theta_i$ is the pullback of $K_X + \Delta$, so it is nef. Theorem 1.1 makes each of these adjoints semiample. There are only finitely many components, hence one common positive Cartier multiple is globally generated on their disjoint union. Thus $\nu^*(K_X + \Delta)$ is semiample. The normalization gluing theorem [26, Theorem 1.5] now gives semiampleness on X . $\square$

Corollary 11.6 (Supported extension for dlt pairs over $\mathbb{C}$). *Let (X, Δ) be a projective dlt pair over $\mathbb{C}$, where Δ is an effective rational divisor, and put $S = \lfloor \Delta \rfloor$. Assume that $K_X + \Delta$ is nef and that, for an effective rational divisor D ,*

$$K_X + \Delta \sim_{\mathbb{Q}} D \geq 0, \quad S \subseteq \text{Supp } D.$$

Then, for every integer $m \geq 2$ for which $m(K_X + \Delta)$ is Cartier, the restriction map

$$H^0(X, \mathcal{O}_X(m(K_X + \Delta))) \longrightarrow H^0(S, \mathcal{O}_X(m(K_X + \Delta))|_S)$$

is surjective.

Proof. Set $B = \Delta - S$. The pair $(X, S + B)$ is dlt, B is an effective rational divisor, and $\lfloor S + B \rfloor = S$. Since a dlt pair is log canonical, Theorem 1.1 makes its nef adjoint semiample. The pair and the displayed effective representative therefore satisfy the hypotheses of [26, Proposition 5.12], which gives the stated restriction surjectivity for every Cartier multiple with $m \geq 2$. This is the supported extension formulation in [26, Conjecture 5.8]. $\square$

The first corollary settles the rational-boundary form of the semi-log-canonical abundance conjecture [26, Conjecture 1.3] in every dimension over $\mathbb{C}$. The second settles the supported dlt extension conjecture [26, Conjectures 1.10 and 5.8] with the precise Cartier-multiple range supplied by Proposition 5.12. Both are applications of the cited implications after normal abundance has been established here. The support condition $S \subseteq \text{Supp } D$ is retained; no arbitrary restriction-surjectivity statement is asserted.

12 Finite generation of rational log canonical rings

The preceding results also settle finite generation for rational lc pairs. The finite-generation conjecture asks whether finitely many rounded pluricanonical sections generate the log canonical ring [28, Conjecture A]. The assertion below concerns the full rounded ring, not merely a sufficiently divisible subring, and requires no nefness assumption. For the projective absolute statement, we combine the ordinary minimal-model theorem with log abundance and keep track of every rounded degree. For the proper relative statement over $\mathbb{C}$, we apply Fujino–Gongyo’s published reduction from good minimal models. In that relative statement the morphism is only required to be proper; the source need not be projective over $\mathbb{C}$.

For an integral Weil divisor G on a normal variety X , $\mathcal{O}_X(G)$ denotes its divisorial sheaf. For a rational divisor D on X and nonnegative integers m, n , the inclusions $\lfloor mD \rfloor + \lfloor nD \rfloor \leq \lfloor (m+n)D \rfloor$ give the usual multiplication on the section algebras below.

Corollary 12.1 (Finite generation for rational lc pairs).

- (i) *Let k be an algebraically closed field of characteristic zero, let (X, Δ) be a normal projective log canonical pair over k , and assume that $\Delta \geq 0$ is rational and $D = K_X + \Delta$ is $\mathbb{Q}$ -Cartier. Then*

$$R(X, D) = \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(\lfloor mD \rfloor))$$

is a finitely generated k -algebra.

- (ii) *Let (X, Δ) be a log canonical pair over $\mathbb{C}$, with $\Delta \geq 0$ rational and $K_X + \Delta$ $\mathbb{Q}$ -Cartier. For every proper morphism $f : X \rightarrow S$ onto an algebraic variety S , the relative log canonical algebra*

$$\mathcal{R}(X/S, K_X + \Delta) = \bigoplus_{m \geq 0} f_* \mathcal{O}_X(\lfloor m(K_X + \Delta) \rfloor)$$

is a finitely generated $\mathcal{O}_S$ -algebra.

Proof. In dimension zero, X is a point, and in (ii) so is S . The boundary and canonical divisor are zero, and the two algebras are $k[t]$ and $\mathcal{O}_S[t]$, respectively, with $\deg t = 1$. We therefore assume $\dim X \geq 1$.

We first prove (i). If D is not pseudo-effective, a nonzero section s in degree $m > 0$ would give

$$\operatorname{div}(s) + mD = (\operatorname{div}(s) + \lfloor mD \rfloor) + (mD - \lfloor mD \rfloor) \geq 0.$$

This would make D pseudo-effective. Thus all positive graded pieces vanish in this case. Since $H^0(X, \mathcal{O}_X) = k$, we have $R(X, D) = k$.

Suppose now that D is pseudo-effective. The ordinary-pair conclusion of the companion minimal-model theorem [53, Corollary 1.2] gives a normal projective lc pair (Y, Δ_Y) and a birational map $\phi : X \dashrightarrow Y$ extracting no divisors, with $\Delta_Y = \phi_* \Delta$ and $D_Y = K_Y + \Delta_Y$ $\mathbb{Q}$ -Cartier and nef. Log discrepancies do not decrease. Theorem 1.1 makes the rational adjoint D_Y semiample.

On a common smooth resolution $p : W \rightarrow X$, $q : W \rightarrow Y$, choose compatible canonical divisors. Then

$$p^* D = q^* D_Y + E, \quad E \geq 0, \quad E \text{ is } q\text{-exceptional.} \quad (67)$$

Indeed, the coefficient of E at a prime divisor F is $a(F; Y, \Delta_Y) - a(F; X, \Delta)$, which is nonnegative by the discrepancy comparison. Suppose F is not q -exceptional. Since ϕ extracts no divisors, F is the strict transform of a divisor on X . The pushed-forward boundary gives equality of these discrepancies. This proves the asserted support.

For any normal variety Z , rational Cartier divisor G , and $m \geq 0$, the nonzero global sections of $\mathcal{O}_Z(\lfloor mG \rfloor)$ are exactly the rational functions s satisfying

$$\operatorname{div}_Z(s) + mG \geq 0.$$

This is equivalent to the usual inequality with $\lfloor mG \rfloor$ because the coefficients of $\operatorname{div}_Z(s)$ are integers. If the inequality holds on X , pull back its effective $\mathbb{Q}$ -Cartier left side to W and push forward by q . Equation (67) and $q_* E = 0$ give the inequality on Y . Conversely, pull back the inequality on Y , add mE , and push forward by p . Thus, inside the common function field,

$$H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) = H^0(Y, \mathcal{O}_Y(\lfloor mD_Y \rfloor)) \quad (m \geq 0).$$

These identifications preserve multiplication, so they identify the full graded rings.

It remains to use semiample on Y . Choose $r > 0$ such that rD_Y is Cartier and globally generated. Its complete linear system defines $\psi : Y \rightarrow \mathbb{P}_k^N$ with $\mathcal{O}_Y(rD_Y) \simeq \psi^*\mathcal{O}_{\mathbb{P}^N}(1)$. Let $A = k[T_0, \dots, T_N]$ act on $R(Y, D_Y)$ by pulling back the coordinate sections, each in degree r . For $0 \leq i < r$, put $\mathcal{F}_i = \psi_*\mathcal{O}_Y(\lfloor iD_Y \rfloor)$, which is coherent on $\mathbb{P}_k^N$ because ψ is proper. Since

$$\lfloor (\ell r + i)D_Y \rfloor = \ell(rD_Y) + \lfloor iD_Y \rfloor \quad (\ell \geq 0),$$

projection formula identifies the part in degrees congruent to i modulo r , as an A -module, with

$$\bigoplus_{\ell \geq 0} H^0(\mathbb{P}_k^N, \mathcal{F}_i(\ell)).$$

This is a finite A -module by the standard finite-generation theorem for section modules [61, Lemma 30.14.1(5)]. There are only r residue classes, so $R(Y, D_Y)$ is finite as an A -module and hence finitely generated as a k -algebra. This proves (i).

For (ii), set $n = \dim X$. In the notation of Fujino–Gongyo, Conjecture $C_{\leq n-1}$ asks for good minimal models of projective $\mathbb{Q}$ -factorial dlt pairs with effective real boundary and pseudo-effective adjoint in dimensions at most $n-1$. Theorem 11.1 supplies this statement. Its good-model convention is the one fixed in Section 2, with a $\mathbb{Q}$ -factorial dlt model as in Proposition 2.5. Theorem 1.1 of Fujino–Gongyo [28] therefore gives their Conjecture B_n : finite generation of the log canonical ring for projective rational plt pairs of dimension n whose adjoint is big and whose boundary has irreducible round-down. Their relative reduction, Corollary 1.4, applies to the given rational lc pair and proper morphism onto S , and gives exactly (ii) without requiring X to be projective over $\mathbb{C}$. Their Theorem 1.1 also gives (i) directly when $k = \mathbb{C}$. $\square$

The good-model input allows real boundaries, but both finite-generation statements retain rational boundaries. In the absolute proof, rationality is what leaves finitely many residue-class section modules. The relative assertion is finite generation over the algebraic base S in the complex setting; it is not a claim about rounded rings for real boundaries or about analytic bases. No uniform degree bound for generators is asserted.

13 Rational curves, cotangent positivity, and covers

We first combine smooth canonical nonvanishing with classical results on rational curves and cotangent tensors. We then use smooth abundance to remove the abundance premise from the quasi-projective-cover results of Claudon, Höring and Kollár. Throughout this section, $\kappa(X) = \kappa(X, K_X)$. We retain the convention $\kappa(\text{pt}) = 0$, and regard a point as rationally connected but not uniruled.

Corollary 13.1 (Uniruledness and Mumford’s criterion). *Let X be a smooth connected projective complex variety.*

- (i) X is uniruled if and only if $\kappa(X) = -\infty$.
- (ii) X is rationally connected if and only if

$$H^0(X, (\Omega_X^1)^{\otimes m}) = 0 \quad \text{for every integer } m > 0.$$

Proof. For positive-dimensional X , the theorem of Boucksom–Demailly–Păun–Peternell [8, Corollary 0.3 and Theorem 2.6] says that X is non-uniruled exactly when K_X is pseudo-effective. Corollary 11.2 then gives $\kappa(X) \geq 0$: the semiample canonical divisor on the good model has a

nonzero section in a positive multiple, and the pluricanonical section spaces agree. Conversely, a nonzero pluricanonical section makes K_X pseudo-effective. This proves (i).

For (ii), rational connectedness forces all positive covariant tensor differentials to vanish, as recalled by Lazić and Peternell. For the converse, their Proposition 2.4 reduces Mumford's criterion in dimension n to tensor nonvanishing for smooth projective varieties of positive dimension at most n whose canonical class is pseudo-effective [46, Proposition 2.4]. If Y is such a variety of dimension d , the preceding good-model argument supplies a nonzero section of mK_Y for some integer $m > 0$. Antisymmetrization embeds Ω_Y^d into $(\Omega_Y^1)^{\otimes d}$ over $\mathbb{C}$; tensoring this injection m times gives the required nonzero tensor differential. This is the observation in [46, Remark 2.5], so their reduction applies in every dimension. Both assertions for a point follow from our conventions. $\square$

For a smooth connected projective complex variety X , let $R(X)$ be a smooth projective model of the base of its maximal rationally connected (MRC) fibration. This base is determined up to birational equivalence and is non-uniruled unless it is a point. We use $\Omega_X^0 = \mathcal{O}_X$.

Corollary 13.2 (Pluriforms and the rational quotient). (i) *For every smooth connected projective complex variety X ,*

$$\dim R(X) = \max_{\substack{p \in \mathbb{Z}_{\geq 0}, k \in \mathbb{Z}_{>0} \\ H^0(X, \text{Sym}^k(\Omega_X^p)) \neq 0}} p.$$

(ii) *A compact connected Kähler manifold X is rationally connected if and only if*

$$H^0(X, \text{Sym}^k(\Omega_X^p)) = 0 \quad \text{for every pair of integers } p, k > 0.$$

Proof. Write $r = \dim R(X)$ in (i). Resolve the MRC map to a morphism $f : X' \rightarrow R(X)$, with X' smooth and projective. For a smooth general fiber F , the cotangent sequence is

$$0 \longrightarrow \mathcal{O}_F^{\oplus r} \longrightarrow \Omega_{X'}^1|_F \longrightarrow \Omega_F^1 \longrightarrow 0.$$

If $p > r$, every graded piece of the induced exterior-power filtration on $\Omega_{X'}^p|_F$ contains a positive exterior power of Ω_F^1 : at most r factors can come from $\mathcal{O}_F^{\oplus r}$. Consequently, for $k > 0$, the graded pieces of the induced filtration on $\text{Sym}^k(\Omega_{X'}^p)|_F$ are direct summands of sums of positive tensor powers of Ω_F^1 . Here we use antisymmetrization and symmetrization over $\mathbb{C}$. Rational connectedness of F makes all their sections vanish, by Corollary 13.1. A global section therefore vanishes on every sufficiently general fiber, hence on X' . Birational invariance of global covariant tensor differentials on smooth projective varieties gives the same vanishing on X . This is the upper-bound argument of [9, Section 3].

If $r > 0$, the smooth non-uniruled base has $\kappa(R(X)) \geq 0$ by Corollary 13.1. A nonzero section of $kK_{R(X)}$ for some $k > 0$ pulls back to a nonzero section of $\text{Sym}^k(\Omega_X^r)$. The pullback is taken on X' and descends to X by the same birational invariance. If $r = 0$, the term $p = 0$ gives a nonzero section and the same equality follows. This is the equality $r = r^-$ predicted in the remark ending [9, Section 3].

For (ii), the forward vanishing is the rational-curve argument recalled by Brunebarbe and Campana in the same section. Conversely, the stated vanishing gives $H^{2,0}(X) = H^0(X, \Omega_X^2) = 0$. Kodaira's projectivity criterion then makes X projective: the real degree-two cohomology is of type $(1, 1)$, so a rational class sufficiently close to a Kähler class is still Kähler, and clearing denominators gives an integral Kähler class. This is the projectivity reduction in [9, Section 3]. Part (i) now applies. Its maximum is zero because every term with $p > 0$ vanishes, so the MRC base is a point and X is rationally connected. $\square$

We next use ordinary cotangent invariants, with no boundary. If $\mathcal{L}$ is a coherent rank-one subsheaf of a vector bundle on a smooth projective variety, its double dual is a line bundle; put $\kappa(\mathcal{L}) = \kappa(\mathcal{L}^{**})$. Define

$$\omega(X) = \max_{\substack{m \geq 0 \\ \mathcal{L} \subset (\Omega_X^1)^{\otimes m} \\ \text{rank } \mathcal{L} = 1}} \kappa(\mathcal{L}),$$

where m is an integer and the value is $-\infty$ if no such sheaf has nonnegative Iitaka dimension. The inclusion of $\mathcal{L}$ extends to its double dual: away from codimension two these sheaves agree, and a map into a locally free sheaf extends across that subset. Taking tensor powers then realizes the systems $H^0(X, (\mathcal{L}^{**})^{\otimes a})$, for integers $a > 0$, inside higher cotangent tensor powers. Thus allowing all m makes this equivalent to Campana's definition using the dimensions of the rational maps defined by $H^0(X, \mathcal{L})$. For positive-dimensional X , also put

$$\kappa^+(X) = \max_{\substack{p \geq 0 \\ 0 \neq \mathcal{F} \subset \Omega_X^p}} \kappa(\det \mathcal{F}).$$

Here p is an integer, $\mathcal{F}$ is coherent, and $\det \mathcal{F} = (\wedge^{\text{rk } \mathcal{F}} \mathcal{F})^{**}$. These are the invariants used in Campana's Appendix A to Taji [60].

Corollary 13.3 (Cotangent invariants and the MRC base). *Let X be a smooth connected projective complex variety. Then*

$$\omega(X) = \omega(R(X)) = \begin{cases} -\infty, & \text{if } R(X) \text{ is a point,} \\ \kappa(R(X)), & \text{if } \dim R(X) > 0. \end{cases}$$

In particular, if X is positive-dimensional and non-uniruled, then $\omega(X) = \kappa(X)$. If X is positive-dimensional and $\kappa(X) \geq 0$, then

$$\kappa^+(X) = \kappa(X).$$

Proof. Campana's birationally invariant fibration properties [60, Appendix A, property 4] give $\omega(X) = \omega(R(X))$: the general MRC fiber is rationally connected and has $\omega = -\infty$. Indeed any rank-one subsheaf with nonnegative Iitaka dimension would give a section of a positive tensor power, contrary to Corollary 13.1; a point has $\omega = -\infty$ by the empty maximum. Birational invariance permits resolving the MRC map. If $R(X)$ is a point, the displayed formula therefore follows without changing our convention $\kappa(\text{pt}) = 0$.

Suppose that $R = R(X)$ has positive dimension. It is non-uniruled, so $\kappa(R) \geq 0$ by Corollary 13.1. To apply Campana's abundance deduction [60, Conjecture A.2 and Remark A.3], take a resolved Iitaka fibration $R' \rightarrow Z$, with R' and Z smooth and projective, R' birational to R , and $\dim Z = \kappa(R)$. Its smooth general fiber F has $\kappa(F) = 0$. If F is positive-dimensional, Corollary 11.2 and Lemma 11.3 give a normal birational good model V_F with $K_{V_F} \sim_{\mathbb{Q}} 0$. In particular its canonical divisor is numerically trivial on its smooth locus. Campana's criterion [60, Appendix A, Examples A.1(2)] says that the existence of such a normal birational model implies $\omega(F) = 0$; it does not require K_F itself to be trivial. Property 4 of the same appendix and birational invariance now give $\omega(R) = \omega(R') \leq \dim Z = \kappa(R)$. If F is a point, this upper bound is simply $\omega(R) \leq \dim R = \kappa(R)$. The reverse inequality $\omega(R) \geq \kappa(R)$ is one of the defining cotangent inequalities recorded in the same appendix. This proves the piecewise formula. A non-uniruled X has $R(X)$ birational to X , giving the first stated specialization.

Campana also records $\omega(X) \geq \kappa^+(X) \geq \kappa(X)$ [60, Appendix A, property 2]. When $\kappa(X) \geq 0$, Corollary 13.1 makes X non-uniruled, and the two outer terms are equal by the formula just proved. Thus $\kappa^+(X) = \kappa(X)$. This is the abundance consequence recalled by Taji [60, Theorem 1.3], who attributes it to Campana [10, Proposition 3.10]. $\square$

Corollary 13.4 (Finite fundamental group in Kodaira dimension zero). *Let X be a positive-dimensional smooth connected projective complex variety. If $\kappa(X) = 0$ and $\chi(X, \mathcal{O}_X) \neq 0$, then its ordinary topological fundamental group $\pi_1(X)$ is finite.*

Proof. Corollary 13.3 gives $\kappa^+(X) = 0$. Campana’s criterion [10, Corollary 5.3], in the form recalled by Taji [60, Theorem 1.6], gives the conclusion. $\square$

13.1 Quasi-projective covers

Here a quasi-projective cover means a complex analytic covering space biholomorphic to a quasi-projective variety; its deck transformations need not be algebraic. The following application describes such covers in terms of bundles over abelian varieties.

Corollary 13.5 (Quasi-projective covering spaces). *Let X be a connected normal projective complex variety.*

- (i) *Its universal cover is biholomorphic to a quasi-projective variety if and only if there is a finite étale Galois cover $X' \rightarrow X$ and a locally trivial holomorphic fiber bundle $X' \rightarrow A$, where A is an abelian variety and the fiber is simply connected and projective.*
- (ii) *If $\tilde{X} \rightarrow X$ is an infinite étale Galois cover and $\tilde{X}$ is biholomorphic to a quasi-projective variety, there are a finite étale Galois cover $X' \rightarrow X$, a locally trivial holomorphic fiber bundle $\alpha : X' \rightarrow A$ over an abelian variety, and an étale cover $\tilde{A} \rightarrow A$ with no positive-dimensional compact analytic subvarieties, such that*

$$\tilde{X} \simeq X' \times_A \tilde{A}$$

as covering spaces over X .

Proof. Theorem 1.1 with zero boundary on a smooth projective complex variety gives precisely Conjecture 1.2 of Claudon–Hörling–Kollár [14]. Their Theorem 1.1 and Corollary 1.5 therefore give (i) and (ii), respectively. $\square$

The simply connected fiber condition belongs to (i), not to the general-cover statement (ii). Local triviality is holomorphic; no global product after a finite cover or compact Kähler extension is asserted.

14 Relative abundance for projective analytic morphisms

The complex rational case of Theorem 1.1 supplies the algebraic fiber input in Fujino’s relative analytic reduction [25, Theorem 1.10 and its proof]. We record only its normal rational case. Throughout this section, complex analytic spaces are Hausdorff and second-countable, and a complex variety means a reduced and irreducible complex analytic space, as in [25, p. 5].

Here a projective analytic morphism is proper and possesses a relatively ample line bundle. Rational Cartierness is local on the source; a single global Cartier index is not part of that definition. A divisor is π -nef over Y if it has nonnegative intersection with every projective integral curve contracted by π , over every point of Y . These are the conventions of [24, Definitions 2.28, 2.32 and 2.48].

Corollary 14.1 (Relative analytic abundance). *Let $\pi : X \rightarrow Y$ be a projective surjective morphism of normal complex analytic varieties. Let (X, Δ) be a log canonical pair with Δ an effective rational*

divisor and $D = K_X + \Delta$ $\mathbb{Q}$ -Cartier. Assume that D is π -nef over all of Y . For every compact subset $W \subset Y$, there exist an analytic open neighborhood $U \supset W$ and a positive integer m such that $mD|_{X_U}$ is Cartier, where $X_U = \pi^{-1}(U)$, and the line bundle

$$L = \mathcal{O}_{X_U}(mD|_{X_U})$$

is generated relative to $\pi_U : X_U \rightarrow U$. Equivalently, the evaluation map

$$\pi_U^*(\pi_U)_*L \longrightarrow L$$

is surjective.

Proof. Since π is proper, $\pi^{-1}(W)$ is compact. Local rational Cartier indices therefore have a common multiple m_0 on an open neighborhood O of $\pi^{-1}(W)$. The image of the closed set $X \setminus O$ is closed and disjoint from W , so there is an open neighborhood $V \supset W$ with $\pi^{-1}(V) \subset O$. Thus m_0D is Cartier over V .

Fix $P \in W$. Following [25, the proof of Theorem 1.10, p. 30], work over a sufficiently small neighborhood of P contained in V and take a crepant dlt blow-up $p : (X', \Delta') \rightarrow (X, \Delta)$. Put $\pi' = \pi \circ p$ and $D' = K_{X'} + \Delta' = p^*D$. For every log canonical stratum S of (X', Δ') , including X' itself, adjunction gives an effective rational boundary Δ_S with

$$K_S + \Delta_S = D'|_S.$$

The strata are normal, and this adjoint is nef over $\pi'(S)$. On each normal projective component F of an analytically sufficiently general fiber of $S \rightarrow \pi'(S)$, restriction gives a log canonical pair (F, Δ_F) with effective rational boundary and nef adjoint $K_F + \Delta_F = (K_S + \Delta_S)|_F$. Theorem 1.1 makes this adjoint semiample, so its Iitaka and numerical dimensions agree. Thus $D'|_S$ is abundant over $\pi'(S)$ for every stratum: this is precisely π' -log abundance in [25, Definitions 2.10 and 2.13].

The nef-and-log-abundant theorem [25, Theorem 1.4] now gives a relatively generated multiple of D' over a neighborhood $U_P \subset V$ of P . Enlarge its integer to a multiple m_P of m_0 ; tensor powers preserve relative generation. Normality gives $p_*\mathcal{O}_{X'} = \mathcal{O}_X$, and projection formula identifies the direct images over U_P of $\mathcal{O}_X(m_P D)$ and $\mathcal{O}_{X'}(m_P D')$. Their evaluation maps are related by pullback. Generation upstairs therefore implies generation downstairs: a base point downstairs would make every pulled-back section vanish on its nonempty fiber.

Choose finitely many of these neighborhoods $U_{P_1}, \dots, U_{P_s}$ covering W , and let m be a common multiple of $m_0, m_{P_1}, \dots, m_{P_s}$. Put $U = \bigcup_i U_{P_i}$. The divisor mD is Cartier over U , and its line bundle is relatively generated on each U_{P_i} . Surjectivity of evaluation is local on the base, so it holds over U , as asserted. $\square$

The base Y need not be algebraic, compact, or Stein, and W need not be a Stein compact subset. The integer m may depend on the pair, the morphism, and W ; no single Cartier index or relatively generated multiple over all of a noncompact base is asserted. The nefness premise is over all of Y , not only over W . Projectivity cannot be replaced here by mere properness or by a Kähler hypothesis: when Y is a point it requires X to be projective. The statement does not include real boundaries or semi-log-canonical analytic pairs.

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